Relations

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This chapter contains some material about relations. Notably, we discuss and explore:

1. The definition of relations (Section 8.1.1).
2. How relations may be viewed as decategorification of profunctors (Section 8.1.2).
3. The various kinds of categories that relations form, namely:
   (a) A category (Section 8.3.2).
   (b) A monoidal category (Section 8.3.3).
   (c) A 2-category (Section 8.3.4).
   (d) A double category (Section 8.3.5).
4. The various categorical properties of the 2-category of relations, including:
   (a) The self-duality of Rel and **Rel** (Proposition 8.4.1.1.1).
   (b) Identifications of equivalences and isomorphisms in **Rel** with bijections (Proposition 8.4.2.1.1).
   (c) Identifications of adjunctions in **Rel** with functions (Proposition 8.4.3.1.1).
   (d) Identifications of monads in **Rel** with preorders (Proposition 8.4.4.1.1).
   (e) Identifications of comonads in **Rel** with subsets (Proposition 8.4.5.1.1).

(f) A description of the monoids and comonoids in **Rel** with respect to the Cartesian product (Remark 8.4.6.1.1).

(g) Characterisations of monomorphisms in Rel (Proposition 8.4.7.1.1).

(h) Characterisations of 2-categorical notions of monomorphisms in **Rel** (Proposition 8.4.8.1.1).

(i) Characterisations of epimorphisms in Rel (Proposition 8.4.9.1.1).

(j) Characterisations of 2-categorical notions of epimorphisms in **Rel** (Proposition 8.4.10.1.1).

(k) The partial co/completeness of Rel (Proposition 8.4.11.1.1).

(l) The existence or non-existence of Kan extensions and Kan lifts in Rel (Remark 8.4.12.1.1).

(m) The closedness of **Rel** (Proposition 8.4.13.1.1).

(n) The identification of **Rel** with the category of free algebras of the powerset monad on Sets (Proposition 8.4.14.1.1).

5. The adjoint pairs

$$R_! \dashv R_{-1} \colon \mathcal{P}(A) \rightleftarrows \mathcal{P}(B),$$
$$R^{-1} \dashv R_* \colon \mathcal{P}(B) \rightleftarrows \mathcal{P}(A)$$

of functors (morphisms of posets) between $\mathcal{P}(A)$ and $\mathcal{P}(B)$ induced by a relation $R \colon A \nrightarrow B$, as well as the properties of $R_!$, R_{-1}, R^{-1}, and R_* (Section 8.5).

Of particular note are the following points:

(a) These two pairs of adjoint functors are the counterpart for relations of the adjoint triple $f_! \dashv f^{-1} \dashv f_*$ induced by a function $f \colon A \to B$ studied in Constructions With Sets, Section 4.6.

(b) We have $R_{-1} = R^{-1}$ iff R is total and functional (Item 8 of Proposition 8.5.2.1.4).

(c) As a consequence of the previous item, when R comes from a function f, the pair of adjunctions

$$R_! \dashv R_{-1} = R^{-1} \dashv R_*$$

reduces to the triple adjunction

$$f_! \dashv f^{-1} \dashv f_*$$

from Constructions With Sets, Section 4.6.

(d) The pairs $R_! \dashv R_{-1}$ and $R^{-1} \dashv R_*$ turn out to be rather important later on, as they appear in the definition and study of continuous, open, and closed relations between topological spaces (Topological Spaces, **??**).

6. A description of two notions of "skew composition" on $\mathbf{Rel}(A, B)$, giving rise to left and right skew monoidal structures analogous to the left skew monoidal structure on $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$ appearing in the definition of a relative monad (Sections 8.6 and 8.7).

This chapter is under revision. TODO:

1. Define apartness composition
2. Examples:
   (a) partial functions
   (b) hyperring examples (tropical, Berkovich, etc.)
3. Revise Section 8.3
4. Replicate Section 8.3 for apartness composition
5. Revise Section 8.4
   (a) Add modules over monads in **Rel**
   (b) internal relations, https://ncatlab.org/nlab/show/internal+relation
   (c) Co/limits in **Rel**.
   (d) Codensity monad $\mathrm{Ran}_J(J)$ of a relation (What about $\mathrm{Rift}_J(J)$?)
      i. Density comonad $\mathrm{Lan}_J(J)$ of a relation when it exists (what about $\mathrm{Lift}_J(J)$?)
   (e) Fibrations in **Rel**, like discrete fibrations and Street fibrations

6. Replicate Section 8.4 for apartness composition
7. Revise Section 8.5
8. Add subsection "A Six Functor Formalism for Sets, Part 2", now with relations, building upon Section 8.5.
9. Replicate Section 8.5 for apartness composition
10. Revise sections on skew monoidal structures on $\mathbf{Rel}(A, B)$
11. Replicate the sections on skew monoidal structures on $\mathbf{Rel}(A, B)$ for apartness composition.
12. Explore relative co/monads in $\mathbf{Rel}$, defined to be co/monoids in $\mathbf{Rel}(A, B)$ with its left/right skew monoidal structures of Relations, Sections 8.6 and 8.7
13. Consider adding the sections
    - The Monoidal Bicategory of Relations
    - The Monoidal Double Category of Relations

    to Relations.
14. functional total relations defined with "satisfying the following equivalent conditions:"

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8.1 Relations

8.1.1 Foundations

Let A and B be sets.

Definition 8.1.1.1.1 ► Relations

A **relation** $R\colon A \nrightarrow B$ **from** A **to** B[1,2] is equivalently:

1. A subset R of $A \times B$.
2. A function from $A \times B$ to $\{\text{true}, \text{false}\}$.
3. A function from A to $\mathcal{P}(B)$.
4. A function from B to $\mathcal{P}(A)$.
5. A cocontinuous morphism of posets from $(\mathcal{P}(A), \subset)$ to $(\mathcal{P}(B), \subset)$.
6. A continuous morphism of posets from $(\mathcal{P}(B), \supset)$ to $(\mathcal{P}(A), \supset)$.

[1] *Further Terminology:* Also called a **multivalued function from** A **to** B.
[2] *Further Terminology:* When $A = B$, we also call $R \subset A \times A$ a **relation on** A.

PROOF 8.1.1.1.2 ▶ PROOF OF THE EQUIVALENCES IN DEFINITION 8.1.1.1.1

(We will prove that Items 1 to 6 are indeed equivalent in a bit.)

REMARK 8.1.1.1.3 ▶ UNWINDING ITEM 1, I

We may think of a relation $R\colon A \nrightarrow B$ as a function from A to B that is *multivalued*, assigning to each element a in A a set $R(a)$ of elements of B, thought of as the *set of values of R at a*.
Note that this includes also the possibility of R having no value at all on a given $a \in A$ when $R(a) = \emptyset$.

REMARK 8.1.1.1.4 ▶ UNWINDING ITEM 2, II

Another way of stating the equivalence between Items 1 to 5 of Definition 8.1.1.1.1 is by saying that we have bijections of sets

$$\begin{aligned}\{\text{relations from } A \text{ to } B\} &\cong \mathcal{P}(A \times B)\\ &\cong \mathsf{Sets}(A \times B, \{\mathsf{true}, \mathsf{false}\})\\ &\cong \mathsf{Sets}(A, \mathcal{P}(B))\\ &\cong \mathsf{Sets}(B, \mathcal{P}(A))\\ &\cong \mathsf{Pos}^{\mathcal{J}}(\mathcal{P}(A), \mathcal{P}(B))\\ &\cong \mathsf{Pos}^{\mathcal{C}}(\mathcal{P}(B), \mathcal{P}(A))\end{aligned}$$

natural in $A, B \in \mathrm{Obj}(\mathsf{Sets})$, where $\mathcal{P}(A)$ and $\mathcal{P}(B)$ are endowed with the poset structure given by inclusion.

PROOF 8.1.1.1.5 ▶ PROOF OF THE EQUIVALENCES IN DEFINITION 8.1.1.1.1

We claim that Items 1 to 5 are indeed equivalent:

- *Item 1* $\Longleftrightarrow$ *Item 2:* This is a special case of Constructions With

Sets, Items 2 and 3 of Proposition 4.5.1.1.4.

- *Item 2* $\iff$ *Item 3*: This follows from the bijections

$$\begin{aligned}\mathsf{Sets}(A \times B, \{\mathsf{true}, \mathsf{false}\}) &\cong \mathsf{Sets}(A, \mathsf{Sets}(B, \{\mathsf{true}, \mathsf{false}\})) \\ &\cong \mathsf{Sets}(A, \mathcal{P}(B)),\end{aligned}$$

  where the last bijection is from Constructions With Sets, Items 2 and 3 of Proposition 4.5.1.1.4.

- *Item 2* $\iff$ *Item 4*: This follows from the bijections

$$\begin{aligned}\mathsf{Sets}(A \times B, \{\mathsf{true}, \mathsf{false}\}) &\cong \mathsf{Sets}(B, \mathsf{Sets}(B, \{\mathsf{true}, \mathsf{false}\})) \\ &\cong \mathsf{Sets}(B, \mathcal{P}(A)),\end{aligned}$$

  where again the last bijection is from Constructions With Sets, Items 2 and 3 of Proposition 4.5.1.1.4.

- *Item 2* $\iff$ *Item 5*: This follows from the universal property of the powerset $\mathcal{P}(X)$ of a set X as the free cocompletion of X via the characteristic embedding

$$\chi_X \colon X \hookrightarrow \mathcal{P}(X)$$

  of X into $\mathcal{P}(X)$, as in Constructions With Sets, Proposition 4.4.5.1.1. In particular, the bijection

$$\mathsf{Sets}(A, \mathcal{P}(B)) \cong \mathsf{Pos}^{\partial}(\mathcal{P}(A), \mathcal{P}(B))$$

  is given by extending each $f \colon A \to \mathcal{P}(B)$ in $\mathsf{Sets}(A, \mathcal{P}(B))$ from A to all of $\mathcal{P}(A)$ by taking its left Kan extension along χ_X, recovering the direct image function $f_! \colon \mathcal{P}(A) \to \mathcal{P}(B)$ of f of Constructions With Sets, Definition 4.6.1.1.1.

- *Item 5* $\iff$ *Item 6*: Omitted.

This finishes the proof. ∎

Notation 8.1.1.1.6 ▶ Further Notation for Relations

Let A and B be sets and let $R\colon \nrightarrow B$ be a relation from A to B.

1. We write $\mathrm{Rel}(A, B)$ for the set of relations from A to B.
2. We write $\mathbf{Rel}(A, B)$ for the sub-poset of $(\mathcal{P}(A \times B), \subset)$ spanned by the relations from A to B.
3. Given $a \in A$ and $b \in B$, we write $a \sim_R b$ to mean $(a, b) \in R$.
4. When viewing R as a function

$$R\colon A \times B \to \{\mathsf{t}, \mathsf{f}\},$$

we write R_a^b for the value of R at (a, b).[1]

[1]The choice to write R_a^b in place of R_b^a is to keep the notation consistent with the notation we will later employ for profunctors in **??**.

Example 8.1.1.1.7 ▶ The Trivial Relation

The **trivial relation on** A **and** B is the relation $\sim_{\mathrm{triv}}$ defined equivalently as follows:

1. As a subset of $A \times B$, we have

$$\sim_{\mathrm{triv}} \stackrel{\mathrm{def}}{=} A \times B.$$

2. As a function from $A \times B$ to $\{\mathsf{true}, \mathsf{false}\}$, the relation $\sim_{\mathrm{triv}}$ is the constant function

$$\Delta_{\mathsf{true}}\colon A \times B \to \{\mathsf{true}, \mathsf{false}\}$$

from $A \times B$ to $\{\mathsf{true}, \mathsf{false}\}$ taking the value true.

3. As a function from A to $\mathcal{P}(B)$, the relation $\sim_{\mathrm{triv}}$ is the function

$$\Delta_{\mathsf{true}}\colon A \to \mathcal{P}(B)$$

defined by

$$\Delta_{\mathsf{true}}(a) \stackrel{\mathrm{def}}{=} B$$

for each $a \in A$.

4. Lastly, it is the unique relation R on A and B such that we have $a \sim_R b$ for each $a \in A$ and each $b \in B$.

Example 8.1.1.1.8 ▶ The Cotrivial Relation

The **cotrivial relation on** A **and** B is the relation $\sim_{\text{cotriv}}$ defined equivalently as follows:

1. As a subset of $A \times B$, we have

$$\sim_{\text{cotriv}} \stackrel{\text{def}}{=} \emptyset.$$

2. As a function from $A \times B$ to $\{\text{true}, \text{false}\}$, the relation $\sim_{\text{cotriv}}$ is the constant function

$$\Delta_{\text{false}} \colon A \times B \to \{\text{true}, \text{false}\}$$

from $A \times B$ to $\{\text{true}, \text{false}\}$ taking the value false.

3. As a function from A to $\mathcal{P}(B)$, the relation $\sim_{\text{cotriv}}$ is the function

$$\Delta_{\text{false}} \colon A \to \mathcal{P}(B)$$

defined by

$$\Delta_{\text{false}}(a) \stackrel{\text{def}}{=} \emptyset$$

for each $a \in A$.

4. Lastly, it is the unique relation R on A and B such that we have $a \nsim_R b$ for each $a \in A$ and each $b \in B$.

Example 8.1.1.1.9 ▶ The Characteristic Relation of a Set

The characteristic relation

$$\chi_X(-_1, -_2) \colon X \times X \to \{\mathsf{t}, \mathsf{f}\}$$

on X of Constructions With Sets, Definition 4.5.3.1.1, defined by

$$\chi_X(x, y) \stackrel{\text{def}}{=} \begin{cases} \text{true} & \text{if } x = y, \\ \text{false} & \text{if } x \neq y \end{cases}$$

for each $x, y \in X$, is another example of a relation.

Proposition 8.1.1.1.10 ▶ Properties of Relations

Let A and B be sets and let $R, S \colon A \nrightarrow B$ be relations.

1. *End Formula for the Set of Inclusions of Relations.* We have

$$\mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, S) \cong \int_{a \in A} \int_{b \in B} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, S_a^b\right).$$

Proof 8.1.1.1.11 ▶ Proof of Proposition 8.1.1.1.10

Item 1: End Formula for the Set of Inclusions of Relations

Unwinding the expression inside the end on the right hand side, we have

$$\int_{a \in A} \int_{b \in B} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, S_a^b\right) \cong \begin{cases} \mathrm{pt} & \text{if, for each } a \in A \text{ and each } b \in B, \\ & \text{we have } \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, S_a^b\right) \cong \mathrm{pt} \\ \emptyset & \text{otherwise.} \end{cases}$$

Since we have $\mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, S_a^b\right) = \{\text{true}\} \cong \mathrm{pt}$ exactly when $R_a^b = \text{false}$ or $R_a^b = S_a^b = \text{true}$, we get

$$\int_{a \in A} \int_{b \in B} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, S_a^b\right) \cong \begin{cases} \mathrm{pt} & \text{if, for each } a \in A \text{ and each } b \in B, \\ & \text{if } a \sim_R b, \text{ then } a \sim_S b, \\ \emptyset & \text{otherwise.} \end{cases}$$

On the left hand-side, we have

$$\mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, S) \cong \begin{cases} \mathrm{pt} & \text{if } R \subset S, \\ \emptyset & \text{otherwise.} \end{cases}$$

Since $(a \sim_R b \implies a \sim_S b)$ iff $R \subset S$, the two sets above are isomorphic. This finishes the proof.

8.1.2 Relations as Decategorifications of Profunctors

Remark 8.1.2.1.1 ► Relations as Decategorifications of Profunctors I

The notion of a relation is a decategorification of that of a profunctor:

1. A profunctor from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a functor

$$\mathfrak{p}\colon \mathcal{D}^{\mathrm{op}} \times \mathcal{C} \longrightarrow \mathsf{Sets}.$$

2. A relation on sets A and B is a function

$$R\colon A \times B \longrightarrow \{\mathsf{true}, \mathsf{false}\}.$$

Here we notice that:

- The opposite X^{op} of a set X is itself, as $(-)^{\mathrm{op}}\colon \mathsf{Cats} \longrightarrow \mathsf{Cats}$ restricts to the identity endofunctor on Sets.
- The values that profunctors and relations take are analogous:
  - A category is enriched over the category

$$\mathsf{Sets} \overset{\mathrm{def}}{=} \mathsf{Cats}_0$$

    of sets, with profunctors taking values on it.
  - A set is enriched over the set

$$\{\mathsf{true}, \mathsf{false}\} \overset{\mathrm{def}}{=} \mathsf{Cats}_{-1}$$

    of classical truth values, with relations taking values on it.

Remark 8.1.2.1.2 ▶ Relations as Decategorifications of Profunctors II

Extending Remark 8.1.2.1.1, the equivalent definitions of relations in Definition 8.1.1.1.1 are also related to the corresponding ones for profunctors (**??**), which state that a profunctor $\mathfrak{p}\colon \mathcal{C} \nrightarrow \mathcal{D}$ is equivalently:

1. A functor $\mathfrak{p}\colon \mathcal{D}^{\mathsf{op}} \times \mathcal{C} \to \mathsf{Sets}$.
2. A functor $\mathfrak{p}\colon \mathcal{C} \to \mathsf{PSh}(\mathcal{D})$.
3. A functor $\mathfrak{p}\colon \mathcal{D}^{\mathsf{op}} \to \mathsf{CoPSh}(\mathcal{C})$.
4. A colimit-preserving functor $\mathfrak{p}\colon \mathsf{PSh}(\mathcal{C}) \to \mathsf{PSh}(\mathcal{D})$.
5. A limit-preserving functor $\mathfrak{p}\colon \mathsf{CoPSh}(\mathcal{D})^{\mathsf{op}} \to \mathsf{CoPSh}(\mathcal{C})^{\mathsf{op}}$.

Indeed:

- The equivalence between Items 1 and 2 (and also that between Items 1 and 3, which is proved analogously) is an instance of currying, both for profunctors as well as for relations, using the isomorphisms

$$\begin{aligned}\mathsf{Sets}(A \times B, \{\mathsf{true}, \mathsf{false}\}) &\cong \mathsf{Sets}(A, \mathsf{Sets}(B, \{\mathsf{true}, \mathsf{false}\}))\\ &\cong \mathsf{Sets}(A, \mathcal{P}(B)),\end{aligned}$$

  and

$$\begin{aligned}\mathsf{Fun}(\mathcal{D}^{\mathsf{op}} \times \mathcal{D}, \mathsf{Sets}) &\cong \mathsf{Fun}(\mathcal{C}, \mathsf{Fun}(\mathcal{D}^{\mathsf{op}}, \mathsf{Sets}))\\ &\cong \mathsf{Fun}(\mathcal{C}, \mathsf{PSh}(\mathcal{D})).\end{aligned}$$

- The equivalence between Items 2 and 4 follows from the universal properties of:
  - The powerset $\mathcal{P}(X)$ of a set X as the free cocompletion of X via the characteristic embedding

$$\chi_{(-)}\colon X \hookrightarrow \mathcal{P}(X)$$

    of X into $\mathcal{P}(X)$, as stated and proved in Constructions With Sets, Proposition 4.4.5.1.1.

- The category $\mathsf{PSh}(\mathcal{C})$ of presheaves on a category $\mathcal{C}$ as the free cocompletion of $\mathcal{C}$ via the Yoneda embedding

$$よ : \mathcal{C} \hookrightarrow \mathsf{PSh}(\mathcal{C})$$

of $\mathcal{C}$ into $\mathsf{PSh}(\mathcal{C})$, as stated and proved in Presheaves and the Yoneda Lemma, **??** of **??**.

- The equivalence between Items 3 and 5 follows from the universal properties of:
  - The powerset $\mathcal{P}(X)$ of a set X as the free completion of X via the characteristic embedding

$$\chi_{(-)} : X \hookrightarrow \mathcal{P}(X)$$

of X into $\mathcal{P}(X)$, as stated and proved in Constructions With Sets, Proposition 4.4.6.1.1.

  - The category $\mathsf{CoPSh}(\mathcal{D})^{\mathsf{op}}$ of copresheaves on a category $\mathcal{D}$ as the free completion of $\mathcal{D}$ via the dual Yoneda embedding

$$ヤ : \mathcal{D} \hookrightarrow \mathsf{CoPSh}(\mathcal{D})^{\mathsf{op}}$$

of $\mathcal{D}$ into $\mathsf{CoPSh}(\mathcal{D})^{\mathsf{op}}$, as stated and proved in Presheaves and the Yoneda Lemma, **??** of **??**.

8.1.3 Composition of Relations

Let A, B, and C be sets and let $R : A \nrightarrow B$ and $S : B \nrightarrow C$ be relations.

Definition 8.1.3.1.1 ▶ Composition of Relations

The **composition of** R **and** S is the relation $S \diamond R$ defined as follows:

- Viewing relations from A to C as subsets of $A \times C$, we define

$$S \diamond R \overset{\text{def}}{=} \left\{ (a, c) \in A \times C \;\middle|\; \begin{array}{l} \text{there exists some } b \in B \text{ such} \\ \text{that } a \sim_R b \text{ and } b \sim_S c \end{array} \right\}.$$

- Viewing relations as functions $A \times B \longrightarrow \{\text{true}, \text{false}\}$, we define

$$\begin{aligned}(S \diamond R)^{-1}_{-2} &\overset{\text{def}}{=} \int^{b \in B} S^{-1}_b \times R^b_{-2} \\ &= \bigvee_{b \in B} S^{-1}_b \times R^b_{-2},\end{aligned}$$

  where the join $\bigvee$ is taken in the poset $(\{\text{true}, \text{false}\}, \preceq)$ of Sets, Definition 3.2.2.1.3.

- Viewing relations as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$S \diamond R \overset{\text{def}}{=} \mathrm{Lan}_{\chi_B}(S) \circ R,$$

$$\begin{tikzcd} & B \arrow[r, "S"] \arrow[d, "\chi_B"'] \arrow[dr, phantom, "\Downarrow"] & \mathcal{P}(C), \\ A \arrow[r, "R"'] & \mathcal{P}(B) \arrow[ur, bend right, "\mathrm{Lan}_{\chi_B}(S)"'] & \end{tikzcd}$$

  where $\mathrm{Lan}_{\chi_B}(S)$ is computed by the formula

$$\begin{aligned}\left[\mathrm{Lan}_{\chi_B}(S)\right](V) &\cong \int^{b \in B} \chi_{\mathcal{P}(B)}(\chi_b, V) \odot S_b \\ &\cong \int^{b \in B} \chi_V(b) \odot S_b \\ &\cong \bigcup_{b \in B} \chi_V(b) \odot S_b \\ &\cong \bigcup_{b \in V} S_b\end{aligned}$$

  for each $V \in \mathcal{P}(B)$, so we have[1]

$$\begin{aligned}[S \diamond R](a) &\overset{\text{def}}{=} S(R(a)) \\ &\overset{\text{def}}{=} \bigcup_{b \in R(a)} S(b).\end{aligned}$$

  for each $a \in A$.

[1]That is: the relation R may send $a \in A$ to a number of elements $\{b_i\}_{i \in I}$ in B, and then the relation S may send the image of each of the b_i's to a number of elements $\{S(b_i)\}_{i \in I} = \left\{\left\{c_{j_i}\right\}_{j_i \in J_i}\right\}_{i \in I}$ in C.

Example 8.1.3.1.2 ► Examples of Composition of Relations

Here are some examples of composition of relations.

1. *Composing Less/Greater Than Equal With Greater/Less Than Equal Signs.* We have

$$\begin{aligned} \leq \diamond \geq &= \sim_{\text{triv}}, \\ \geq \diamond \leq &= \sim_{\text{triv}}. \end{aligned}$$

2. *Composing Less/Greater Than Equal Signs With Less/Greater Than Equal Signs.* We have

$$\begin{aligned} \leq \diamond \leq &= \leq, \\ \geq \diamond \geq &= \geq. \end{aligned}$$

Proposition 8.1.3.1.3 ► Properties of Composition of Relations

Let $R\colon A \nrightarrow B$, $S\colon B \nrightarrow C$, and $T\colon C \nrightarrow D$ be relations.

1. *Interaction With Ranges and Domains.* We have

$$\begin{aligned} \operatorname{dom}(S \diamond R) &\subset \operatorname{dom}(R), \\ \operatorname{range}(S \diamond R) &\subset \operatorname{range}(S). \end{aligned}$$

2. *Associativity.* We have

$$(T \diamond S) \diamond R = T \diamond (S \diamond R).$$

3. *Unitality.* We have

$$\begin{aligned} \chi_B \diamond R &= R, \\ R \diamond \chi_A &= R. \end{aligned}$$

4. *Interaction With Converses.* We have

$$(S \diamond R)^{\dagger} = R^{\dagger} \diamond S^{\dagger}.$$

5. *Interaction With Composition.* We have

$$\chi_B \subset R \diamond R^\dagger,$$
$$\chi_A \subset R^\dagger \diamond R.$$

Proof 8.1.3.1.4 ▶ Proof of Proposition 8.1.3.1.3

Item 1: Interaction With Ranges and Domains

Clear.

Item 2: Associativity

Indeed, we have

$$\begin{aligned}
(T \diamond S) \diamond R &\overset{\text{def}}{=} \left(\int^{c\in C} T_c^{-1} \times S^c_{-_2}\right) \diamond R \\
&\overset{\text{def}}{=} \int^{b\in B} \left(\int^{c\in C} T_c^{-1} \times S^c_b\right) \diamond R^b_{-_2} \\
&= \int^{b\in B} \int^{c\in C} \left(T_c^{-1} \times S^c_b\right) \diamond R^b_{-_2} \\
&= \int^{c\in C} \int^{b\in B} \left(T_c^{-1} \times S^c_b\right) \diamond R^b_{-_2} \\
&= \int^{c\in C} \int^{b\in B} T_c^{-1} \times \left(S^c_b \diamond R^b_{-_2}\right) \\
&= \int^{c\in C} T_c^{-1} \times \left(\int^{b\in B} S^c_b \diamond R^b_{-_2}\right) \\
&\overset{\text{def}}{=} \int^{c\in C} T_c^{-1} \times (S \diamond R)^c_{-_2} \\
&\overset{\text{def}}{=} T \diamond (S \diamond R).
\end{aligned}$$

In the language of relations, given $a \in A$ and $d \in D$, the stated equality witnesses the equivalence of the following two statements:

- We have $a \sim_{(T\diamond S)\diamond R} d$, i.e. there exists some $b \in B$ such that:

  - We have $a \sim_R b$;
  - We have $b \sim_{T \diamond S} d$, i.e. there exists some $c \in C$ such that:
    * We have $b \sim_S c$;
    * We have $c \sim_T d$;

- We have $a \sim_{T \diamond (S \diamond R)} d$, i.e. there exists some $c \in C$ such that:
  - We have $a \sim_{S \diamond R} c$, i.e. there exists some $b \in B$ such that:
    * We have $a \sim_R b$;
    * We have $b \sim_S c$;
  - We have $c \sim_T d$;

both of which are equivalent to the statement

- There exist $b \in B$ and $c \in C$ such that $a \sim_R b \sim_S c \sim_T d$.

Item 3: Unitality

Indeed, we have

$$\begin{aligned}
\chi_B \diamond R &\overset{\text{def}}{=} \int^{x \in B} (\chi_B)^{-1}_x \times R^x_{-_2} \\
&= \bigvee_{x \in B} (\chi_B)^{-1}_x \times R^x_{-_2} \\
&= \bigvee_{\substack{x \in B \\ x = -_1}} R^x_{-_2} \\
&= R^{-1}_{-_2},
\end{aligned}$$

and

$$\begin{aligned}
R \diamond \chi_A &\overset{\text{def}}{=} \int^{x \in A} R^{-1}_x \times (\chi_A)^x_{-_2} \\
&= \bigvee_{x \in B} R^{-1}_x \times (\chi_A)^x_{-_2}
\end{aligned}$$

$$= \bigvee_{\substack{x \in B \\ x = -_2}} R_x^{-1}$$
$$= R_{-_2}^{-1}.$$

In the language of relations, given $a \in A$ and $b \in B$:

- The equality
$$\chi_B \diamond R = R$$
witnesses the equivalence of the following two statements:
  - We have $a \sim_b B$.
  - There exists some $b' \in B$ such that:
    * We have $a \sim_R b'$
    * We have $b' \sim_{\chi_B} b$, i.e. $b' = b$.
- The equality
$$R \diamond \chi_A = R$$
witnesses the equivalence of the following two statements:
  - There exists some $a' \in A$ such that:
    * We have $a \sim_{\chi_B} a'$, i.e. $a = a'$.
    * We have $a' \sim_R b$
  - We have $a \sim_b B$.

Item 4: Interaction With Converses

Clear.

Item 5: Interaction With Composition

Clear. ∎

8.1.4 Apartness Composition of Relations

Let A, B, and C be sets and let $R\colon A \nrightarrow B$ and $S\colon B \nrightarrow C$ be relations.

Definition 8.1.4.1.1 ▶ Apartness Composition of Relations

The **apartness composition of** R **and** S is the relation $S \mathbin{\square} R$ defined as follows:

- Viewing relations from A to C as subsets of $A \times C$, we define

$$S \mathbin{\square} R \overset{\text{def}}{=} \left\{ (a, c) \in A \times C \;\middle|\; \begin{array}{l} \text{for each } b \in B, \text{ we have} \\ a \sim_R b \text{ or } b \sim_S c \end{array} \right\}.$$

- Viewing relations as functions $A \times B \longrightarrow \{\text{true}, \text{false}\}$, we define

$$\begin{aligned} (S \mathbin{\square} R)^{-1}_{-2} &\overset{\text{def}}{=} \int_{b \in B} S^{-1}_b \coprod R^b_{-2} \\ &= \bigwedge_{b \in B} S^{-1}_b \coprod R^b_{-2}, \end{aligned}$$

  where the meet $\bigwedge$ is taken in the poset $(\{\text{true}, \text{false}\}, \preceq)$ of Sets, Definition 3.2.2.1.3.

- Viewing relations as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$[S \mathbin{\square} R](a) \overset{\text{def}}{=} \bigcap_{b \in B \setminus R(a)} S(b)$$

8.1.5 The Converse of a Relation

Let A, B, and C be sets and let $R \subset A \times B$ be a relation.

Definition 8.1.5.1.1 ▶ The Converse of a Relation

The **converse of** R[1] is the relation $R^\dagger$ defined as follows:

- Viewing relations as subsets, we define

$$R^\dagger \overset{\text{def}}{=} \{(b, a) \in B \times A \mid \text{we have } b \sim_R a\}.$$

- Viewing relations as functions $A \times B \longrightarrow \{\text{true}, \text{false}\}$, we define

$$\left[R^\dagger\right]^a_b \overset{\text{def}}{=} R^b_a$$

for each $(b, a) \in B \times A$.

- Viewing relations as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$\begin{aligned}\left[R^{\dagger}\right](b) &\stackrel{\text{def}}{=} R^{\dagger}(\{b\}) \\ &\stackrel{\text{def}}{=} \{a \in A \mid b \in R(a)\}\end{aligned}$$

for each $b \in B$, where $R^{\dagger}(\{b\})$ is the fibre of R over $\{b\}$.

[1] *Further Terminology:* Also called the **opposite of** R or the **transpose of** R.

Example 8.1.5.1.2 ▶ Examples of Converses of Relations

Here are some examples of converses of relations.

1. *Less Than Equal Signs.* We have $(\leq)^{\dagger} = \geq$.
2. *Greater Than Equal Signs.* Dually to Item 1, we have $(\geq)^{\dagger} = \leq$.
3. *Functions.* Let $f\colon A \longrightarrow B$ be a function. We have

$$\begin{aligned}\mathrm{Gr}(f)^{\dagger} &= f^{-1}, \\ \left(f^{-1}\right)^{\dagger} &= \mathrm{Gr}(f).\end{aligned}$$

Proposition 8.1.5.1.3 ▶ Properties of Converses of Relations

Let $R\colon A \nrightarrow B$ and $S\colon B \nrightarrow C$ be relations.

1. *Functoriality.* The assignment $R \longmapsto R^{\dagger}$ defines a functor (i.e. morphism of posets)

$$(-)^{\dagger}\colon \mathbf{Rel}(A, B) \longrightarrow \mathbf{Rel}(B, A).$$

In particular, given relations $R, S\colon A \rightrightarrows B$, we have:

($\star$) If $R \subset S$, then $R^{\dagger} \subset S^{\dagger}$.

2. *Interaction With Ranges and Domains.* We have

$$\mathrm{dom}\left(R^{\dagger}\right) = \mathrm{range}(R),$$

$$\operatorname{range}\left(R^{\dagger}\right) = \operatorname{dom}(R).$$

3. *Interaction With Composition I.* We have

$$(S \diamond R)^{\dagger} = R^{\dagger} \diamond S^{\dagger}.$$

4. *Interaction With Composition II.* We have

$$\chi_B \subset R \diamond R^{\dagger},$$
$$\chi_A \subset R^{\dagger} \diamond R.$$

5. *Invertibility.* We have

$$\left(R^{\dagger}\right)^{\dagger} = R.$$

6. *Identity.* We have

$$\chi_A^{\dagger} = \chi_A.$$

Proof 8.1.5.1.4 ► Proof of Proposition 8.1.5.1.3

Item 1: Functoriality

Clear.

Item 2: Interaction With Ranges and Domains

Clear.

Item 3: Interaction With Composition I

Clear.

Item 4: Interaction With Composition II

Clear.

Item 5: Invertibility

Clear.

Item 6: Identity

Clear.

8.2 Examples of Relations

8.2.1 Elementary Examples of Relations

EXAMPLE 8.2.1.1.1 ▶ SQUARE ROOTS

Square roots are examples of relations:

1. *Square Roots in* $\mathbb{R}$. The assignment $x \mapsto \sqrt{x}$ defines a relation

$$\sqrt{-}\colon \mathbb{R} \to \mathcal{P}(\mathbb{R})$$

from $\mathbb{R}$ to itself, being explicitly given by

$$\sqrt{x} \overset{\text{def}}{=} \begin{cases} 0 & \text{if } x = 0, \\ \left\{-\sqrt{|x|}, \sqrt{|x|}\right\} & \text{if } x \neq 0. \end{cases}$$

2. *Square Roots in* $\mathbb{Q}$. Square roots in $\mathbb{Q}$ are similar to square roots in $\mathbb{R}$, though now additionally it may also occur that $\sqrt{-}\colon \mathbb{Q} \to \mathcal{P}(\mathbb{Q})$ sends a rational number x (e.g. 2) to the empty set (since $\sqrt{2} \notin \mathbb{Q}$).

EXAMPLE 8.2.1.1.2 ▶ COMPLEX LOGARITHMS

The complex logarithm defines a relation

$$\log\colon \mathbb{C} \to \mathcal{P}(\mathbb{C})$$

from $\mathbb{C}$ to itself, where we have

$$\log(a + bi) \overset{\text{def}}{=} \left\{\log\left(\sqrt{a^2 + b^2}\right) + i \arg(a + bi) + (2\pi i)k \,\middle|\, k \in \mathbb{Z}\right\}$$

for each $a + bi \in \mathbb{C}$.

Example 8.2.1.1.3 ▶ More Examples of Relations

See [Wik25] for more examples of relations, such as antiderivation, inverse trigonometric functions, and inverse hyperbolic functions.

8.2.2 The Graph of a Function

Let $f\colon A \to B$ be a function.

Definition 8.2.2.1.1 ▶ The Graph of a Function

The **graph of** f is the relation $\mathrm{Gr}(f)\colon A \nrightarrow B$ defined as follows:[1]

- Viewing relations from A to B as subsets of $A \times B$, we define

$$\mathrm{Gr}(f) \overset{\text{def}}{=} \{(a, f(a)) \in A \times B \mid a \in A\}.$$

- Viewing relations from A to B as functions $A \times B \to \{\text{true}, \text{false}\}$, we define

$$[\mathrm{Gr}(f)](a, b) \overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } b = f(a), \\ \text{false} & \text{otherwise} \end{cases}$$

  for each $(a, b) \in A \times B$.

- Viewing relations from A to B as functions $A \to \mathcal{P}(B)$, we define

$$[\mathrm{Gr}(f)](a) \overset{\text{def}}{=} \{f(a)\}$$

  for each $a \in A$, i.e. we define $\mathrm{Gr}(f)$ as the composition

$$A \xrightarrow{f} B \xrightarrow{\chi_B} \mathcal{P}(B).$$

[1] *Further Notation:* We write $\mathrm{Gr}(A)$ for $\mathrm{Gr}(\mathrm{id}_A)$, and call it the **graph** of A.

Proposition 8.2.2.1.2 ▶ Properties of Graphs of Functions

Let $f\colon A \to B$ be a function.

1. *Functoriality.* The assignment $A \mapsto \mathrm{Gr}(A)$ defines a functor

$$\mathrm{Gr}\colon \mathsf{Sets} \longrightarrow \mathsf{Rel}$$

where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\mathrm{Gr}(A) \overset{\text{def}}{=} A.$$

- *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathsf{Sets})$, the action on Hom-sets

$$\mathrm{Gr}_{A,B}\colon \mathsf{Sets}(A, B) \longrightarrow \underbrace{\mathsf{Rel}(\mathrm{Gr}(A), \mathrm{Gr}(B))}_{\overset{\text{def}}{=}\mathsf{Rel}(A,B)}$$

of Gr at (A, B) is defined by

$$\mathrm{Gr}_{A,B}(f) \overset{\text{def}}{=} \mathrm{Gr}(f),$$

where $\mathrm{Gr}(f)$ is the graph of f as in Definition 8.2.2.1.1.

In particular:

- *Preservation of Identities.* We have

$$\mathrm{Gr}(\mathrm{id}_A) = \chi_A$$

for each $A \in \mathrm{Obj}(\mathsf{Sets})$.

- *Preservation of Composition.* We have

$$\mathrm{Gr}(g \circ f) = \mathrm{Gr}(g) \diamond \mathrm{Gr}(f)$$

for each pair of functions $f\colon A \longrightarrow B$ and $g\colon B \longrightarrow C$.

2. *Adjointness Inside* **Rel**. We have an adjunction

$$\big(\mathrm{Gr}(f) \dashv f^{-1}\big)\colon \quad A \underset{f^{-1}}{\overset{\mathrm{Gr}(f)}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} B$$

in **Rel**, where f^{-1} is the converse of f of **??**.

3. *Adjointness.* We have an adjunction

$$(\mathrm{Gr} \dashv \mathcal{P}_!)\colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left, "\mathrm{Gr}"] \arrow[r, phantom, "\perp"] & \mathsf{Rel}, \arrow[l, bend left, "\mathcal{P}_!"] \end{tikzcd}$$

witnessed by a bijection of sets

$$\mathrm{Rel}(\mathrm{Gr}(A), B) \cong \mathsf{Sets}(A, \mathcal{P}(B))$$

natural in $A \in \mathrm{Obj}(\mathsf{Sets})$ and $B \in \mathrm{Obj}(\mathrm{Rel})$.

4. *Interaction With Converses.* We have

$$\begin{aligned} \mathrm{Gr}(f)^\dagger &= f^{-1}, \\ \left(f^{-1}\right)^\dagger &= \mathrm{Gr}(f). \end{aligned}$$

5. *Cocontinuity.* The functor $\mathrm{Gr}\colon \mathsf{Sets} \to \mathrm{Rel}$ of Item 1 preserves colimits.

6. *Characterisations.* Let $R\colon A \nrightarrow B$ be a relation. The following conditions are equivalent:

   (a) There exists a function $f\colon A \to B$ such that $R = \mathrm{Gr}(f)$.

   (b) The relation R is total and functional.

   (c) The weak and strong inverse images of R agree, i.e. we have $R^{-1} = R_{-1}$.

   (d) The relation R has a right adjoint $R^\dagger$ in Rel.

Proof 8.2.2.1.3 ▶ Proof of Proposition 8.2.2.1.2

Item 1: Functoriality

Clear.

Item 2: Adjointness Inside Rel

We need to check that there are inclusions

$$\begin{aligned} \chi_A &\subset f^{-1} \diamond \mathrm{Gr}(f), \\ \mathrm{Gr}(f) \diamond f^{-1} &\subset \chi_B. \end{aligned}$$

These correspond respectively to the following conditions:

1. For each $a \in A$, there exists some $b \in B$ such that $a \sim_{\mathrm{Gr}(f)} b$ and $b \sim_{f^{-1}} a$.
2. For each $a, b \in A$, if $a \sim_{\mathrm{Gr}(f)} b$ and $b \sim_{f^{-1}} a$, then $a = b$.

In other words, the first condition states that the image of any $a \in A$ by f is nonempty, whereas the second condition states that f is not multivalued. As f is a function, both of these statements are true, and we are done.

Item 3: Adjointness

The stated bijection follows from Relations, **??**, with naturality being clear.

Item 4: Interaction With Converses

Clear.

Item 5: Cocontinuity

Omitted.

Item 6: Characterisations

We claim that Items 6a to 6d are indeed equivalent:

- *Item 6a* $\Longleftrightarrow$ *Item 6b*. This is shown in the proof of **??** of **??**.
- *Item 6b* $\Longrightarrow$ *Item 6c*. If R is total and functional, then, for each $a \in A$, the set $R(a)$ is a singleton, implying that

$$R^{-1}(V) \stackrel{\text{def}}{=} \{a \in A \mid R(a) \cap V \neq \emptyset\},$$

$$R_{-1}(V) \stackrel{\text{def}}{=} \{a \in A \mid R(a) \subset V\}$$

are equal for all $V \in \mathcal{P}(B)$, as the conditions $R(a) \cap V \neq \emptyset$ and $R(a) \subset V$ are equivalent when $R(a)$ is a singleton.

- *Item 6c* $\Longrightarrow$ *Item 6b*. We claim that R is indeed total and functional:
  - *Totality.* If we had $R(a) = \emptyset$ for some $a \in A$, then we would have $a \in R_{-1}(\emptyset)$, so that $R_{-1}(\emptyset) \neq \emptyset$. But since $R^{-1}(\emptyset) = \emptyset$, this would imply $R_{-1}(\emptyset) \neq R^{-1}(\emptyset)$, a contradiction. Thus $R(a) \neq \emptyset$ for all $a \in A$ and R is total.
  - *Functionality.* If $R^{-1} = R_{-1}$, then we have

    $$\begin{aligned}\{a\} &= R^{-1}(\{b\}) \\ &= R_{-1}(\{b\})\end{aligned}$$

    for each $b \in R(a)$ and each $a \in A$, and thus $R(a) \subset \{b\}$. But since R is total, we must have $R(a) = \{b\}$, and thus we see that R is functional.
- *Item 6a* $\Longleftrightarrow$ *Item 6d*. This follows from Relations, Proposition 8.4.3.1.1.

This finishes the proof. ∎

8.2.3 The Inverse of a Function

Let $f\colon A \to B$ be a function.

Definition 8.2.3.1.1 ▶ The Inverse of a Function

The **inverse of** f is the relation $f^{-1}\colon B \nrightarrow A$ defined as follows:

- Viewing relations from B to A as subsets of $B \times A$, we define

  $$f^{-1} \stackrel{\text{def}}{=} \left\{\left(b, f^{-1}(b)\right) \in B \times A \,\middle|\, a \in A\right\},$$

  where

  $$f^{-1}(b) \stackrel{\text{def}}{=} \{a \in A \mid f(a) = b\}$$

for each $b \in B$.

- Viewing relations from B to A as functions $B \times A \longrightarrow \{\text{true}, \text{false}\}$, we define

$$f^{-1}(b, a) \stackrel{\text{def}}{=} \begin{cases} \text{true} & \text{if there exists } a \in A \text{ with } f(a) = b, \\ \text{false} & \text{otherwise} \end{cases}$$

for each $(b, a) \in B \times A$.

- Viewing relations from B to A as functions $B \longrightarrow \mathcal{P}(A)$, we define

$$f^{-1}(b) \stackrel{\text{def}}{=} \{a \in A \mid f(a) = b\}$$

for each $b \in B$.

Proposition 8.2.3.1.2 ▶ Properties of Inverses of Functions

Let $f \colon A \longrightarrow B$ be a function.

1. *Functoriality.* The assignment $A \mapsto A, f \mapsto f^{-1}$ defines a functor

$$(-)^{-1} \colon \mathsf{Sets} \longrightarrow \mathsf{Rel}$$

where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\left[(-)^{-1}\right](A) \stackrel{\text{def}}{=} A.$$

- *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathsf{Sets})$, the action on Hom-sets

$$(-)^{-1}_{A,B} \colon \mathsf{Sets}(A, B) \longrightarrow \mathsf{Rel}(A, B)$$

of $(-)^{-1}$ at (A, B) is defined by

$$(-)^{-1}_{A,B}(f) \stackrel{\text{def}}{=} \left[(-)^{-1}\right](f),$$

where f^{-1} is the inverse of f as in Definition 8.2.3.1.1.

In particular:

- *Preservation of Identities.* We have

$$\mathrm{id}_A^{-1} = \chi_A$$

for each $A \in \mathrm{Obj}(\mathsf{Sets})$.

- *Preservation of Composition.* We have

$$(g \circ f)^{-1} = g^{-1} \diamond f^{-1}$$

for pair of functions $f\colon A \to B$ and $g\colon B \to C$.

2. *Adjointness Inside* **Rel**. We have an adjunction

$$\left(\mathrm{Gr}(f) \dashv f^{-1}\right)\colon \quad \begin{tikzcd} A \arrow[r, bend left, "\mathrm{Gr}(f)"] \arrow[r, phantom, "\perp"] & B \arrow[l, bend left, "f^{-1}"] \end{tikzcd}$$

in **Rel**.

3. *Interaction With Converses of Relations.* We have

$$\begin{aligned} \left(f^{-1}\right)^{\dagger} &= \mathrm{Gr}(f), \\ \mathrm{Gr}(f)^{\dagger} &= f^{-1}. \end{aligned}$$

Proof 8.2.3.1.3 ▶ Proof of Proposition 8.2.3.1.2

Item 1: Functoriality

Clear.

Item 2: Adjointness Inside Rel

This is proved in Item 2 of Proposition 8.2.2.1.2.

Item 3: Interaction With Converses of Relations

Clear. ∎

8.2.4 Representable Relations

Let A and B be sets.

Definition 8.2.4.1.1 ▶ Representable Relations

Let $f\colon A \to B$ and $g\colon B \to A$ be functions.[1]

1. The **representable relation associated to** f is the relation $\chi_f\colon A \nrightarrow B$ defined as the composition

$$A \times B \xrightarrow{f \times \mathrm{id}_B} B \times B \xrightarrow{\chi_B} \{\mathsf{true}, \mathsf{false}\},$$

   i.e. given by declaring $a \sim_{\chi_f} b$ iff $f(a) = b$.

2. The **corepresentable relation associated to** g is the relation $\chi^g\colon B \nrightarrow A$ defined as the composition

$$B \times A \xrightarrow{g \times \mathrm{id}_A} A \times A \xrightarrow{\chi_A} \{\mathsf{true}, \mathsf{false}\},$$

   i.e. given by declaring $b \sim_{\chi^g} a$ iff $g(b) = a$.

[1] More generally, given functions

$$f\colon A \to C,$$
$$g\colon B \to D$$

and a relation $B \nrightarrow D$, we may consider the composite relation

$$A \times B \xrightarrow{f \times g} C \times D \xrightarrow{R} \{\mathsf{true}, \mathsf{false}\},$$

for which we have $a \sim_{R \circ (f \times g)} b$ iff $f(a) \sim_R g(b)$.

8.3 Categories of Relations

8.3.1 The Category of Relations Between Two Sets

Definition 8.3.1.1.1 ▶ The Category of Relations Between Two Sets

The **category of relations from** A **to** B is the category $\mathbf{Rel}(A, B)$ defined by[1]

$$\mathbf{Rel}(A, B) \stackrel{\text{def}}{=} \mathbf{Rel}(A, B)_{\text{pos}},$$

where $\mathbf{Rel}(A, B)_{\text{pos}}$ is the posetal category associated to the poset $\mathbf{Rel}(A, B)$ of Item 2 of Notation 8.1.1.1.6 and Categories, Definition 11.2.7.1.1.

[1] Here we choose to abuse notation by writing $\mathbf{Rel}(A, B)$ instead of $\mathbf{Rel}(A, B)_{\text{pos}}$ for the posetal category of relations from A to B, even though the same notation is used for the poset of relations from A to B.

8.3.2 The Category of Relations

Definition 8.3.2.1.1 ▶ The Category of Relations

The **category of relations** is the category Rel where

- *Objects.* The objects of Rel are sets.
- *Morphisms.* For each $A, B \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$\mathrm{Rel}(A, B) \stackrel{\text{def}}{=} \mathrm{Rel}(A, B).$$

- *Identities.* For each $A \in \mathrm{Obj}(\mathrm{Rel})$, the unit map

$$\mathbb{1}_A^{\mathrm{Rel}} \colon \mathrm{pt} \longrightarrow \mathrm{Rel}(A, A)$$

  of Rel at A is defined by

$$\mathrm{id}_A^{\mathrm{Rel}} \stackrel{\text{def}}{=} \chi_A(-_1, -_2),$$

  where $\chi_A(-_1, -_2)$ is the characteristic relation of A of Constructions With Sets, **??** of **??**.

- *Composition.* For each $A, B, C \in \mathrm{Obj}(\mathsf{Rel})$, the composition map

$$\circ^{\mathsf{Rel}}_{A,B,C} \colon \mathrm{Rel}(B, C) \times \mathrm{Rel}(A, B) \longrightarrow \mathrm{Rel}(A, C)$$

of Rel at (A, B, C) is defined by

$$S \circ^{\mathsf{Rel}}_{A,B,C} R \overset{\text{def}}{=} S \diamond R$$

for each $(S, R) \in \mathrm{Rel}(B, C) \times \mathrm{Rel}(A, B)$, where $S \diamond R$ is the composition of S and R of Constructions With Relations, **??**.

8.3.3 The Closed Symmetric Monoidal Category of Relations

8.3.3.1 The Monoidal Product

Definition 8.3.3.1.1 ► The Monoidal Product of Rel

The **monoidal product of** Rel is the functor

$$\times \colon \mathsf{Rel} \times \mathsf{Rel} \longrightarrow \mathsf{Rel}$$

where

- *Action on Objects.* For each $A, B \in \mathrm{Obj}(\mathsf{Rel})$, we have

$$\times(A, B) \overset{\text{def}}{=} A \times B,$$

where $A \times B$ is the Cartesian product of sets of Constructions With Sets, Definition 4.1.3.1.1.

- *Action on Morphisms.* For each $(A, C), (B, D) \in \mathrm{Obj}(\mathsf{Rel} \times \mathsf{Rel})$, the action on morphisms

$$\times_{(A,C),(B,D)} \colon \mathrm{Rel}(A, B) \times \mathrm{Rel}(C, D) \longrightarrow \mathrm{Rel}(A \times C, B \times D)$$

of $\times$ is given by sending a pair of morphisms (R, S) of the form

$$R \colon A \nrightarrow B,$$

$$S\colon C \nrightarrow D$$

to the relation

$$R \times S\colon A \times C \nrightarrow B \times D$$

of Constructions With Relations, Definition 9.3.6.1.1.

8.3.3.2 The Monoidal Unit

Definition 8.3.3.2.1 ▶ The Monoidal Unit of Rel

The **monoidal unit of** Rel is the functor

$$\mathbb{1}^{\mathsf{Rel}}\colon \mathsf{pt} \to \mathsf{Rel}$$

picking the set

$$\mathbb{1}_{\mathsf{Rel}} \overset{\text{def}}{=} \mathsf{pt}$$

of Rel.

8.3.3.3 The Associator

Definition 8.3.3.3.1 ▶ The Associator of Rel

The **associator of** Rel is the natural isomorphism

$$\alpha^{\mathsf{Rel}}\colon \times \circ ((\times) \times \mathsf{id}) \overset{\sim}{\Longrightarrow} \times \circ (\mathsf{id} \times (\times)) \circ \boldsymbol{\alpha}^{\mathsf{Cats}}_{\mathsf{Rel},\mathsf{Rel},\mathsf{Rel}},$$

as in the diagram

$$\begin{tikzcd}
 & \mathsf{Rel} \times (\mathsf{Rel} \times \mathsf{Rel}) \arrow[dr, "\mathsf{id}\times(\times)"] \arrow[dl, dashed, "\boldsymbol{\alpha}^{\mathsf{Cats}}_{\mathsf{Rel},\mathsf{Rel},\mathsf{Rel}}"'] \arrow[d, phantom, "\sim"] & \\
(\mathsf{Rel} \times \mathsf{Rel}) \times \mathsf{Rel} \arrow[dr, "(\times)\times\mathsf{id}"'] \arrow[rr, phantom, "\overset{\alpha^{\mathsf{Rel}}}{\Longrightarrow}"] & & \mathsf{Rel} \times \mathsf{Rel} \arrow[dl, "\times"] \\
 & \mathsf{Rel} \times \mathsf{Rel} \arrow[r, "\times"'] & \mathsf{Rel},
\end{tikzcd}$$

whose component

$$\alpha^{\mathsf{Rel}}_{A,B,C} \colon (A \times B) \times C \nrightarrow A \times (B \times C)$$

at $A, B, C \in \mathsf{Obj}(\mathsf{Rel})$ is the relation defined by declaring

$$((a, b), c) \sim_{\alpha^{\mathsf{Rel}}_{A,B,C}} (a', (b', c'))$$

iff $a = a'$, $b = b'$, and $c = c'$.

8.3.3.4 The Left Unitor

Definition 8.3.3.4.1 ▶ The Left Unitor of Rel

The **left unitor of** Rel is the natural isomorphism

$$\lambda^{\mathsf{Rel}} \colon \times \circ \left(\mathbb{1}^{\mathsf{Rel}} \times \mathsf{id}\right) \overset{\sim}{\Longrightarrow} \boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathsf{Rel}},$$

$$\begin{tikzcd}
\mathsf{pt} \times \mathsf{Rel} \arrow[r, "\mathbb{1}^{\mathsf{Rel}} \times \mathsf{id}"] \arrow[dr, dashed, "\boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathsf{Rel}}"'] & \mathsf{Rel} \times \mathsf{Rel}, \arrow[d, "\times"] \arrow[dl, Rightarrow, "\lambda^{\mathsf{Rel}}"] \\
& \mathsf{Rel}
\end{tikzcd}$$

whose component

$$\lambda^{\mathsf{Rel}}_A \colon \mathbb{1}_{\mathsf{Rel}} \times A \nrightarrow A$$

at A is defined by declaring

$$(\star, a) \sim_{\lambda^{\mathsf{Rel}}_A} b$$

iff $a = b$.

8.3.3.5 The Right Unitor

Definition 8.3.3.5.1 ▶ The Right Unitor of Rel

The **right unitor of** Rel is the natural isomorphism

$$\rho^{\mathsf{Rel}} \colon \times \circ \left(\mathrm{id} \times \mathbb{1}^{\mathsf{Rel}}\right) \overset{\sim}{\Longrightarrow} \boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathsf{Rel}},$$

$$\begin{tikzcd}
\mathsf{Rel} \times \mathsf{pt} \arrow[r, "\mathrm{id}\times\mathbb{1}^{\mathsf{Rel}}"] \arrow[dr, dashed, bend right, "\boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathsf{Rel}}"'] & \mathsf{Rel} \times \mathsf{Rel}, \arrow[d, "\times"] \arrow[dl, phantom, "\overset{\rho^{\mathsf{Rel}}}{\Swarrow}"] \\
& \mathsf{Rel}
\end{tikzcd}$$

whose component

$$\rho^{\mathsf{Rel}}_A \colon A \times \mathbb{1}_{\mathsf{Rel}} \nrightarrow A$$

at A is defined by declaring

$$(a, \star) \sim_{\rho^{\mathsf{Rel}}_A} b$$

iff $a = b$.

8.3.3.6 The Symmetry

Definition 8.3.3.6.1 ▶ The Symmetry of Rel

The **symmetry of** Rel is the natural isomorphism

$$\sigma^{\mathsf{Rel}} \colon \times \Longrightarrow \times \circ \boldsymbol{\sigma}^{\mathsf{Cats}_2}_{\mathsf{Rel},\mathsf{Rel}},$$

$$\begin{tikzcd}
\mathsf{Rel} \times \mathsf{Rel} \arrow[rr, "\times"] \arrow[dr, bend right, "\boldsymbol{\sigma}^{\mathsf{Cats}_2}_{\mathsf{Rel},\mathsf{Rel}}"'] \arrow[rr, phantom, shift right=3ex, "\overset{\sigma^{\mathsf{Rel}}}{\Downarrow}"] & & \mathsf{Rel}, \\
& \mathsf{Rel} \times \mathsf{Rel} \arrow[ur, bend right, "\times"'] &
\end{tikzcd}$$

whose component

$$\sigma^{\mathsf{Rel}}_{A,B} \colon A \times B \longrightarrow B \times A$$

at (A, B) is defined by declaring

$$(a, b) \sim_{\sigma^{\mathsf{Rel}}_{A,B}} (b', a')$$

iff $a = a'$ and $b = b'$.

8.3.3.7 The Internal Hom

Definition 8.3.3.7.1 ▶ The Internal Hom of Rel

The **internal Hom of** Rel is the functor

$$\mathsf{Rel}\colon \mathsf{Rel}^{\mathsf{op}} \times \mathsf{Rel} \longrightarrow \mathsf{Rel}$$

defined

- On objects by sending $A, B \in \mathrm{Obj}(\mathsf{Rel})$ to the set $\mathrm{Rel}(A, B)$ of **??** of **??**.
- On morphisms by pre/post-composition defined as in Constructions With Relations, **??**.

Proposition 8.3.3.7.2 ▶ Properties of the Internal Hom of Rel

Let $A, B, C \in \mathrm{Obj}(\mathsf{Rel})$.

1. *Adjointness.* We have adjunctions

$$(A \times - \dashv \mathrm{Rel}(A, -))\colon \quad \begin{tikzcd} \mathsf{Rel} \arrow[r, bend left, "A\times-"] \arrow[r, phantom, "\perp"] & \mathsf{Rel}, \arrow[l, bend left, "\mathrm{Rel}(A,-)"] \end{tikzcd}$$

$$(- \times B \dashv \mathrm{Rel}(B, -))\colon \quad \begin{tikzcd} \mathsf{Rel} \arrow[r, bend left, "-\times B"] \arrow[r, phantom, "\perp"] & \mathsf{Rel}, \arrow[l, bend left, "\mathrm{Rel}(B,-)"] \end{tikzcd}$$

witnessed by bijections

$$\mathrm{Rel}(A \times B, C) \cong \mathrm{Rel}(A, \mathrm{Rel}(B, C)),$$

$$\mathrm{Rel}(A \times B, C) \cong \mathrm{Rel}(B, \mathrm{Rel}(A, C)),$$

natural in $A, B, C \in \mathrm{Obj}(\mathsf{Rel})$.

Proof 8.3.3.7.3 ▶ Proof of Proposition 8.3.3.7.2

Item 1: Adjointness

Indeed, we have

$$\begin{aligned}\mathrm{Rel}(A \times B, C) &\overset{\text{def}}{=} \mathsf{Sets}(A \times B \times C, \{\mathsf{true}, \mathsf{false}\}) \\ &\overset{\text{def}}{=} \mathrm{Rel}(A, B \times C) \\ &\overset{\text{def}}{=} \mathrm{Rel}(A, \mathrm{Rel}(B, C)),\end{aligned}$$

and similarly for the bijection $\mathrm{Rel}(A \times B, C) \cong \mathrm{Rel}(B, \mathrm{Rel}(A, C))$.

8.3.3.8 The Closed Symmetric Monoidal Category of Relations

Proposition 8.3.3.8.1 ▶ The Closed Symmetric Monoidal Category of Relations

The category Rel admits a closed symmetric monoidal category structure consisting of[1]

- *The Underlying Category.* The category Rel of sets and relations of Definition 8.3.2.1.1.
- *The Monoidal Product.* The functor

$$\times \colon \mathsf{Rel} \times \mathsf{Rel} \longrightarrow \mathsf{Rel}$$

  of Definition 8.3.3.1.1.
- *The Internal Hom.* The internal Hom functor

$$\mathbf{Rel} \colon \mathsf{Rel}^{\mathsf{op}} \times \mathsf{Rel} \longrightarrow \mathsf{Rel}$$

  of Definition 8.3.3.7.1.

- *The Monoidal Unit.* The functor

$$\mathbb{1}^{\mathrm{Rel}} \colon \mathsf{pt} \longrightarrow \mathsf{Rel}$$

of Definition 8.3.3.2.1.

- *The Associators.* The natural isomorphism

$$\alpha^{\mathrm{Rel}} \colon \times \circ (\times \times \mathrm{id}_{\mathrm{Rel}}) \overset{\sim}{\Longrightarrow} \times \circ (\mathrm{id}_{\mathrm{Rel}} \times \times) \circ \boldsymbol{\alpha}^{\mathsf{Cats}}_{\mathrm{Rel},\mathrm{Rel},\mathrm{Rel}}$$

of Definition 8.3.3.3.1.

- *The Left Unitors.* The natural isomorphism

$$\lambda^{\mathrm{Rel}} \colon \times \circ \left(\mathbb{1}^{\mathrm{Rel}} \times \mathrm{id}_{\mathrm{Rel}}\right) \overset{\sim}{\Longrightarrow} \boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathrm{Rel}}$$

of Definition 8.3.3.4.1.

- *The Right Unitors.* The natural isomorphism

$$\rho^{\mathrm{Rel}} \colon \times \circ \left(\mathrm{id} \times \mathbb{1}^{\mathrm{Rel}}\right) \overset{\sim}{\Longrightarrow} \boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathrm{Rel}}$$

of Definition 8.3.3.5.1.

- *The Symmetry.* The natural isomorphism

$$\sigma^{\mathrm{Rel}} \colon \times \overset{\sim}{\Longrightarrow} \times \circ \boldsymbol{\sigma}^{\mathsf{Cats}_2}_{\mathrm{Rel},\mathrm{Rel}}$$

of Definition 8.3.3.6.1.

---

[1] *Warning:* This is not a Cartesian monoidal structure, as the product on Rel is in fact given by the disjoint union of sets; see Constructions With Relations, **??**.

Proof 8.3.3.8.2 ▶ Proof of Proposition 8.3.3.8.1

Omitted. ∎

8.3.4 The 2-Category of Relations

Definition 8.3.4.1.1 ► The 2-Category of Relations

The 2-**category of relations** is the locally posetal 2-category $\mathbf{Rel}$ where

- *Objects.* The objects of $\mathbf{Rel}$ are sets.
- $\mathbf{Hom}$-*Objects.* For each $A, B \in \mathrm{Obj}(\mathrm{Sets})$, we have

$$\begin{aligned}\mathrm{Hom}_{\mathbf{Rel}}(A, B) &\stackrel{\text{def}}{=} \mathbf{Rel}(A, B)\\ &\stackrel{\text{def}}{=} (\mathrm{Rel}(A, B), \subset).\end{aligned}$$

- *Identities.* For each $A \in \mathrm{Obj}(\mathbf{Rel})$, the unit map

$$\mathbb{1}_A^{\mathbf{Rel}} \colon \mathrm{pt} \to \mathbf{Rel}(A, A)$$

  of $\mathbf{Rel}$ at A is defined by

$$\mathrm{id}_A^{\mathbf{Rel}} \stackrel{\text{def}}{=} \chi_A(-_1, -_2),$$

  where $\chi_A(-_1, -_2)$ is the characteristic relation of A of Constructions With Sets, **??** of **??**.

- *Composition.* For each $A, B, C \in \mathrm{Obj}(\mathbf{Rel})$, the composition map[1]

$$\circ_{A,B,C}^{\mathbf{Rel}} \colon \mathbf{Rel}(B, C) \times \mathbf{Rel}(A, B) \to \mathbf{Rel}(A, C)$$

  of $\mathbf{Rel}$ at (A, B, C) is defined by

$$S \circ_{A,B,C}^{\mathbf{Rel}} R \stackrel{\text{def}}{=} S \diamond R$$

  for each $(S, R) \in \mathbf{Rel}(B, C) \times \mathbf{Rel}(A, B)$, where $S \diamond R$ is the composition of S and R of Constructions With Relations, **??**.

[1] Note that this is indeed a morphism of posets: given relations $R_1, R_2 \in \mathbf{Rel}(A, B)$ and $S_1, S_2 \in \mathbf{Rel}(B, C)$ such that

$$\begin{aligned}R_1 &\subset R_2,\\ S_1 &\subset S_2,\end{aligned}$$

we have also $S_1 \diamond R_1 \subset S_2 \diamond R_2$.

8.3.5 The Double Category of Relations

8.3.5.1 The Double Category of Relations

Definition 8.3.5.1.1 ▶ The Double Category of Relations

The **double category of relations** is the locally posetal double category $\mathsf{Rel}^{\mathsf{dbl}}$ where

- *Objects.* The objects of $\mathsf{Rel}^{\mathsf{dbl}}$ are sets.
- *Vertical Morphisms.* The vertical morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$ are maps of sets $f\colon A \to B$.
- *Horizontal Morphisms.* The horizontal morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$ are relations $R\colon A \nrightarrow X$.
- *2-Morphisms.* A 2-cell

$$\begin{tikzcd} A \arrow[r, "R", "|" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow\alpha"] & B \arrow[d, "g"] \\ X \arrow[r, "S"', "|" marking] & Y \end{tikzcd}$$

of $\mathsf{Rel}^{\mathsf{dbl}}$ is either non-existent or an inclusion of relations of the form

$$R \subset S \circ (f \times g), \qquad \begin{tikzcd} A \times B \arrow[r, "R"] \arrow[d, "f\times g"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ X \times Y \arrow[r, "S"'] & \{\mathsf{true}, \mathsf{false}\}. \end{tikzcd}$$

- *Horizontal Identities.* The horizontal unit functor of $\mathsf{Rel}^{\mathsf{dbl}}$ is the functor of Definition 8.3.5.2.1.
- *Vertical Identities.* For each $A \in \mathrm{Obj}\left(\mathsf{Rel}^{\mathsf{dbl}}\right)$, we have

$$\mathrm{id}_A^{\mathsf{Rel}^{\mathsf{dbl}}} \overset{\mathrm{def}}{=} \mathrm{id}_A\,.$$

- *Identity 2-Morphisms.* For each horizontal morphism $R\colon A \nrightarrow B$ of $\mathsf{Rel}^{\mathsf{dbl}}$, the identity 2-morphism

$$\begin{tikzcd} A \arrow[r, "R", "|" marking] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\Downarrow \mathrm{id}_R"] & B \arrow[d, "\mathrm{id}_B"] \\ A \arrow[r, "R"', "|" marking] & B \end{tikzcd}$$

of R is the identity inclusion

$$R \subset R, \qquad \begin{tikzcd} B \times A \arrow[r, "R"] \arrow[d, "\mathrm{id}_B \times \mathrm{id}_A"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ B \times A \arrow[r, "R"'] & \{\mathsf{true}, \mathsf{false}\}. \end{tikzcd}$$

- *Horizontal Composition.* The horizontal composition functor of $\mathsf{Rel}^{\mathsf{dbl}}$ is the functor of Definition 8.3.5.3.1.

- *Vertical Composition of 1-Morphisms.* For each composable pair $A \xrightarrow{F} B \xrightarrow{G} C$ of vertical morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$, i.e. maps of sets, we have

$$g \circ^{\mathsf{Rel}^{\mathsf{dbl}}} f \overset{\text{def}}{=} g \circ f.$$

- *Vertical Composition of 2-Morphisms.* The vertical composition of 2-morphisms in $\mathsf{Rel}^{\mathsf{dbl}}$ is defined as in Definition 8.3.5.4.1.

- *Associators.* The associators of $\mathsf{Rel}^{\mathsf{dbl}}$ is defined as in Definition 8.3.5.5.1.

- *Left Unitors.* The left unitors of $\mathsf{Rel}^{\mathsf{dbl}}$ is defined as in Definition 8.3.5.6.1.

- *Right Unitors.* The right unitors of $\mathsf{Rel}^{\mathsf{dbl}}$ is defined as in Definition 8.3.5.7.1.

8.3.5.2 Horizontal Identities

Definition 8.3.5.2.1 ▶ The Horizontal Identities of $\mathsf{Rel}^{\mathsf{dbl}}$

The **horizontal unit functor** of $\mathsf{Rel}^{\mathsf{dbl}}$ is the functor

$$\mathbb{1}^{\mathsf{Rel}^{\mathsf{dbl}}} \colon \mathsf{Rel}_0^{\mathsf{dbl}} \to \mathsf{Rel}_1^{\mathsf{dbl}}$$

of $\mathsf{Rel}^{\mathsf{dbl}}$ is the functor where

- *Action on Objects.* For each $A \in \mathrm{Obj}\left(\mathsf{Rel}_0^{\mathsf{dbl}}\right)$, we have

$$\mathbb{1}_A \overset{\text{def}}{=} \chi_A(-_1, -_2).$$

- *Action on Morphisms.* For each vertical morphism $f \colon A \to B$ of $\mathsf{Rel}^{\mathsf{dbl}}$, i.e. each map of sets f from A to B, the identity 2-morphism

$$\begin{tikzcd}
A \arrow[r, "\mathbb{1}_A", "|" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow \mathbb{1}_f"] & A \arrow[d, "f"] \\
B \arrow[r, "\mathbb{1}_B"', "|" marking] & B
\end{tikzcd}$$

of f is the inclusion

$$\chi_B \circ (f \times f) \subset \chi_A, \qquad \begin{tikzcd}
A \times A \arrow[r, "\chi_A(-_1,-_2)"] \arrow[d, "f \times f"'] \arrow[dr, phantom, "\subset"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\
B \times B \arrow[r, "\chi_B(-_1,-_2)"'] & \{\mathsf{true}, \mathsf{false}\}
\end{tikzcd}$$

of Constructions With Sets, Item 1 of Proposition 4.5.3.1.3.

8.3.5.3 Horizontal Composition

Definition 8.3.5.3.1 ► The Horizontal Composition of $\mathsf{Rel}^{\mathsf{dbl}}$

The **horizontal composition functor** of $\mathsf{Rel}^{\mathsf{dbl}}$ is the functor

$$\odot^{\mathsf{Rel}^{\mathsf{dbl}}} : \mathsf{Rel}_1^{\mathsf{dbl}} \underset{\mathsf{Rel}_0^{\mathsf{dbl}}}{\times} \mathsf{Rel}_1^{\mathsf{dbl}} \to \mathsf{Rel}_1^{\mathsf{dbl}}$$

of $\mathsf{Rel}^{\mathsf{dbl}}$ is the functor where

- *Action on Objects.* For each composable pair $A \overset{R}{\nrightarrow} B \overset{S}{\nrightarrow} C$ of horizontal morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$, we have

  $$S \odot R \overset{\text{def}}{=} S \diamond R,$$

  where $S \diamond R$ is the composition of R and S of Constructions With Relations, **??**.

- *Action on Morphisms.* For each horizontally composable pair

  $$\begin{tikzcd} A \arrow[r, "R", "|" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow\alpha"] & B \arrow[d, "g"] \\ X \arrow[r, "T"', "|" marking] & Y \end{tikzcd} \qquad \begin{tikzcd} B \arrow[r, "S", "|" marking] \arrow[d, "g"'] \arrow[dr, phantom, "\Downarrow\beta"] & C \arrow[d, "h"] \\ Y \arrow[r, "U"', "|" marking] & Z \end{tikzcd}$$

  of 2-morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$, i.e. for each pair

  $$\begin{tikzcd} A \times B \arrow[r, "R"] \arrow[d, "f \times g"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ X \times Y \arrow[r, "T"'] & \{\mathsf{true}, \mathsf{false}\} \end{tikzcd} \qquad \begin{tikzcd} B \times C \arrow[r, "S"] \arrow[d, "g \times h"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ Y \times Z \arrow[r, "U"'] & \{\mathsf{true}, \mathsf{false}\} \end{tikzcd}$$

  of inclusions of relations, the horizontal composition

  $$\begin{tikzcd} A \arrow[r, "S \odot R", "|" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow\beta\odot\alpha"] & C \arrow[d, "h"] \\ X \arrow[r, "U \odot T"', "|" marking] & Z \end{tikzcd}$$

of α and β is the inclusion of relations[1]

$$(U \diamond T) \circ (f \times h) \subset (S \diamond R) \qquad \begin{tikzcd} A \times C \arrow[r, "S \diamond R"] \arrow[d, "f \times h"'] \arrow[dr, phantom, "\circlearrowright"] & \{\text{true}, \text{false}\} \arrow[d, "\mathrm{id}_{\{\text{true},\text{false}\}}"] \\ X \times Z \arrow[r, "U \diamond T"'] & \{\text{true}, \text{false}\}. \end{tikzcd}$$

[1]This is justified by noting that, given $(a, c) \in A \times C$, the statement

- We have $a \sim_{(U \diamond T) \circ (f \times h)} c$, i.e. $f(a) \sim_{U \diamond T} h(c)$, i.e. there exists some $y \in Y$ such that:
  - We have $f(a) \sim_T y$.
  - We have $y \sim_U h(c)$.

is implied by the statement

- We have $a \sim_{S \diamond R} c$, i.e. there exists some $b \in B$ such that:
  - We have $a \sim_R b$.
  - We have $b \sim_S c$.

since:

- If $a \sim_R b$, then $f(a) \sim_T g(b)$, as $T \circ (f \times g) \subset R$;
- If $b \sim_S c$, then $g(b) \sim_U h(c)$, as $U \circ (g \times h) \subset S$.

8.3.5.4 Vertical Composition of 2-Morphisms

Definition 8.3.5.4.1 ▶ The Vertical Composition of 2-Morphisms in $\mathsf{Rel}^{\mathsf{dbl}}$

The **vertical composition** in $\mathsf{Rel}^{\mathsf{dbl}}$ is defined as follows: for each vertically composable pair

$$\begin{tikzcd} A \arrow[r, "R", "|" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow \alpha"] & X \arrow[d, "g"] \\ B \arrow[r, "S"', "|" marking] & Y \end{tikzcd} \qquad \begin{tikzcd} B \arrow[r, "S", "|" marking] \arrow[d, "h"'] \arrow[dr, phantom, "\Downarrow \beta"] & Y \arrow[d, "k"] \\ C \arrow[r, "T"', "|" marking] & Z \end{tikzcd}$$

of 2-morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$, i.e. for each each pair

$$\begin{tikzcd} A \times X \arrow[r, "R"] \arrow[d, "f \times g"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ B \times Y \arrow[r, "S"'] & \{\mathsf{true}, \mathsf{false}\} \end{tikzcd} \qquad \begin{tikzcd} B \times Y \arrow[r, "S"] \arrow[d, "h \times k"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ C \times Z \arrow[r, "T"'] & \{\mathsf{true}, \mathsf{false}\} \end{tikzcd}$$

of inclusions of relations, we define the vertical composition

$$\begin{tikzcd} A \arrow[r, "R", "|" marking] \arrow[d, "h \circ f"'] \arrow[dr, phantom, "\Vert \; \beta \circ \alpha \; \Downarrow"] & X \arrow[d, "k \circ g"] \\ C \arrow[r, "T"', "|" marking] & Z \end{tikzcd}$$

of α and β as the inclusion of relations

$$T \circ [(h \circ f) \times (k \circ g)] \subset R, \qquad \begin{tikzcd} A \times X \arrow[r, "R"] \arrow[d, "(h \circ f) \times (k \circ g)"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ C \times Z \arrow[r, "T"'] & \{\mathsf{true}, \mathsf{false}\} \end{tikzcd}$$

given by the pasting of inclusions[1]

$$\begin{tikzcd} A \times X \arrow[r, "R"] \arrow[d, "f \times g"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ B \times Y \arrow[r, "S"] \arrow[d, "h \times k"'] \arrow[dr, phantom, "\circlearrowright"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ C \times Z \arrow[r, "T"'] & \{\mathsf{true}, \mathsf{false}\}. \end{tikzcd}$$

[1]This is justified by noting that, given $(a, x) \in A \times X$, the statement

- We have $h(f(a)) \sim_T k(g(x))$;

is implied by the statement

- We have $a \sim_R x$;

since

- If $a \sim_R x$, then $f(a) \sim_S g(x)$, as $S \circ (f \times g) \subset R$;
- If $b \sim_S y$, then $h(b) \sim_T k(y)$, as $T \circ (h \times k) \subset S$, and thus, in particular:
  - If $f(a) \sim_S g(x)$, then $h(f(a)) \sim_T k(g(x))$.

8.3.5.5 The Associators

Definition 8.3.5.5.1 ▶ The Associators of $\mathsf{Rel}^{\mathsf{dbl}}$

For each composable triple

$$A \overset{R}{\nrightarrow} B \overset{S}{\nrightarrow} C \overset{T}{\nrightarrow} D$$

of horizontal morphisms of $\mathsf{Rel}^{\mathsf{dbl}}$, the component

$$\alpha^{\mathsf{Rel}^{\mathsf{dbl}}}_{T,S,R} : (T \odot S) \odot R \overset{\sim}{\Longrightarrow} T \odot (S \odot R),$$

$$\begin{tikzcd}
A \arrow[r, "R", "\shortmid"{marking}] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\Downarrow \alpha^{\mathsf{Rel}^{\mathsf{dbl}}}_{T,S,R}"] & B \arrow[r, "S", "\shortmid"{marking}] & C \arrow[r, "T", "\shortmid"{marking}] & D \arrow[d, "\mathrm{id}_D"] \\
A \arrow[r, "R"', "\shortmid"{marking}] & B \arrow[r, "S"', "\shortmid"{marking}] & C \arrow[r, "T"', "\shortmid"{marking}] & D
\end{tikzcd}$$

of the associator of $\mathsf{Rel}^{\mathsf{dbl}}$ at (R, S, T) is the identity inclusion[1]

$$(T \diamond S) \diamond R = T \diamond (S \diamond R)$$

$$\begin{tikzcd}
A \times B \arrow[r, "(T \diamond S) \diamond R"] \arrow[d, equal] \arrow[dr, phantom, "\neq"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\
A \times B \arrow[r, "T \diamond (S \diamond R)"'] & \{\mathsf{true}, \mathsf{false}\}.
\end{tikzcd}$$

[1] This is justified by Constructions With Relations, **??** of **??**.

8.3.5.6 The Left Unitors

Definition 8.3.5.6.1 ▶ The Left Unitors of $\mathsf{Rel}^{\mathsf{dbl}}$

For each horizontal morphism $R\colon A \nrightarrow B$ of $\mathsf{Rel}^{\mathsf{dbl}}$, the component

$$\lambda_R^{\mathsf{Rel}^{\mathsf{dbl}}} : \mathbb{1}_B \odot R \overset{\sim}{\Longrightarrow} R,$$

$$\begin{tikzcd} A \arrow[r, "R", "|" marking] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\Downarrow \lambda_R^{\mathsf{Rel}^{\mathsf{dbl}}}"] & B \arrow[r, "\mathbb{1}_B", "|" marking] \arrow[d, "\mathrm{id}_B"] & B \arrow[d, "\mathrm{id}_B"] \\ A \arrow[rr, "R"', "|" marking] & & B \end{tikzcd}$$

of the left unitor of $\mathsf{Rel}^{\mathsf{dbl}}$ at R is the identity inclusion[1]

$$R = \chi_B \diamond R,$$

$$\begin{tikzcd} A \times B \arrow[r, "\chi_B \diamond R"] \arrow[d, equal] \arrow[dr, phantom, "=="] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ A \times B \arrow[r, "R"'] & \{\mathsf{true}, \mathsf{false}\}. \end{tikzcd}$$

[1]This is justified by Constructions With Relations, **??** of **??**.

8.3.5.7 The Right Unitors

Definition 8.3.5.7.1 ▶ The Right Unitors of $\mathsf{Rel}^{\mathsf{dbl}}$

For each horizontal morphism $R\colon A \nrightarrow B$ of $\mathsf{Rel}^{\mathsf{dbl}}$, the component

$$\rho_R^{\mathsf{Rel}^{\mathsf{dbl}}} : R \odot \mathbb{1}_A \overset{\sim}{\Longrightarrow} R,$$

$$\begin{tikzcd} A \arrow[r, "\mathbb{1}_A", "|" marking] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\Downarrow \rho_R^{\mathsf{Rel}^{\mathsf{dbl}}}"] & A \arrow[r, "R", "|" marking] & B \arrow[d, "\mathrm{id}_B"] \\ A \arrow[rr, "R"', "|" marking] & & B \end{tikzcd}$$

of the right unitor of $\mathsf{Rel}^{\mathsf{dbl}}$ at R is the identity inclusion[1]

$$R = R \diamond \chi_A, \qquad \begin{tikzcd} A \times B \arrow[r, "R \diamond \chi_A"] \arrow[d, equal] \arrow[dr, phantom, "\parallel"] & \{\mathsf{true}, \mathsf{false}\} \arrow[d, "\mathrm{id}_{\{\mathsf{true},\mathsf{false}\}}"] \\ A \times B \arrow[r, "R"'] & \{\mathsf{true}, \mathsf{false}\}. \end{tikzcd}$$

[1]This is justified by Constructions With Relations, **??** of **??**.

8.4 Properties of the 2-Category of Relations

8.4.1 Self-Duality

Proposition 8.4.1.1.1 ▶ Self-Duality for the (2-)Category of Relations

The (2-)category of relations is self-dual:

1. *Self-Duality I.* We have an isomorphism

$$\mathsf{Rel}^{\mathsf{op}} \overset{\text{eq.}}{\cong} \mathsf{Rel}$$

of categories.

2. *Self-Duality II.* We have a 2-isomorphism

$$\mathbf{Rel}^{\mathsf{op}} \overset{\text{eq.}}{\cong} \mathbf{Rel}$$

of 2-categories.

Proof 8.4.1.1.2 ▶ Proof of Proposition 8.4.1.1.1

Item 1: Self-Duality I

We claim that the functor

$$F\colon \mathsf{Rel}^{\mathsf{op}} \longrightarrow \mathsf{Rel}$$

given by the identity on objects and by $R \mapsto R^{\dagger}$ on morphisms is an isomorphism of categories.
By Categories, Item 1 of Proposition 11.6.8.1.3, it suffices to show that F is bijective on objects (which is clear) and fully faithful. Indeed, the map

$$(-)^{\dagger}\colon \mathrm{Rel}(A, B) \longrightarrow \mathrm{Rel}(B, A)$$

defined by the assignment $R \mapsto R^{\dagger}$ is a bijection by Constructions With Relations, **??** of **??**, showing F to be fully faithful.

Item 2: Self-Duality II

We claim that the 2-functor

$$F\colon \mathsf{Rel}^{\mathsf{op}} \longrightarrow \mathsf{Rel}$$

given by the identity on objects, by $R \mapsto R^{\dagger}$ on morphisms, and by preserving inclusions on 2-morphisms via Constructions With Relations, **??** of **??**, is an isomorphism of categories.
By **??**, **??** of **??**, it suffices to show that F is:

- Bijective on objects, which is clear.
- Bijective on 1-morphisms, which was shown in Item 1.
- Bijective on 2-morphisms, which follows from Constructions With Relations, **??** of **??**.

Thus F is indeed a 2-isomorphism of categories.

8.4.2 Isomorphisms and Equivalences

Let $R\colon A \nrightarrow B$ be a relation from A to B.

Proposition 8.4.2.1.1 ▶ Isomorphisms and Equivalences in Rel

The following conditions are equivalent:

1. The relation $R\colon A \nrightarrow B$ is an equivalence in **Rel**, i.e.:

   ($\star$) There exists a relation $R^{-1}\colon B \nrightarrow A$ from B to A together with isomorphisms

   $$R^{-1} \diamond R \cong \chi_A,$$
   $$R \diamond R^{-1} \cong \chi_B.$$

2. The relation $R\colon A \nrightarrow B$ is an isomorphism in Rel, i.e.:

   ($\star$) There exists a relation $R^{-1}\colon B \nrightarrow A$ from B to A such that we have

   $$R^{-1} \diamond R = \chi_A,$$
   $$R \diamond R^{-1} = \chi_B.$$

3. There exists a bijection $f\colon A \xrightarrow{\sim} B$ with $R = \mathrm{Gr}(f)$.

Proof 8.4.2.1.2 ▶ Proof of Proposition 8.4.2.1.1

We claim that Items 1 to 3 are indeed equivalent:

- *Item 1* $\iff$ *Item 2:* This follows from the fact that **Rel** is locally posetal, so that natural isomorphisms and equalities of 1-morphisms in **Rel** coincide.

- *Item 2* $\implies$ *Item 3:* The equalities in Item 2 imply $R \dashv R^{-1}$, and thus by Proposition 8.4.3.1.1, there exists a function $f_R\colon A \to B$ associated to R, where, for each $a \in A$, the image $f_R(a)$ of a by f_R is the unique element of $R(a)$, which implies $R = \mathrm{Gr}(f_R)$ in particular. Furthermore, we have $R^{-1} = f_R^{-1}$ (as in Constructions With Relations, **??**). The conditions from Item 2 then become the

following:

$$f_R^{-1} \diamond f_R = \chi_A,$$
$$f_R \diamond f_R^{-1} = \chi_B.$$

All that is left is to show then is that f_R is a bijection:

- *The Function f_R Is Injective.* Let $a, b \in A$ and suppose that $f_R(a) = f_R(b)$. Since $a \sim_R f_R(a)$ and $f_R(a) = f_R(b) \sim_{R^{-1}} b$, the condition $f_R^{-1} \diamond f_R = \chi_A$ implies that $a = b$, showing f_R to be injective.
- *The Function f_R Is Surjective.* Let $b \in B$. Applying the condition $f_R \diamond f_R^{-1} = \chi_B$ to (b, b), it follows that there exists some $a \in A$ such that $f_R^{-1}(b) = a$ and $f_R(a) = b$. This shows f_R to be surjective.

- *Item 3 $\implies$ Item 2:* By Constructions With Relations, **??** of **??**, we have an adjunction $\mathrm{Gr}(f) \dashv f^{-1}$, giving inclusions

$$\chi_A \subset f^{-1} \diamond \mathrm{Gr}(f),$$
$$\mathrm{Gr}(f) \diamond f^{-1} \subset \chi_B.$$

We claim the reverse inclusions are also true:

- $f^{-1} \diamond Gr(f) \subset \chi_A$: This is equivalent to the statement that if $f(a) = b$ and $f^{-1}(b) = a'$, then $a = a'$, which follows from the injectivity of f.
- $\chi_B \subset Gr(f) \diamond f^{-1}$: This is equivalent to the statement that given $b \in B$ there exists some $a \in A$ such that $f^{-1}(b) = a$ and $f(a) = b$, which follows from the surjectivity of f.

This finishes the proof.

8.4.3 Internal Adjunctions

Let A and B be sets.

Proposition 8.4.3.1.1 ► Adjunctions in Rel

We have a natural bijection

$$\left\{\begin{matrix}\text{Adjunctions in } \mathbf{Rel} \\ \text{from } A \text{ to } B\end{matrix}\right\} \cong \left\{\begin{matrix}\text{Functions} \\ \text{from } A \text{ to } B\end{matrix}\right\},$$

with every adjunction in **Rel** being of the form $\mathrm{Gr}(f) \dashv f^{-1}$ for some function f.

Proof 8.4.3.1.2 ► Proof of Proposition 8.4.3.1.1

We proceed step by step:

1. *From Adjunctions in* **Rel** *to Functions.* An adjunction in **Rel** from A to B consists of a pair of relations

$$\begin{aligned} R &\colon A \nrightarrow B, \\ S &\colon B \nrightarrow A, \end{aligned}$$

together with inclusions

$$\begin{aligned} \chi_A &\subset S \diamond R, \\ R \diamond S &\subset \chi_B. \end{aligned}$$

We claim that these conditions imply that R is total and functional, i.e. that $R(a)$ is a singleton for each $a \in A$:

(a) *$R(a)$ Has an Element.* Given $a \in A$, since $\chi_A \subset S \diamond R$, we must have $\{a\} \subset S(R(a))$, implying that there exists some $b \in B$ such that $a \sim_R b$ and $b \sim_S a$, and thus $R(a) \neq \emptyset$, as $b \in R(a)$.

(b) *$R(a)$ Has No More Than One Element.* Suppose that we have $a \sim_R b$ and $a \sim_R b'$ for $b, b' \in B$. We claim that $b = b'$:

i. Since $\chi_A \subset S \diamond R$, there exists some $k \in B$ such that $a \sim_R k$ and $k \sim_S a$.

ii. Since $R \diamond S \subset \chi_B$, if $b'' \sim_S a'$ and $a' \sim_R b'''$, then $b'' = b'''$.

iii. Applying the above to $b'' = k$, $b''' = b$, and $a' = a$, since $k \sim_S a$ and $a \sim_R b'$, we have $k = b$.

iv. Similarly $k = b'$.

v. Thus $b = b'$.

Together, the above two items show $R(a)$ to be a singleton, being thus given by $\mathrm{Gr}(f)$ for some function $f\colon A \to B$, which gives a map

$$\left\{\begin{matrix}\text{Adjunctions in } \mathbf{Rel} \\ \text{from } A \text{ to } B\end{matrix}\right\} \to \left\{\begin{matrix}\text{Functions} \\ \text{from } A \text{ to } B\end{matrix}\right\}.$$

Moreover, by uniqueness of adjoints (Internal Adjunctions, **??** of **??**), this implies also that $S = f^{-1}$.

2. *From Functions to Adjunctions in* **Rel**. By Constructions With Relations, **??** of **??**, every function $f\colon A \to B$ gives rise to an adjunction $\mathrm{Gr}(f) \dashv f^{-1}$ in Rel, giving a map

$$\left\{\begin{matrix}\text{Functions} \\ \text{from } A \text{ to } B\end{matrix}\right\} \to \left\{\begin{matrix}\text{Adjunctions in } \mathbf{Rel} \\ \text{from } A \text{ to } B\end{matrix}\right\}.$$

3. *Invertibility: From Functions to Adjunctions Back to Functions.* We need to show that starting with a function $f\colon A \to B$, passing to $\mathrm{Gr}(f) \dashv f^{-1}$, and then passing again to a function gives f again. This is clear however, since we have $a \sim_{\mathrm{Gr}(f)} b$ iff $f(a) = b$.

4. *Invertibility: From Adjunctions to Functions Back to Adjunctions.* We need to show that, given an adjunction $R \dashv S$ in **Rel** giving rise to a function $f_{R,S}\colon A \to B$, we have

$$\begin{aligned}\mathrm{Gr}\left(f_{R,S}\right) &= R, \\ f_{R,S}^{-1} &= S.\end{aligned}$$

We check these explicitly:

- $Gr(f_{R,S}) = R$. We have

$$\begin{aligned}\mathrm{Gr}(f_{R,S}) &\overset{\text{def}}{=} \left\{(a, f_{R,S}(a)) \in A \times B \,\middle|\, a \in A\right\}\\ &\overset{\text{def}}{=} \{(a, R(a)) \in A \times B \mid a \in A\}\\ &= R.\end{aligned}$$

- $f_{R,S}^{-1} = S$. We first claim that, given $a \in A$ and $b \in B$, the following conditions are equivalent:
  - We have $a \sim_R b$.
  - We have $b \sim_S a$.

  Indeed:
  - *If $a \sim_R b$, then $b \sim_S a$:* Since $\chi_A \subset S \diamond R$, there exists $k \in B$ such that $a \sim_R k$ and $k \sim_S a$, but since $a \sim_R b$ and R is functional, we have $k = b$ and thus $b \sim_S a$.
  - *If $b \sim_S a$, then $a \sim_R b$:* First note that since R is total we have $a \sim_R b'$ for some $b' \in B$. Now, since $R \diamond S \subset \chi_B$, $b \sim_S a$, and $a \sim_R b'$, we have $b = b'$, and thus $a \sim_R b$.

  Having show this, we now have

$$\begin{aligned}f_{R,S}^{-1}(b) &\overset{\text{def}}{=} \left\{a \in A \,\middle|\, f_{R,S}(a) = b\right\}\\ &\overset{\text{def}}{=} \{a \in A \mid a \sim_R b\}\\ &= \{a \in A \mid b \sim_S a\}\\ &\overset{\text{def}}{=} S(b).\end{aligned}$$

  for each $b \in B$, showing $f_{R,S}^{-1} = S$.

This finishes the proof. ∎

8.4.4 Internal Monads

Let A be a set.

Proposition 8.4.4.1.1 ▶ Monads in Rel

We have a natural identification[1]

$$\left\{\begin{matrix}\text{Monads in}\\ \textbf{Rel}\text{ on } A\end{matrix}\right\} \cong \{\text{Preorders on } A\}.$$

[1]See also **??** for an extension of this correspondence to "relative monads in **Rel**".

Proof 8.4.4.1.2 ▶ Proof of Proposition 8.4.4.1.1

A monad in **Rel** on A consists of a relation $R\colon A \nrightarrow A$ together with maps

$$\begin{aligned}\mu_R &\colon R \diamond R \subset R,\\ \eta_R &\colon \chi_A \subset R\end{aligned}$$

making the diagrams

$$\begin{tikzcd}
\chi_A \diamond R \arrow[r, "\eta_R \diamond \mathrm{id}_R", "\shortmid"{marking}] \arrow[dr, equal, "\lambda_R^{\mathbf{Rel}(A,B)}"'] & R \diamond R \arrow[d, "\mu_R", "\shortmid"{marking}] \\
& R
\end{tikzcd}
\qquad
\begin{tikzcd}
& R \diamond (R \diamond R) \arrow[dl, equal, "\alpha_{R,R,R}^{\mathbf{Rel}(A,B)}"'] \arrow[dr, "\mathrm{id}_R \diamond \mu_R", "\not\shortmid"{marking}] & \\
(R \diamond R) \diamond R \arrow[d, "\mu_R \diamond \mathrm{id}_R", "\not\shortmid"{marking}] & & R \diamond R \arrow[d, "\mu_R", "\not\shortmid"{marking}] \\
R \diamond R \arrow[rr, "\mu_R", "\shortmid"{marking}] & & R
\end{tikzcd}
\qquad
\begin{tikzcd}
R \diamond \chi_A \arrow[r, "\mathrm{id}_R \diamond \eta_R", "\shortmid"{marking}] \arrow[dr, equal, "\rho_R^{\mathbf{Rel}(A,B)}"'] & R \diamond R \arrow[d, "\mu_R", "\shortmid"{marking}] \\
& R
\end{tikzcd}$$

commute. However, since all morphisms involved are inclusions, the commutativity of the above diagrams is automatic (Categories, Item 4 of Proposition 11.2.7.1.2), and hence all that is left is the data of the two maps μ_R and η_R, which correspond respectively to the following conditions:

1. For each $a, b, c \in A$, if $a \sim_R b$ and $b \sim_R c$, then $a \sim_R c$.
2. For each $a \in A$, we have $a \sim_R a$.

These are exactly the requirements for R to be a preorder (**??**, **??**). Conversely, any preorder $\preceq$ gives rise to a pair of maps $\mu_\preceq$ and $\eta_\preceq$, forming a monad on A. ∎

8.4.5 Internal Comonads

Let A be a set.

Proposition 8.4.5.1.1 ▶ Comonads in Rel

We have a natural identification

$$\left\{\begin{matrix}\text{Comonads in}\\ \textbf{Rel}\text{ on } A\end{matrix}\right\} \cong \{\text{Subsets of } A\}.$$

Proof 8.4.5.1.2 ▶ Proof of Proposition 8.4.5.1.1

A comonad in **Rel** on A consists of a relation $R\colon A \nrightarrow A$ together with maps

$$\begin{aligned}\Delta_R &\colon R \subset R \diamond R,\\ \epsilon_R &\colon R \subset \chi_A\end{aligned}$$

making the diagrams

$$\begin{tikzcd}
R \arrow[r, "\Delta_R", "|"] \arrow[dr, equal, "\lambda_R^{\mathbf{Rel}(A,B),-1}"'] & R \diamond R \arrow[d, "|", "\epsilon_R \diamond \mathrm{id}_R"] \\
& \chi_A \diamond R
\end{tikzcd}
\qquad
\begin{tikzcd}
& R \diamond R \arrow[dr, "|", "\mathrm{id}_R \diamond \Delta_R"] & \\
R \arrow[ur, "\Delta_R", "|"] \arrow[dr, "\Delta_R"', "|"] & & R \diamond (R \diamond R) \arrow[d, equal, "\alpha_{R,R,R}^{\mathbf{Rel}(A,B),-1}"] \\
& R \diamond R \arrow[r, "|", "\Delta_R \diamond \mathrm{id}_R"'] & (R \diamond R) \diamond R
\end{tikzcd}
\qquad
\begin{tikzcd}
R \arrow[r, "\Delta_R", "|"] \arrow[dr, equal, "\rho_R^{\mathbf{Rel}(A,B),-1}"'] & R \diamond R \arrow[d, "|", "\mathrm{id}_R \diamond \epsilon_R"] \\
& R \diamond \chi_A
\end{tikzcd}$$

commute. However, since all morphisms involved are inclusions, the commutativity of the above diagrams is automatic (Categories, Item 4 of Proposition 11.2.7.1.2), and hence all that is left is the data of the two maps Δ_R and ϵ_R, which correspond respectively to the following conditions:

1. For each $a, b \in A$, if $a \sim_R b$, then there exists some $k \in A$ such that $a \sim_R k$ and $k \sim_R b$.

2. For each $a, b \in A$, if $a \sim_R b$, then $a = b$.

Taking $k = b$ in the first condition above shows it to be trivially satisfied, while the second condition implies $R \subset \Delta_A$, i.e. R must be a subset of A. Conversely, any subset U of A satisfies $U \subset \Delta_A$, defining a comonad as above.

8.4.6 Co/Monoids

Remark 8.4.6.1.1 ► Co/Monoids in Rel

The monoids in **Rel** with respect to the Cartesian monoidal structure of Proposition 8.3.3.8.1 are called *hypermonoids*, and their theory is explored in **??**. Similarly, the comonoids in **Rel** are called *hypercomonoids*, and they are defined and studied in **??**.

8.4.7 Monomorphisms

In this section we characterise the epimorphisms in the category Rel, following **??**, **??**.

Proposition 8.4.7.1.1 ► Characterisations of Monomorphisms in Rel

Let $R\colon A \nrightarrow B$ be a relation. The following conditions are equivalent:

1. The relation R is a monomorphism in Rel.
2. The direct image function
$$R_!\colon \mathcal{P}(A) \to \mathcal{P}(B)$$
associated to R is injective.
3. The codirect image function
$$R_*\colon \mathcal{P}(A) \to \mathcal{P}(B)$$
associated to R is injective.

Moreover, if R is a monomorphism, then it satisfies the following condition, and the converse holds if R is total:

($\star$) For each $a, a' \in A$, if there exists some $b \in B$ such that

$$\begin{aligned} a &\sim_R b, \\ a' &\sim_R b, \end{aligned}$$

then $a = a'$.

Proof 8.4.7.1.2 ▶ Proof of Proposition 8.4.7.1.1

Firstly note that Items 2 and 3 are equivalent by Constructions With Relations, **??** of **??**. We then claim that Items 1 and 2 are also equivalent:

- *Item 1* $\Longrightarrow$ *Item 2*: Let $U, V \in \mathcal{P}(A)$ and consider the diagram

$$\mathrm{pt} \underset{V}{\overset{U}{\rightrightarrows}} A \xrightarrow{R} B.$$

  By Constructions With Relations, **??**, we have

$$\begin{aligned} R_!(U) &= R \diamond U, \\ R_!(V) &= R \diamond V. \end{aligned}$$

  Now, if $R \diamond U = R \diamond V$, i.e. $R_!(U) = R_!(V)$, then $U = V$ since R is assumed to be a monomorphism, showing $R_!$ to be injective.

- *Item 2* $\Longrightarrow$ *Item 1*: Conversely, suppose that $R_!$ is injective, consider the diagram

$$X \underset{T}{\overset{S}{\rightrightarrows}} A \xrightarrow{R} B,$$

  and suppose that $R \diamond S = R \diamond T$. Note that, since $R_!$ is injective, given a diagram of the form

$$\mathrm{pt} \underset{V}{\overset{U}{\rightrightarrows}} A \xrightarrow{R} B,$$

if $R_!(U) = R \diamond U = R \diamond V = R_!(V)$, then $U = V$. In particular, for each $x \in X$, we may consider the diagram

$$\begin{tikzcd} \mathrm{pt} \arrow[r, "{[x]}", "\shortmid"{marking}] & X \arrow[r, shift left, "S"{name=U}, "\shortmid"{marking}] \arrow[r, shift right, "T"'{name=D}, "\shortmid"{marking}] & A \arrow[r, "R", "\shortmid"{marking}] & B, \end{tikzcd}$$

for which we have $R \diamond S \diamond [x] = R \diamond T \diamond [x]$, implying that we have

$$S(x) = S \diamond [x] = T \diamond [x] = T(x)$$

for each $x \in X$, implying $S = T$, and thus R is a monomorphism.

We can also prove this in a more abstract way, following [MSE 350788]:

- *Item 1* $\Longrightarrow$ *Item 2*: Assume that R is a monomorphism.
  - We first notice that the functor $\mathrm{Rel}(\mathrm{pt}, -) : \mathrm{Rel} \longrightarrow \mathrm{Sets}$ maps R to $R_!$ by Constructions With Relations, **??**.
  - Since $\mathrm{Rel}(\mathrm{pt}, -)$ preserves all limits by Limits and Colimits, **??** of **??**, it follows by **??**, **??** of **??** that $\mathrm{Rel}(\mathrm{pt}, -)$ also preserves monomorphisms.
  - Since R is a monomorphism and $\mathrm{Rel}(\mathrm{pt}, -)$ maps R to $R_!$, it follows that $R_!$ is also a monomorphism.
  - Since the monomorphisms in Sets are precisely the injections (**??**, **??** of **??**), it follows that $R_!$ is injective.
- *Item 2* $\Longrightarrow$ *Item 1*: Assume that $R_!$ is injective.
  - We first notice that the functor $\mathrm{Rel}(\mathrm{pt}, -) : \mathrm{Rel} \longrightarrow \mathrm{Sets}$ maps R to $R_!$ by Constructions With Relations, **??**.
  - Since the monomorphisms in Sets are precisely the injections (**??**, **??** of **??**), it follows that $R_!$ is a monomorphism.
  - Since $\mathrm{Rel}(\mathrm{pt}, -)$ is faithful, it follows by **??**, **??** of **??** that $\mathrm{Rel}(\mathrm{pt}, -)$ reflects monomorphisms.

– Since $R_!$ is a monomorphism and $\mathrm{Rel}(\mathrm{pt}, -)$ maps R to $R_!$, it follows that R is also a monomorphism.

Finally, we prove the second part of the statement. Assume that R is a monomorphism, let $a, a' \in A$ such that $a \sim_R b$ and $a' \sim_R b$ for some $b \in B$, and consider the diagram

$$\begin{tikzcd} \mathrm{pt} \arrow[r, shift left, "{[a]}"] \arrow[r, shift right, "{[a']}"'] & A \arrow[r, "R"] & B. \end{tikzcd}$$

Since $\star \sim_{[a]} a$ and $a \sim_R b$, we have $\star \sim_{R \diamond [a]} b$. Similarly, $\star \sim_{R \diamond [a']} b$. Thus $R \diamond [a] = R \diamond [a']$, and since R is a monomorphism, we have $[a] = [a']$, i.e. $a = a'$.
Conversely, assume the condition

($\star$) For each $a, a' \in A$, if there exists some $b \in B$ such that

$$a \sim_R b,$$
$$a' \sim_R b,$$

then $a = a'$.

consider the diagram

$$\begin{tikzcd} X \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] & A \arrow[r, "R"] & B, \end{tikzcd}$$

and let $(x, a) \in S$. Since R is total and $a \in A$, there exists some $b \in B$ such that $a \sim_R b$. In this case, we have $x \sim_{R \diamond S} b$, and since $R \diamond S = R \diamond T$, we have also $x \sim_{R \diamond T} b$. Thus there must exist some $a' \in A$ such that $x \sim_T a'$ and $a' \sim_R b$. However, since $a, a' \sim_R b$, we must have $a = a'$, and thus $(x, a) \in T$ as well.
A similar argument shows that if $(x, a) \in T$, then $(x, a) \in S$, and thus $S = T$ and it follows that R is a monomorphism.

8.4.8 2-Categorical Monomorphisms

In this section we characterise (for now, some of) the 2-categorical monomorphisms in **Rel**, following Types of Morphisms in Bicategories, Section 13.1.

Proposition 8.4.8.1.1 ▶ 2-Categorical Monomorphisms in Rel

Let $R\colon A \nrightarrow B$ be a relation.

1. *Representably Faithful Morphisms in* **Rel**. Every morphism of **Rel** is a representably faithful morphism.

2. *Representably Full Morphisms in* **Rel**. The following conditions are equivalent:

   (a) The morphism $R\colon A \nrightarrow B$ is a representably full morphism.

   (b) For each pair of relations $S, T\colon X \rightrightarrows A$, the following condition is satisfied:

   ($\star$) If $R \diamond S \subset R \diamond T$, then $S \subset T$.

   (c) The functor

   $$R_!\colon (\mathcal{P}(A), \subset) \to (\mathcal{P}(B), \subset)$$

   is full.

   (d) For each $U, V \in \mathcal{P}(A)$, if $R_!(U) \subset R_!(V)$, then $U \subset V$.

   (e) The functor

   $$R_*\colon (\mathcal{P}(A), \subset) \to (\mathcal{P}(B), \subset)$$

   is full.

   (f) For each $U, V \in \mathcal{P}(A)$, if $R_*(U) \subset R_*(V)$, then $U \subset V$.

3. *Representably Fully Faithful Morphisms in* **Rel**. Every representably full morphism in **Rel** is a representably fully faithful morphism.

PROOF 8.4.8.1.2 ▶ PROOF OF PROPOSITION 8.4.8.1.1

Item 1: Representably Faithful Morphisms in Rel

The relation R is a representably faithful morphism in **Rel** iff, for each $X \in \mathrm{Obj}(\mathbf{Rel})$, the functor

$$R_! \colon \mathbf{Rel}(X, A) \to \mathbf{Rel}(X, B)$$

is faithful, i.e. iff the morphism

$$R_{*|S,T} \colon \mathrm{Hom}_{\mathbf{Rel}(X,A)}(S, T) \to \mathrm{Hom}_{\mathbf{Rel}(X,B)}(R \diamond S, R \diamond T)$$

is injective for each $S, T \in \mathrm{Obj}(\mathbf{Rel}(X, A))$. However, $\mathrm{Hom}_{\mathbf{Rel}(X,A)}(S, T)$ is either empty or a singleton, in either case of which the map $R_{*|S,T}$ is necessarily injective.

Item 2: Representably Full Morphisms in Rel

We claim Items 2a to 2f are indeed equivalent:

- *Item 2a* $\Longleftrightarrow$ *Item 2b:* This is simply a matter of unwinding definitions: The relation R is a representably full morphism in **Rel** iff, for each $X \in \mathrm{Obj}(\mathbf{Rel})$, the functor

  $$R_! \colon \mathbf{Rel}(X, A) \to \mathbf{Rel}(X, B)$$

  is full, i.e. iff the morphism

  $$R_{*|S,T} \colon \mathrm{Hom}_{\mathbf{Rel}(X,A)}(S, T) \to \mathrm{Hom}_{\mathbf{Rel}(X,B)}(R \diamond S, R \diamond T)$$

  is surjective for each $S, T \in \mathrm{Obj}(\mathbf{Rel}(X, A))$, i.e. iff, whenever $R \diamond S \subset R \diamond T$, we also have $S \subset T$.

- *Item 2c* $\Longleftrightarrow$ *Item 2d:* This is also simply a matter of unwinding definitions: The functor

  $$R_! \colon (\mathcal{P}(A), \subset) \to (\mathcal{P}(B), \subset)$$

  is full iff, for each $U, V \in \mathcal{P}(A)$, the morphism

  $$R_{*|U,V} \colon \mathrm{Hom}_{\mathcal{P}(A)}(U, V) \to \mathrm{Hom}_{\mathcal{P}(B)}(R_!(U), R_!(V))$$

  is surjective, i.e. iff whenever $R_!(U) \subset R_!(V)$, we also necessarily have $U \subset V$.

- *Item 2e* $\Longleftrightarrow$ *Item 2f:* This is once again simply a matter of unwinding definitions, and proceeds exactly in the same way as in the proof of the equivalence between Items 2c and 2d given above.

- *Item 2d* $\Longrightarrow$ *Item 2f:* Suppose that the following condition is true:

  ($\star$) For each $U, V \in \mathcal{P}(A)$, if $R_!(U) \subset R_!(V)$, then $U \subset V$.

  We need to show that the condition

  ($\star$) For each $U, V \in \mathcal{P}(A)$, if $R_*(U) \subset R_*(V)$, then $U \subset V$.

  is also true. We proceed step by step:

  - Suppose we have $U, V \in \mathcal{P}(A)$ with $R_*(U) \subset R_*(V)$.
  - By Constructions With Relations, **??** of **??**, we have

    $$\begin{aligned} R_*(U) &= B \setminus R_!(A \setminus U), \\ R_*(V) &= B \setminus R_!(A \setminus V). \end{aligned}$$

  - By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 we have $R_!(A \setminus V) \subset R_!(A \setminus U)$.
  - By assumption, we then have $A \setminus V \subset A \setminus U$.
  - By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 again, we have $U \subset V$.

- *Item 2f* $\Longrightarrow$ *Item 2d:* Suppose that the following condition is true:

  ($\star$) For each $U, V \in \mathcal{P}(A)$, if $R_*(U) \subset R_*(V)$, then $U \subset V$.

  We need to show that the condition

  ($\star$) For each $U, V \in \mathcal{P}(A)$, if $R_!(U) \subset R_!(V)$, then $U \subset V$.

  is also true. We proceed step by step:

  - Suppose we have $U, V \in \mathcal{P}(A)$ with $R_!(U) \subset R_!(V)$.

- By Constructions With Relations, **??** of **??**, we have

$$R_!(U) = B \setminus R_*(A \setminus U),$$
$$R_!(V) = B \setminus R_*(A \setminus V).$$

- By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 we have $R_*(A \setminus V) \subset R_*(A \setminus U)$.
- By assumption, we then have $A \setminus V \subset A \setminus U$.
- By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 again, we have $U \subset V$.

- *Item 2b* $\Longrightarrow$ *Item 2d:* Consider the diagram

$$\begin{tikzcd} X \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] \arrow[r, phantom, "\mid"] & A \arrow[r, "R", "\mid"'] & B, \end{tikzcd}$$

and suppose that $R \diamond S \subset R \diamond T$. Note that, by assumption, given a diagram of the form

$$\begin{tikzcd} \mathrm{pt} \arrow[r, shift left, "U"] \arrow[r, shift right, "V"'] \arrow[r, phantom, "\mid"] & A \arrow[r, "R", "\mid"'] & B, \end{tikzcd}$$

if $R_!(U) = R \diamond U \subset R \diamond V = R_!(V)$, then $U \subset V$. In particular, for each $x \in X$, we may consider the diagram

$$\begin{tikzcd} \mathrm{pt} \arrow[r, "{[x]}", "\mid"'] & X \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] \arrow[r, phantom, "\mid"] & A \arrow[r, "R", "\mid"'] & B, \end{tikzcd}$$

for which we have $R \diamond S \diamond [x] \subset R \diamond T \diamond [x]$, implying that we have

$$S(x) = S \diamond [x] \subset T \diamond [x] = T(x)$$

for each $x \in X$, implying $S \subset T$.

- *Item 2d* $\Longrightarrow$ *Item 2b:* Let $U, V \in \mathcal{P}(A)$ and consider the diagram

$$\begin{tikzcd} \mathrm{pt} \arrow[r, shift left, "U"] \arrow[r, shift right, "V"'] \arrow[r, phantom, "\nrightarrow"] & A \arrow[r, "R", "\nrightarrow"'] & B. \end{tikzcd}$$

By Remark 8.5.1.1.2, we have

$$\begin{aligned} R_!(U) &= R \diamond U, \\ R_!(V) &= R \diamond V. \end{aligned}$$

Now, if $R_!(U) \subset R_!(V)$, i.e. $R \diamond U \subset R \diamond V$, then $U \subset V$ by assumption.

Item 3: Representably Fully Faithful Morphisms in Rel

This follows from Items 1 and 2. ∎

Question 8.4.8.1.3 ▶ Better Characterisations of Representably Full Morphisms in Rel

Item 2 of Proposition 8.4.8.1.1 gives a characterisation of the representably full morphisms in **Rel**.
Are there other nice characterisations of these?
This question also appears as [MO 467527].

8.4.9 Epimorphisms

In this section we characterise the epimorphisms in the category Rel, following **??**, **??**.

Proposition 8.4.9.1.1 ▶ Characterisations of Epimorphisms in Rel

Let $R\colon A \nrightarrow B$ be a relation. The following conditions are equivalent:

1. The relation R is an epimorphism in Rel.
2. The weak inverse image function

$$R^{-1}\colon \mathcal{P}(B) \to \mathcal{P}(A)$$

associated to R is injective.

3. The strong inverse image function

$$R_{-1}\colon \mathcal{P}(B) \to \mathcal{P}(A)$$

associated to R is injective.

4. The function $R\colon A \to \mathcal{P}(B)$ is "surjective on singletons":

   ($\star$) For each $b \in B$, there exists some $a \in A$ such that $R(a) = \{b\}$.

Moreover, if R is total and an epimorphism, then it satisfies the following equivalent conditions:

1. For each $b \in B$, there exists some $a \in A$ such that $a \sim_R b$.
2. We have $\mathrm{Im}(R) = B$.

Proof 8.4.9.1.2 ► Proof of Proposition 8.4.9.1.1

Firstly note that Items 2 and 3 are equivalent by Constructions With Relations, **??** of **??**. We then claim that Items 1 and 2 are also equivalent:

- *Item 1* $\Longrightarrow$ *Item 2*: Let $U, V \in \mathcal{P}(A)$ and consider the diagram

$$\begin{tikzcd} A \arrow[r, "R", "\shortmid"{marking}] & B \arrow[r, shift left, "U"] \arrow[r, shift right, "V"'] & \mathrm{pt}. \end{tikzcd}$$

  By Constructions With Relations, **??**, we have

$$\begin{aligned} R^{-1}(U) &= U \diamond R, \\ R^{-1}(V) &= V \diamond R. \end{aligned}$$

  Now, if $U \diamond R = V \diamond R$, i.e. $R^{-1}(U) = R^{-1}(V)$, then $U = V$ since R is assumed to be an epimorphism, showing R^{-1} to be injective.

- *Item 2* $\Longrightarrow$ *Item 1*: Conversely, suppose that R^{-1} is injective, consider the diagram

$$\begin{tikzcd} A \arrow[r, mapsto, "R"] & B \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] & X, \end{tikzcd}$$

and suppose that $S \diamond R = T \diamond R$. Note that, since R^{-1} is injective, given a diagram of the form

$$\begin{tikzcd} A \arrow[r, mapsto, "R"] & B \arrow[r, shift left, "U"] \arrow[r, shift right, "V"'] & \mathrm{pt}, \end{tikzcd}$$

if $R^{-1}(U) = U \diamond R = V \diamond R = R^{-1}(V)$, then $U = V$. In particular, for each $x \in X$, we may consider the diagram

$$\begin{tikzcd} A \arrow[r, mapsto, "R"] & B \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] & X \arrow[r, mapsto, "{[x]}"] & \mathrm{pt}, \end{tikzcd}$$

for which we have $[x] \diamond S \diamond R = [x] \diamond T \diamond R$, implying that we have

$$S^{-1}(x) = [x] \diamond S = [x] \diamond T = T^{-1}(x)$$

for each $x \in X$, implying $S = T$, and thus R is an epimorphism.

We can also prove this in a more abstract way, following [MSE 350788]:

- *Item 1* $\Longrightarrow$ *Item 2*: Assume that R is an epimorphism.
  - We first notice that the functor $\mathrm{Rel}(-, \mathrm{pt}) \colon \mathrm{Rel}^{\mathrm{op}} \longrightarrow \mathrm{Sets}$ maps R to R^{-1} by Constructions With Relations, **??**.
  - Since $\mathrm{Rel}(-, \mathrm{pt})$ preserves limits by Limits and Colimits, **??** of **??**, it follows by **??**, **??** of **??** that $\mathrm{Rel}(-, \mathrm{pt})$ also preserves monomorphisms.
  - That is: $\mathrm{Rel}(-, \mathrm{pt})$ sends monomorphisms in $\mathrm{Rel}^{\mathrm{op}}$ to monomorphisms in Sets.

- The monomorphisms $\mathrm{Rel}^{\mathrm{op}}$ are precisely the epimorphisms in Rel by **??**, **??** of **??**.
- Since R is an epimorphism and $\mathrm{Rel}(-, \mathrm{pt})$ maps R to R^{-1}, it follows that R^{-1} is a monomorphism.
- Since the monomorphisms in Sets are precisely the injections (**??**, **??** of **??**), it follows that R^{-1} is injective.

- *Item 2* $\implies$ *Item 1:* Assume that R^{-1} is injective.
  - We first notice that the functor $\mathrm{Rel}(-, \mathrm{pt}) \colon \mathrm{Rel}^{\mathrm{op}} \to \mathrm{Sets}$ maps R to R^{-1} by Constructions With Relations, **??**.
  - Since the monomorphisms in Sets are precisely the injections (**??**, **??** of **??**), it follows that R^{-1} is a monomorphism.
  - Since $\mathrm{Rel}(-, \mathrm{pt})$ is faithful, it follows by **??**, **??** of **??** that $\mathrm{Rel}(, \mathrm{pt})$ reflects monomorphisms.
  - That is: $\mathrm{Rel}(-, \mathrm{pt})$ reflects monomorphisms in Sets to monomorphisms in $\mathrm{Rel}^{\mathrm{op}}$.
  - The monomorphisms $\mathrm{Rel}^{\mathrm{op}}$ are precisely the epimorphisms in Rel by **??**, **??** of **??**.
  - Since R^{-1} is a monomorphism and $\mathrm{Rel}(-, \mathrm{pt})$ maps R to R^{-1}, it follows that R is an epimorphism.

We also claim that Items 2 and 4 are equivalent, following [MO 350788]:

- *Item 2* $\implies$ *Item 4:* Since $B \setminus \{b\} \subset B$ and R^{-1} is injective, we have $R^{-1}(B \setminus \{b\}) \subsetneq R^{-1}(B)$. So taking some $a \in R^{-1}(B) \setminus R^{-1}(B \setminus \{b\})$ we get an element of A such that $R(a) = \{b\}$.

- *Item 4* $\implies$ *Item 2:* Let $U, V \subset B$ with $U \neq V$. Without loss of generality, we can assume $U \setminus V \neq \emptyset$; otherwise just swap U and V. Let then $b \in U \setminus V$. By assumption, there exists an $a \in A$ with $R(a) = \{b\}$. Then $a \in R^{-1}(U)$ but $a \notin R^{-1}(V)$, and thus $R^{-1}(U) \neq R^{-1}(V)$, showing R^{-1} to be injective.

Finally, we prove the second part of the statement. So assume R is a total epimorphism in Rel and consider the diagram

$$A \overset{R}{\longrightarrow\!\!\!\!\!\!+\;\;} B \underset{T}{\overset{S}{\rightrightarrows}} \{0, 1\},$$

where $b \sim_S 0$ for each $b \in B$ and where we have

$$b \sim_T \begin{cases} 0 & \text{if } b \in \operatorname{Im}(R), \\ 1 & \text{otherwise} \end{cases}$$

for each $b \in B$. Since R is total, we have $a \sim_{S \diamond R} 0$ and $a \sim_{T \diamond R} 0$ for all $a \in A$, and no element of A is related to 1 by $S \diamond R$ or $T \diamond R$. Thus $S \diamond R = T \diamond R$, and since R is an epimorphism, we have $S = T$. But by the definition of T, this implies $\operatorname{Im}(R) = B$. ∎

8.4.10 2-Categorical Epimorphisms

In this section we characterise (for now, some of) the 2-categorical epimorphisms in **Rel**, following Types of Morphisms in Bicategories, Section 13.2.

Proposition 8.4.10.1.1 ▶ 2-Categorical Epimorphisms in Rel

Let $R\colon A \nrightarrow B$ be a relation.

1. *Corepresentably Faithful Morphisms in* **Rel**. Every morphism of **Rel** is a corepresentably faithful morphism.

2. *Corepresentably Full Morphisms in* **Rel**. The following conditions are equivalent:

   (a) The morphism $R\colon A \nrightarrow B$ is a corepresentably full morphism.

   (b) For each pair of relations $S, T\colon X \rightrightarrows A$, the following condition is satisfied:

   ($\star$) If $S \diamond R \subset T \diamond R$, then $S \subset T$.

(c) The functor

$$R^{-1}\colon (\mathcal{P}(B), \subset) \to (\mathcal{P}(A), \subset)$$

is full.

(d) For each $U, V \in \mathcal{P}(B)$, if $R^{-1}(U) \subset R^{-1}(V)$, then $U \subset V$.

(e) The functor

$$R_{-1}\colon (\mathcal{P}(B), \subset) \to (\mathcal{P}(A), \subset)$$

is full.

(f) For each $U, V \in \mathcal{P}(B)$, if $R_{-1}(U) \subset R_{-1}(V)$, then $U \subset V$.

3. *Corepresentably Fully Faithful Morphisms in* **Rel**. Every corepresentably full morphism of **Rel** is a corepresentably fully faithful morphism.

Proof 8.4.10.1.2 ► Proof of Proposition 8.4.10.1.1

Item 1: Corepresentably Faithful Morphisms in Rel

The relation R is a corepresentably faithful morphism in **Rel** iff, for each $X \in \mathrm{Obj}(\mathbf{Rel})$, the functor

$$R^*\colon \mathbf{Rel}(B, X) \to \mathbf{Rel}(A, X)$$

is faithful, i.e. iff the morphism

$$R^*_{S,T}\colon \mathrm{Hom}_{\mathbf{Rel}(B,X)}(S, T) \to \mathrm{Hom}_{\mathbf{Rel}(A,X)}(S \diamond R, T \diamond R)$$

is injective for each $S, T \in \mathrm{Obj}(\mathbf{Rel}(B, X))$. However, $\mathrm{Hom}_{\mathbf{Rel}(B,X)}(S, T)$ is either empty or a singleton, in either case of which the map $R^*_{S,T}$ is necessarily injective.

Item 2: Corepresentably Full Morphisms in Rel

We claim Items 2a to 2f are indeed equivalent:

- *Item 2a* $\Longleftrightarrow$ *Item 2b:* This is simply a matter of unwinding definitions: The relation R is a corepresentably full morphism in **Rel** iff, for each $X \in \mathrm{Obj}(\mathbf{Rel})$, the functor

$$R^* \colon \mathbf{Rel}(B, X) \to \mathbf{Rel}(A, X)$$

is full, i.e. iff the morphism

$$R^*_{S,T} \colon \mathrm{Hom}_{\mathbf{Rel}(B,X)}(S, T) \to \mathrm{Hom}_{\mathbf{Rel}(A,X)}(S \diamond R, T \diamond R)$$

is surjective for each $S, T \in \mathrm{Obj}(\mathbf{Rel}(B, X))$, i.e. iff, whenever $S \diamond R \subset T \diamond R$, we also have $S \subset T$.

- *Item 2c* $\Longleftrightarrow$ *Item 2d:* This is also simply a matter of unwinding definitions: The functor

$$R^{-1} \colon (\mathcal{P}(B), \subset) \to (\mathcal{P}(A), \subset)$$

is full iff, for each $U, V \in \mathcal{P}(A)$, the morphism

$$R^{-1}_{U,V} \colon \mathrm{Hom}_{\mathcal{P}(B)}(U, V) \to \mathrm{Hom}_{\mathcal{P}(A)}\big(R^{-1}(U), R^{-1}(V)\big)$$

is surjective, i.e. iff whenever $R^{-1}(U) \subset R^{-1}(V)$, we also necessarily have $U \subset V$.

- *Item 2e* $\Longleftrightarrow$ *Item 2f:* This is once again simply a matter of unwinding definitions, and proceeds exactly in the same way as in the proof of the equivalence between Items 2c and 2d given above.

- *Item 2d* $\Longrightarrow$ *Item 2f:* Suppose that the following condition is true:

  ($\star$) For each $U, V \in \mathcal{P}(B)$, if $R^{-1}(U) \subset R^{-1}(V)$, then $U \subset V$.

  We need to show that the condition

  ($\star$) For each $U, V \in \mathcal{P}(B)$, if $R_{-1}(U) \subset R_{-1}(V)$, then $U \subset V$.

  is also true. We proceed step by step:

- Suppose we have $U, V \in \mathcal{P}(B)$ with $R_{-1}(U) \subset R_{-1}(V)$.
- By Constructions With Relations, **??** of **??**, we have

$$\begin{aligned} R_{-1}(U) &= B \setminus R^{-1}(A \setminus U), \\ R_{-1}(V) &= B \setminus R^{-1}(A \setminus V). \end{aligned}$$

- By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 we have $R^{-1}(A \setminus V) \subset R^{-1}(A \setminus U)$.
- By assumption, we then have $A \setminus V \subset A \setminus U$.
- By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 again, we have $U \subset V$.

- *Item 2f* $\Longrightarrow$ *Item 2d:* Suppose that the following condition is true:

  ($\star$) For each $U, V \in \mathcal{P}(B)$, if $R_{-1}(U) \subset R_{-1}(V)$, then $U \subset V$.

  We need to show that the condition

  ($\star$) For each $U, V \in \mathcal{P}(B)$, if $R^{-1}(U) \subset R^{-1}(V)$, then $U \subset V$.

  is also true. We proceed step by step:

  - Suppose we have $U, V \in \mathcal{P}(B)$ with $R^{-1}(U) \subset R^{-1}(V)$.
  - By Constructions With Relations, **??** of **??**, we have

$$\begin{aligned} R^{-1}(U) &= B \setminus R_{-1}(A \setminus U), \\ R^{-1}(V) &= B \setminus R_{-1}(A \setminus V). \end{aligned}$$

  - By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 we have $R_{-1}(A \setminus V) \subset R_{-1}(A \setminus U)$.
  - By assumption, we then have $A \setminus V \subset A \setminus U$.
  - By Constructions With Sets, Item 1 of Proposition 4.3.10.1.2 again, we have $U \subset V$.

- *Item 2b* $\Longrightarrow$ *Item 2d:* Consider the diagram

$$\begin{tikzcd} A \arrow[r, "R", "\shortmid"{marking}] & B \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] \arrow[r, phantom, "\shortmid"] & X, \end{tikzcd}$$

and suppose that $S \diamond R \subset T \diamond R$. Note that, by assumption, given a diagram of the form

$$\begin{tikzcd} A \arrow[r, "R", "\shortmid"{marking}] & B \arrow[r, shift left, "U"] \arrow[r, shift right, "V"'] \arrow[r, phantom, "\shortmid"] & \mathrm{pt}, \end{tikzcd}$$

if $R^{-1}(U) = R \diamond U \subset R \diamond V = R^{-1}(V)$, then $U \subset V$. In particular, for each $x \in X$, we may consider the diagram

$$\begin{tikzcd} A \arrow[r, "R", "\shortmid"{marking}] & B \arrow[r, shift left, "S"] \arrow[r, shift right, "T"'] \arrow[r, phantom, "\shortmid"] & X \arrow[r, "{[x]}", "\shortmid"{marking}] & \mathrm{pt}, \end{tikzcd}$$

for which we have $[x] \diamond S \diamond R \subset [x] \diamond T \diamond R$, implying that we have

$$S^{-1}(x) = [x] \diamond S \subset [x] \diamond T = T^{-1}(x)$$

for each $x \in X$, implying $S \subset T$.

- *Item 2d* $\Longrightarrow$ *Item 2b:* Let $U, V \in \mathcal{P}(B)$ and consider the diagram

$$\begin{tikzcd} A \arrow[r, "R", "\shortmid"{marking}] & B \arrow[r, shift left, "U"] \arrow[r, shift right, "V"'] \arrow[r, phantom, "\shortmid"] & \mathrm{pt}. \end{tikzcd}$$

By Remark 8.5.1.1.2, we have

$$\begin{aligned} R^{-1}(U) &= U \diamond R, \\ R^{-1}(V) &= V \diamond R. \end{aligned}$$

Now, if $R^{-1}(U) \subset R^{-1}(V)$, i.e. $U \diamond R \subset V \diamond R$, then $U \subset V$ by assumption.

Item 3: Corepresentably Fully Faithful Morphisms in Rel

This follows from Items 1 and 2.

Question 8.4.10.1.3 ► Better Characterisations of Corepresentably Full Morphisms in Rel

Item 2 of Proposition 8.4.10.1.1 gives a characterisation of the corepresentably full morphisms in **Rel**.
Are there other nice characterisations of these?
This question also appears as [MO 467527].

8.4.11 Co/Limits

Proposition 8.4.11.1.1 ► Co/Limits in Rel

This will be properly written later on.

Proof 8.4.11.1.2 ► Proof of Proposition 8.4.11.1.1

Omitted. ∎

8.4.12 Internal Kan Extensions and Lifts

Remark 8.4.12.1.1 ► Kan Extensions and Kan Lifts in Rel

The 2-category **Rel** admits all right Kan extensions and right Kan lifts, though not all left Kan extensions and neither does it admit all left Kan lifts. See Constructions With Relations, Section 9.2 for a detailed discussion of this.

8.4.13 Closedness

Proposition 8.4.13.1.1 ▶ Closedness of Rel

The 2-category **Rel** is a closed bicategory, there being, for each $R\colon A \nrightarrow B$ and set X, a pair of adjunctions

$$(R^* \dashv \mathrm{Ran}_R)\colon \quad \begin{tikzcd} \mathrm{Rel}(B,X) \arrow[r, bend left, "R^*"] \arrow[r, phantom, "\perp"] & \mathrm{Rel}(A,X), \arrow[l, bend left, "\mathrm{Ran}_R"] \end{tikzcd}$$

$$(R_! \dashv \mathrm{Rift}_R)\colon \quad \begin{tikzcd} \mathrm{Rel}(X,A) \arrow[r, bend left, "R_!"] \arrow[r, phantom, "\perp"] & \mathrm{Rel}(X,B), \arrow[l, bend left, "\mathrm{Rift}_R"] \end{tikzcd}$$

witnessed by bijections

$$\begin{aligned} \mathbf{Rel}(S \diamond R, T) &\cong \mathbf{Rel}(S, \mathrm{Ran}_R(T)), \\ \mathbf{Rel}(R \diamond U, V) &\cong \mathbf{Rel}(U, \mathrm{Rift}_R(V)), \end{aligned}$$

natural in $S \in \mathrm{Rel}(B,X)$, $T \in \mathrm{Rel}(A,X)$, $U \in \mathrm{Rel}(X,A)$, and $V \in \mathrm{Rel}(X,B)$.

Proof 8.4.13.1.2 ▶ Proof of Proposition 8.4.13.1.1

This follows from Constructions With Relations, Propositions 9.2.3.1.1 and 9.2.4.1.1. ∎

8.4.14 Rel as a Category of Free Algebras

Proposition 8.4.14.1.1 ▶ Rel as a Category of Free Algebras

We have an isomorphism of categories

$$\mathsf{Rel} \cong \mathsf{FreeAlg}_{\mathcal{P}_!}(\mathsf{Sets}),$$

where $\mathcal{P}_!$ is the powerset monad of **??**, **??**.

Proof 8.4.14.1.2 ▶ Proof of Proposition 8.4.14.1.1

Omitted. ∎

8.5 The Adjoint Pairs $R_! \dashv R_{-1}$ and $R^{-1} \dashv R_*$

8.5.1 Direct Images

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation.

Definition 8.5.1.1.1 ▶ Direct Images

The **direct image function associated to** R is the function

$$R_!\colon \mathcal{P}(A) \to \mathcal{P}(B)$$

defined by[1,2,3]

$$\begin{aligned} R_!(U) &\overset{\text{def}}{=} \bigcup_{a \in U} R(a) \\ &= \left\{ b \in B \;\middle|\; \begin{array}{l} \text{there exists some } a \in U \\ \text{such that } b \in R(a) \end{array} \right\} \end{aligned}$$

for each $U \in \mathcal{P}(A)$.

[1] *Further Notation:* The set $R_!(U)$ is also denoted $R(U)$.
[2] *Further Terminology:* The set $R(U)$ is called the **direct image of** U **by** R.
[3] We also have

$$R_!(U) = B \setminus R_*(A \setminus U);$$

see Item 7 of Proposition 8.5.1.1.3.

Remark 8.5.1.1.2 ▶ Unwinding Definition 8.5.1.1.1

Identifying subsets of A with relations from pt to A via Constructions With Sets, **??** of **??**, we see that the direct image function associated to R

is equivalently the function

$$R_!\colon \underbrace{\mathcal{P}(A)}_{\cong \mathrm{Rel}(\mathrm{pt},A)} \to \underbrace{\mathcal{P}(B)}_{\cong \mathrm{Rel}(\mathrm{pt},B)}$$

defined by

$$R_!(U) \overset{\text{def}}{=} R \diamond U$$

for each $U \in \mathcal{P}(A)$, where $R \diamond U$ is the composition

$$\mathrm{pt} \overset{U}{\nrightarrow} A \overset{R}{\nrightarrow} B.$$

Proposition 8.5.1.1.3 ▶ Properties of Direct Image Functions

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functoriality.* The assignment $U \mapsto R_!(U)$ defines a functor

$$R_!\colon (\mathcal{P}(A), \subset) \to (\mathcal{P}(B), \subset)$$

where

- *Action on Objects.* For each $U \in \mathcal{P}(A)$, we have

$$[R_!](U) \overset{\text{def}}{=} R_!(U).$$

- *Action on Morphisms.* For each $U, V \in \mathcal{P}(A)$:
  - If $U \subset V$, then $R_!(U) \subset R_!(V)$.

2. *Adjointness.* We have an adjunction

$$(R_! \dashv R_{-1})\colon \quad \mathcal{P}(A) \underset{R_{-1}}{\overset{R_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathcal{P}(B),$$

witnessed by a bijections of sets

$$\mathrm{Hom}_{\mathcal{P}(A)}(R_!(U), V) \cong \mathrm{Hom}_{\mathcal{P}(A)}(U, R_{-1}(V)),$$

natural in $U \in \mathcal{P}(A)$ and $V \in \mathcal{P}(B)$, i.e. such that:

($\star$) The following conditions are equivalent:

- We have $R_!(U) \subset V$.
- We have $U \subset R_{-1}(V)$.

3. *Preservation of Colimits.* We have an equality of sets

$$R_!\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} R_!(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(A)^{\times I}$. In particular, we have equalities

$$\begin{aligned} R_!(U) \cup R_!(V) &= R_!(U \cup V), \\ R_!(\emptyset) &= \emptyset, \end{aligned}$$

natural in $U, V \in \mathcal{P}(A)$.

4. *Oplax Preservation of Limits.* We have an inclusion of sets

$$R_!\left(\bigcap_{i\in I} U_i\right) \subset \bigcap_{i\in I} R_!(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(A)^{\times I}$. In particular, we have inclusions

$$\begin{aligned} R_!(U \cap V) &\subset R_!(U) \cap R_!(V), \\ R_!(A) &\subset B, \end{aligned}$$

natural in $U, V \in \mathcal{P}(A)$.

5. *Symmetric Strict Monoidality With Respect to Unions.* The direct image function of Item 1 has a symmetric strict monoidal structure

$$\left(R_!, R_!^{\otimes}, R_{*|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(A), \cup, \emptyset) \to (\mathcal{P}(B), \cup, \emptyset),$$

being equipped with equalities

$$\begin{aligned} R_{*|U,V}^{\otimes} &\colon R_!(U) \cup R_!(V) \xrightarrow{=} R_!(U \cup V), \\ R_{*|\mathbb{1}}^{\otimes} &\colon \emptyset \xrightarrow{=} \emptyset, \end{aligned}$$

natural in $U, V \in \mathcal{P}(A)$.

6. *Symmetric Oplax Monoidality With Respect to Intersections.* The direct image function of Item 1 has a symmetric oplax monoidal structure

$$\left(R_!, R_!^{\otimes}, R_{*|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(A), \cap, A) \to (\mathcal{P}(B), \cap, B),$$

being equipped with inclusions

$$R_{*|U,V}^{\otimes} \colon R_!(U \cap V) \subset R_!(U) \cap R_!(V),$$
$$R_{*|\mathbb{1}}^{\otimes} \colon R_!(A) \subset B,$$

natural in $U, V \in \mathcal{P}(A)$.

7. *Relation to Codirect Images.* We have

$$R_!(U) = B \setminus R_*(A \setminus U)$$

for each $U \in \mathcal{P}(A)$.

Proof 8.5.1.1.4 ▶ Proof of Proposition 8.5.1.1.3

Item 1: Functoriality

Clear.

Item 2: Adjointness

This follows from Kan Extensions, **??** of **??**.

Item 3: Preservation of Colimits

This follows from Item 2 and **??**, **??** of **??**.

Item 4: Oplax Preservation of Limits

Omitted.

Item 5: Symmetric Strict Monoidality With Respect to Unions

This follows from Item 3.

Item 6: Symmetric Oplax Monoidality With Respect to Intersections

This follows from Item 4.

Item 7: Relation to Codirect Images

The proof proceeds in the same way as in the case of functions (Constructions With Sets, Item 17 of Proposition 4.6.1.1.5): applying Item 7 of Proposition 8.5.4.1.4 to $A \setminus U$, we have

$$\begin{aligned} R_*(A \setminus U) &= B \setminus R_!(A \setminus (A \setminus U)) \\ &= B \setminus R_!(U). \end{aligned}$$

Taking complements, we then obtain

$$\begin{aligned} R_!(U) &= B \setminus (B \setminus R_!(U)), \\ &= B \setminus R_*(A \setminus U), \end{aligned}$$

which finishes the proof.

Proposition 8.5.1.1.5 ▶ Properties of the Direct Image Function Operation

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functionality I.* The assignment $R \mapsto R_!$ defines a function
$$(-)_!\colon \mathrm{Rel}(A, B) \longrightarrow \mathsf{Sets}(\mathcal{P}(A), \mathcal{P}(B)).$$
2. *Functionality II.* The assignment $R \mapsto R_!$ defines a function
$$(-)_!\colon \mathrm{Rel}(A, B) \longrightarrow \mathsf{Pos}((\mathcal{P}(A), \subset), (\mathcal{P}(B), \subset)).$$
3. *Interaction With Identities.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have[1]
$$(\chi_A)_! = \mathrm{id}_{\mathcal{P}(A)}.$$
4. *Interaction With Composition.* For each pair of composable relations $R\colon A \nrightarrow B$ and $S\colon B \nrightarrow C$, we have[2]
$$(S \diamond R)_! = S_! \circ R_!, \qquad \begin{tikzcd} \mathcal{P}(A) \arrow[r, "R_!"] \arrow[dr, "(S\diamond R)_!"'] & \mathcal{P}(B) \arrow[d, "S_!"] \\ & \mathcal{P}(C). \end{tikzcd}$$

[1]That is, the postcomposition function

$$(\chi_A)_! \colon \mathrm{Rel}(\mathrm{pt}, A) \longrightarrow \mathrm{Rel}(\mathrm{pt}, A)$$

is equal to $\mathrm{id}_{\mathrm{Rel}(\mathrm{pt},A)}$.

[2]That is, we have

$$(S \diamond R)_! = S_! \circ R_!,$$

$$\begin{tikzcd} \mathrm{Rel}(\mathrm{pt}, A) \arrow[r, "R_!"] \arrow[dr, "(S\diamond R)_!"'] & \mathrm{Rel}(\mathrm{pt}, B) \arrow[d, "S_!"] \\ & \mathrm{Rel}(\mathrm{pt}, C). \end{tikzcd}$$

Proof 8.5.1.1.6 ▶ Proof of Proposition 8.5.1.1.5

Item 1: Functionality I

Clear.

Item 2: Functionality II

Clear.

Item 3: Interaction With Identities

Indeed, we have

$$\begin{aligned} (\chi_A)_!(U) &\overset{\text{def}}{=} \bigcup_{a \in U} \chi_A(a) \\ &\overset{\text{def}}{=} \bigcup_{a \in U} \{a\} \\ &= U \\ &\overset{\text{def}}{=} \mathrm{id}_{\mathcal{P}(A)}(U) \end{aligned}$$

for each $U \in \mathcal{P}(A)$. Thus $(\chi_A)_! = \mathrm{id}_{\mathcal{P}(A)}$.

Item 4: Interaction With Composition

Indeed, we have

$$(S \diamond R)_!(U) \overset{\text{def}}{=} \bigcup_{a \in U} [S \diamond R](a)$$

$$
\begin{aligned}
&\stackrel{\text{def}}{=} \bigcup_{a\in U} S(R(a)) \\
&\stackrel{\text{def}}{=} \bigcup_{a\in U} S_!(R(a)) \\
&= S_!\left(\bigcup_{a\in U} R(a)\right) \\
&\stackrel{\text{def}}{=} S_!(R_!(U)) \\
&\stackrel{\text{def}}{=} [S_! \circ R_!](U)
\end{aligned}
$$

for each $U \in \mathcal{P}(A)$, where we used Item 3 of Proposition 8.5.1.1.3. Thus $(S \diamond R)_! = S_! \circ R_!$.

8.5.2 Strong Inverse Images

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation.

Definition 8.5.2.1.1 ▶ Strong Inverse Images

The **strong inverse image function associated to** R is the function

$$R_{-1}\colon \mathcal{P}(B) \to \mathcal{P}(A)$$

defined by[1]

$$R_{-1}(V) \stackrel{\text{def}}{=} \{a \in A \mid R(a) \subset V\}$$

for each $V \in \mathcal{P}(B)$.

[1] *Further Terminology:* The set $R_{-1}(V)$ is called the **strong inverse image of** V **by** R.

Remark 8.5.2.1.2 ▶ Unwinding Definition 8.5.2.1.1

Identifying subsets of B with relations from pt to B via Constructions With Sets, **??** of **??**, we see that the inverse image function associated to R is equivalently the function

$$R_{-1}\colon \underbrace{\mathcal{P}(B)}_{\cong \mathrm{Rel}(\mathrm{pt},B)} \to \underbrace{\mathcal{P}(A)}_{\cong \mathrm{Rel}(\mathrm{pt},A)}$$

defined by

$$R_{-1}(V) \overset{\text{def}}{=} \mathrm{Rift}_R(V), \qquad \begin{tikzcd} & A \arrow[d, "R"] \arrow[dl, Rightarrow] \\ \mathrm{pt} \arrow[ur, dashed, "\mathrm{Rift}_R(V)"] \arrow[r, "V"'] & B, \end{tikzcd}$$

and being explicitly computed by

$$\begin{aligned} R_{-1}(V) &\overset{\text{def}}{=} \mathrm{Rift}_R(V) \\ &\cong \int_{b \in B} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-1}, V^b_{-2}\right), \end{aligned}$$

where we have used **??**.

Proof 8.5.2.1.3 ► Proof of Remark 8.5.2.1.2

We have

$$\begin{aligned} \mathrm{Rift}_R(V) &\cong \int_{b \in B} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-1}, V^b_{-2}\right) \\ &= \left\{ a \in A \,\middle|\, \int_{b \in B} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, V^b_\star\right) = \mathsf{true} \right\} \\ &= \left\{ a \in A \,\middle|\, \begin{array}{l} \text{for each } b \in B\text{, at least one of the} \\ \text{following conditions hold:} \\ \\ \quad 1.\ \text{We have } R^b_a = \mathsf{false} \\ \quad 2.\ \text{The following conditions hold:} \\ \\ \qquad (a)\ \text{We have } R^b_a = \mathsf{true} \\ \qquad (b)\ \text{We have } V^b_\star = \mathsf{true} \end{array} \right\} \end{aligned}$$

$$
\begin{aligned}
&= \left\{ a \in A \;\middle|\; \begin{array}{l} \text{for each } b \in B\text{, at least one of the} \\ \text{following conditions hold:} \\ \\ \quad 1.\ \text{We have } b \notin R(a) \\ \quad 2.\ \text{The following conditions hold:} \\ \\ \qquad (a)\ \text{We have } b \in R(a) \\ \qquad (b)\ \text{We have } b \in V \end{array} \right\} \\
&= \{a \in A \mid \text{for each } b \in R(a)\text{, we have } b \in V\} \\
&= \{a \in A \mid R(a) \subset V\} \\
&\overset{\text{def}}{=} R_{-1}(V).
\end{aligned}
$$

This finishes the proof. ∎

Proposition 8.5.2.1.4 ► Properties of Strong Inverse Images

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functoriality.* The assignment $V \mapsto R_{-1}(V)$ defines a functor

$$R_{-1}\colon (\mathcal{P}(B), \subset) \longrightarrow (\mathcal{P}(A), \subset)$$

where

- *Action on Objects.* For each $V \in \mathcal{P}(B)$, we have

$$[R_{-1}](V) \overset{\text{def}}{=} R_{-1}(V).$$

- *Action on Morphisms.* For each $U, V \in \mathcal{P}(B)$:
  - If $U \subset V$, then $R_{-1}(U) \subset R_{-1}(V)$.

2. *Adjointness.* We have an adjunction

$$(R_! \dashv R_{-1})\colon \quad \mathcal{P}(A) \underset{R_{-1}}{\overset{R_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathcal{P}(B),$$

witnessed by a bijections of sets

$$\mathrm{Hom}_{\mathcal{P}(A)}(R_!(U), V) \cong \mathrm{Hom}_{\mathcal{P}(A)}(U, R_{-1}(V)),$$

natural in $U \in \mathcal{P}(A)$ and $V \in \mathcal{P}(B)$, i.e. such that:

($\star$) The following conditions are equivalent:
- We have $R_!(U) \subset V$.
- We have $U \subset R_{-1}(V)$.

3. *Lax Preservation of Colimits.* We have an inclusion of sets

$$\bigcup_{i \in I} R_{-1}(U_i) \subset R_{-1}\left(\bigcup_{i \in I} U_i\right),$$

natural in $\{U_i\}_{i \in I} \in \mathcal{P}(B)^{\times I}$. In particular, we have inclusions

$$R_{-1}(U) \cup R_{-1}(V) \subset R_{-1}(U \cup V),$$
$$\emptyset \subset R_{-1}(\emptyset),$$

natural in $U, V \in \mathcal{P}(B)$.

4. *Preservation of Limits.* We have an equality of sets

$$R_{-1}\left(\bigcap_{i \in I} U_i\right) = \bigcap_{i \in I} R_{-1}(U_i),$$

natural in $\{U_i\}_{i \in I} \in \mathcal{P}(B)^{\times I}$. In particular, we have equalities

$$R_{-1}(U \cap V) = R_{-1}(U) \cap R_{-1}(V),$$
$$R_{-1}(B) = B,$$

natural in $U, V \in \mathcal{P}(B)$.

5. *Symmetric Lax Monoidality With Respect to Unions.* The codirect image function of Item 1 has a symmetric lax monoidal structure

$$\left(R_{-1}, R_{-1}^{\otimes}, R_{-1|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(A), \cup, \emptyset) \to (\mathcal{P}(B), \cup, \emptyset),$$

being equipped with inclusions

$$R^{\otimes}_{-1|U,V} \colon R_{-1}(U) \cup R_{-1}(V) \subset R_{-1}(U \cup V),$$
$$R^{\otimes}_{-1|\mathbb{1}} \colon \emptyset \subset R_{-1}(\emptyset),$$

natural in $U, V \in \mathcal{P}(B)$.

6. *Symmetric Strict Monoidality With Respect to Intersections.* The direct image function of Item 1 has a symmetric strict monoidal structure

$$\left(R_{-1}, R^{\otimes}_{-1}, R^{\otimes}_{-1|\mathbb{1}}\right) \colon (\mathcal{P}(A), \cap, A) \to (\mathcal{P}(B), \cap, B),$$

being equipped with equalities

$$R^{\otimes}_{-1|U,V} \colon R_{-1}(U \cap V) \xrightarrow{=} R_{-1}(U) \cap R_{-1}(V),$$
$$R^{\otimes}_{-1|\mathbb{1}} \colon R_{-1}(A) \xrightarrow{=} B,$$

natural in $U, V \in \mathcal{P}(B)$.

7. *Interaction With Weak Inverse Images I.* We have

$$R_{-1}(V) = A \setminus R^{-1}(B \setminus V)$$

for each $V \in \mathcal{P}(B)$.

8. *Interaction With Weak Inverse Images II.* Let $R \colon A \nrightarrow B$ be a relation from A to B.

   (a) If R is a total relation, then we have an inclusion of sets

   $$R_{-1}(V) \subset R^{-1}(V)$$

   natural in $V \in \mathcal{P}(B)$.

   (b) If R is total and functional, then the above inclusion is in fact an equality.

   (c) Conversely, if we have $R_{-1} = R^{-1}$, then R is total and functional.

Proof 8.5.2.1.5 ▶ Proof of Proposition 8.5.2.1.4

Item 1: Functoriality

Clear.

Item 2: Adjointness

This follows from Kan Extensions, **??** of **??**.

Item 3: Lax Preservation of Colimits

Omitted.

Item 4: Preservation of Limits

This follows from Item 2 and **??**, **??** of **??**.

Item 5: Symmetric Lax Monoidality With Respect to Unions

This follows from Item 3.

Item 6: Symmetric Strict Monoidality With Respect to Intersections

This follows from Item 4.

Item 7: Interaction With Weak Inverse Images I

We claim we have an equality

$$R_{-1}(B \setminus V) = A \setminus R^{-1}(V).$$

Indeed, we have

$$\begin{aligned} R_{-1}(B \setminus V) &= \{a \in A \mid R(a) \subset B \setminus V\}, \\ A \setminus R^{-1}(V) &= \{a \in A \mid R(a) \cap V = \emptyset\}. \end{aligned}$$

Taking $V = B \setminus V$ then implies the original statement.

Item 8: Interaction With Weak Inverse Images II

Item 8a is clear, while Items 8b and 8c follow from Item 6 of Proposition 8.2.2.1.2. ∎

Proposition 8.5.2.1.6 ► Properties of the Strong Inverse Image Function Operation

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functionality I.* The assignment $R \mapsto R_{-1}$ defines a function

$$(-)_{-1}\colon \mathsf{Sets}(A, B) \to \mathsf{Sets}(\mathcal{P}(A), \mathcal{P}(B)).$$

2. *Functionality II.* The assignment $R \mapsto R_{-1}$ defines a function

$$(-)_{-1}\colon \mathsf{Sets}(A, B) \to \mathsf{Pos}((\mathcal{P}(A), \subset), (\mathcal{P}(B), \subset)).$$

3. *Interaction With Identities.* For each $A \in \mathrm{Obj}(\mathsf{Sets})$, we have

$$(\mathrm{id}_A)_{-1} = \mathrm{id}_{\mathcal{P}(A)}\,.$$

4. *Interaction With Composition.* For each pair of composable relations $R\colon A \nrightarrow B$ and $S\colon B \nrightarrow C$, we have

$$(S \diamond R)_{-1} = R_{-1} \circ S_{-1}, \qquad \begin{tikzcd} \mathcal{P}(C) \arrow[r, "S_{-1}"] \arrow[dr, "(S\diamond R)_{-1}"'] & \mathcal{P}(B) \arrow[d, "R_{-1}"] \\ & \mathcal{P}(A). \end{tikzcd}$$

Proof 8.5.2.1.7 ► Proof of Proposition 8.5.2.1.6

Item 1: Functionality I

Clear.

Item 2: Functionality II

Clear.

Item 3: Interaction With Identities

Indeed, we have

$$\begin{aligned}(\chi_A)_{-1}(U) &\overset{\text{def}}{=} \{a \in A \mid \chi_A(a) \subset U\} \\ &\overset{\text{def}}{=} \{a \in A \mid \{a\} \subset U\} \\ &= U\end{aligned}$$

for each $U \in \mathcal{P}(A)$. Thus $(\chi_A)_{-1} = \mathrm{id}_{\mathcal{P}(A)}$.

Item 4: Interaction With Composition

Indeed, we have

$$\begin{aligned}(S \diamond R)_{-1}(U) &\overset{\text{def}}{=} \{a \in A \mid [S \diamond R](a) \subset U\} \\ &\overset{\text{def}}{=} \{a \in A \mid S(R(a)) \subset U\} \\ &\overset{\text{def}}{=} \{a \in A \mid S_!(R(a)) \subset U\} \\ &= \{a \in A \mid R(a) \subset S_{-1}(U)\} \\ &\overset{\text{def}}{=} R_{-1}(S_{-1}(U)) \\ &\overset{\text{def}}{=} [R_{-1} \circ S_{-1}](U)\end{aligned}$$

for each $U \in \mathcal{P}(C)$, where we used Item 2 of Proposition 8.5.2.1.4, which implies that the conditions

- We have $S_!(R(a)) \subset U$.
- We have $R(a) \subset S_{-1}(U)$.

are equivalent. Thus $(S \diamond R)_{-1} = R_{-1} \circ S_{-1}$.

8.5.3 Weak Inverse Images

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation.

Definition 8.5.3.1.1 ▶ Weak Inverse Images

The **weak inverse image function associated to** R[1] is the function

$$R^{-1}\colon \mathcal{P}(B) \to \mathcal{P}(A)$$

defined by[2]

$$R^{-1}(V) \overset{\text{def}}{=} \{a \in A \mid R(a) \cap V \neq \emptyset\}$$

for each $V \in \mathcal{P}(B)$.

[1] *Further Terminology:* Also called simply the **inverse image function associated to** R.

[2] *Further Terminology:* The set $R^{-1}(V)$ is called the **weak inverse image of** V **by** R or simply the **inverse image of** V **by** R.

Remark 8.5.3.1.2 ► Unwinding Definition 8.5.3.1.1

Identifying subsets of B with relations from B to pt via Constructions With Sets, **??** of **??**, we see that the weak inverse image function associated to R is equivalently the function

$$R^{-1}\colon \underbrace{\mathcal{P}(B)}_{\cong \mathrm{Rel}(B,\mathrm{pt})} \to \underbrace{\mathcal{P}(A)}_{\cong \mathrm{Rel}(A,\mathrm{pt})}$$

defined by

$$R^{-1}(V) \overset{\text{def}}{=} V \diamond R$$

for each $V \in \mathcal{P}(A)$, where $R \diamond V$ is the composition

$$A \overset{R}{\nrightarrow} B \overset{V}{\nrightarrow} \mathrm{pt}.$$

Explicitly, we have

$$\begin{aligned} R^{-1}(V) &\overset{\text{def}}{=} V \diamond R \\ &\overset{\text{def}}{=} \int^{b \in B} V_b^{-1} \times R_{-2}^b. \end{aligned}$$

Proof 8.5.3.1.3 ▶ Proof of Remark 8.5.3.1.2

We have

$$
\begin{aligned}
V \diamond R &\overset{\text{def}}{=} \int^{b \in B} V_b^{-1} \times R_{-_2}^b \\
&= \left\{ a \in A \;\middle|\; \int^{b \in B} V_b^\star \times R_a^b = \mathsf{true} \right\} \\
&= \left\{ a \in A \;\middle|\; \begin{array}{l} \text{there exists } b \in B \text{ such that the} \\ \text{following conditions hold:} \\ \quad 1.\ \text{We have } V_b^\star = \mathsf{true} \\ \quad 2.\ \text{We have } R_a^b = \mathsf{true} \end{array} \right\} \\
&= \left\{ a \in A \;\middle|\; \begin{array}{l} \text{there exists } b \in B \text{ such that the} \\ \text{following conditions hold:} \\ \quad 1.\ \text{We have } b \in V \\ \quad 2.\ \text{We have } b \in R(a) \end{array} \right\} \\
&= \{ a \in A \mid \text{there exists } b \in V \text{ such that } b \in R(a) \} \\
&= \{ a \in A \mid R(a) \cap V \neq \emptyset \} \\
&\overset{\text{def}}{=} R^{-1}(V)
\end{aligned}
$$

This finishes the proof. ∎

Proposition 8.5.3.1.4 ▶ Properties of Weak Inverse Image Functions

Let $R \colon A \nrightarrow B$ be a relation.

1. *Functoriality.* The assignment $V \mapsto R^{-1}(V)$ defines a functor

$$R^{-1} \colon (\mathcal{P}(B), \subset) \to (\mathcal{P}(A), \subset)$$

where

- *Action on Objects.* For each $V \in \mathcal{P}(B)$, we have

$$\left[R^{-1}\right](V) \overset{\text{def}}{=} R^{-1}(V).$$

- *Action on Morphisms.* For each $U, V \in \mathcal{P}(B)$:
  - If $U \subset V$, then $R^{-1}(U) \subset R^{-1}(V)$.

2. *Adjointness.* We have an adjunction

$$\left(R^{-1} \dashv R_*\right): \quad \mathcal{P}(B) \underset{R_*}{\overset{R^{-1}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathcal{P}(A),$$

witnessed by a bijections of sets

$$\mathrm{Hom}_{\mathcal{P}(A)}\left(R^{-1}(U), V\right) \cong \mathrm{Hom}_{\mathcal{P}(A)}(U, R_*(V)),$$

natural in $U \in \mathcal{P}(A)$ and $V \in \mathcal{P}(B)$, i.e. such that:

($\star$) The following conditions are equivalent:
  - We have $R^{-1}(U) \subset V$.
  - We have $U \subset R_*(V)$.

3. *Preservation of Colimits.* We have an equality of sets

$$R^{-1}\left(\bigcup_{i \in I} U_i\right) = \bigcup_{i \in I} R^{-1}(U_i),$$

natural in $\{U_i\}_{i \in I} \in \mathcal{P}(B)^{\times I}$. In particular, we have equalities

$$R^{-1}(U) \cup R^{-1}(V) = R^{-1}(U \cup V),$$
$$R^{-1}(\emptyset) = \emptyset,$$

natural in $U, V \in \mathcal{P}(B)$.

4. *Oplax Preservation of Limits.* We have an inclusion of sets

$$R^{-1}\left(\bigcap_{i \in I} U_i\right) \subset \bigcap_{i \in I} R^{-1}(U_i),$$

natural in $\{U_i\}_{i \in I} \in \mathcal{P}(B)^{\times I}$. In particular, we have inclusions

$$R^{-1}(U \cap V) \subset R^{-1}(U) \cap R^{-1}(V),$$
$$R^{-1}(A) \subset B,$$

natural in $U, V \in \mathcal{P}(B)$.

5. *Symmetric Strict Monoidality With Respect to Unions.* The direct image function of Item 1 has a symmetric strict monoidal structure

$$\left(R^{-1}, R^{-1,\otimes}, R_{\mathbb{1}}^{-1,\otimes}\right) \colon (\mathcal{P}(A), \cup, \emptyset) \to (\mathcal{P}(B), \cup, \emptyset),$$

being equipped with equalities

$$R_{U,V}^{-1,\otimes} \colon R^{-1}(U) \cup R^{-1}(V) \xrightarrow{=} R^{-1}(U \cup V),$$
$$R_{\mathbb{1}}^{-1,\otimes} \colon \emptyset \xrightarrow{=} \emptyset,$$

natural in $U, V \in \mathcal{P}(B)$.

6. *Symmetric Oplax Monoidality With Respect to Intersections.* The direct image function of Item 1 has a symmetric oplax monoidal structure

$$\left(R^{-1}, R^{-1,\otimes}, R_{\mathbb{1}}^{-1,\otimes}\right) \colon (\mathcal{P}(A), \cap, A) \to (\mathcal{P}(B), \cap, B),$$

being equipped with inclusions

$$R_{U,V}^{-1,\otimes} \colon R^{-1}(U \cap V) \subset R^{-1}(U) \cap R^{-1}(V),$$
$$R_{\mathbb{1}}^{-1,\otimes} \colon R^{-1}(A) \subset B,$$

natural in $U, V \in \mathcal{P}(B)$.

7. *Interaction With Strong Inverse Images I.* We have

$$R^{-1}(V) = A \setminus R_{-1}(B \setminus V)$$

for each $V \in \mathcal{P}(B)$.

8. *Interaction With Strong Inverse Images II.* Let $R \colon A \nrightarrow B$ be a relation from A to B.

   (a) If R is a total relation, then we have an inclusion of sets

   $$R_{-1}(V) \subset R^{-1}(V)$$

   natural in $V \in \mathcal{P}(B)$.

   (b) If R is total and functional, then the above inclusion is in fact an equality.

   (c) Conversely, if we have $R_{-1} = R^{-1}$, then R is total and functional.

Proof 8.5.3.1.5 ▶ Proof of Proposition 8.5.3.1.4

Item 1: Functoriality

Clear.

Item 2: Adjointness

This follows from Kan Extensions, **??** of **??**.

Item 3: Preservation of Colimits

This follows from Item 2 and **??**, **??** of **??**.

Item 4: Oplax Preservation of Limits

Omitted.

Item 5: Symmetric Strict Monoidality With Respect to Unions

This follows from Item 3.

Item 6: Symmetric Oplax Monoidality With Respect to Intersections

This follows from Item 4.

Item 7: Interaction With Strong Inverse Images I

This follows from Item 7 of Proposition 8.5.2.1.4.

Item 8: Interaction With Strong Inverse Images II

This was proved in Item 8 of Proposition 8.5.2.1.4.

Proposition 8.5.3.1.6 ▶ Properties of the Weak Inverse Image Function Operation

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functionality I.* The assignment $R \mapsto R^{-1}$ defines a function

$$(-)^{-1}\colon \mathsf{Rel}(A,B) \longrightarrow \mathsf{Sets}(\mathcal{P}(A),\mathcal{P}(B)).$$

2. *Functionality II.* The assignment $R \mapsto R^{-1}$ defines a function

$$(-)^{-1}\colon \mathsf{Rel}(A,B) \longrightarrow \mathsf{Pos}((\mathcal{P}(A),\subset),(\mathcal{P}(B),\subset)).$$

3. *Interaction With Identities.* For each $A \in \mathsf{Obj}(\mathsf{Sets})$, we have[1]

$$(\chi_A)^{-1} = \mathrm{id}_{\mathcal{P}(A)}.$$

4. *Interaction With Composition.* For each pair of composable relations $R\colon A \nrightarrow B$ and $S\colon B \nrightarrow C$, we have[2]

$$(S \diamond R)^{-1} = R^{-1} \circ S^{-1},$$

$$\begin{tikzcd} \mathcal{P}(C) \arrow[r, "S^{-1}"] \arrow[dr, "(S\diamond R)^{-1}"'] & \mathcal{P}(B) \arrow[d, "R^{-1}"] \\ & \mathcal{P}(A). \end{tikzcd}$$

[1]That is, the postcomposition

$$(\chi_A)^{-1}\colon \mathrm{Rel}(\mathrm{pt}, A) \longrightarrow \mathrm{Rel}(\mathrm{pt}, A)$$

is equal to $\mathrm{id}_{\mathrm{Rel}(\mathrm{pt},A)}$.

[2]That is, we have

$$(S \diamond R)^{-1} = R^{-1} \circ S^{-1},$$

$$\begin{tikzcd} \mathrm{Rel}(\mathrm{pt}, C) \arrow[r, "R^{-1}"] \arrow[dr, "(S\diamond R)^{-1}"'] & \mathrm{Rel}(\mathrm{pt}, B) \arrow[d, "S^{-1}"] \\ & \mathrm{Rel}(\mathrm{pt}, A). \end{tikzcd}$$

Proof 8.5.3.1.7 ▶ Proof of Proposition 8.5.3.1.6

Item 1: Functionality I

Clear.

Item 2: Functionality II

Clear.

Item 3: Interaction With Identities

This follows from Categories, Item 5 of Proposition 11.1.4.1.2.

Item 4: Interaction With Composition

This follows from Categories, Item 2 of Proposition 11.1.4.1.2. ∎

8.5.4 Codirect Images

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation.

Definition 8.5.4.1.1 ▶ Codirect Images

The **codirect image function associated to** R is the function

$$R_* \colon \mathcal{P}(A) \to \mathcal{P}(B)$$

defined by[1,2]

$$\begin{aligned} R_*(U) &\overset{\text{def}}{=} \left\{ b \in B \;\middle|\; \begin{array}{l} \text{for each } a \in A, \text{ if we have} \\ b \in R(a), \text{ then } a \in U \end{array} \right\} \\ &= \left\{ b \in B \;\middle|\; R^{-1}(b) \subset U \right\} \end{aligned}$$

for each $U \in \mathcal{P}(A)$.

[1] *Further Terminology:* The set $R_*(U)$ is called the **codirect image of** U **by** R.

[2] We also have

$$R_*(U) = B \setminus R_!(A \setminus U);$$

see Item 7 of Proposition 8.5.4.1.4.

Remark 8.5.4.1.2 ▶ Unwinding Definition 8.5.4.1.1

Identifying subsets of B with relations from pt to B via Constructions With Sets, **??** of **??**, we see that the codirect image function associated to R is equivalently the function

$$R_* \colon \underbrace{\mathcal{P}(A)}_{\cong \mathrm{Rel}(A,\mathrm{pt})} \to \underbrace{\mathcal{P}(B)}_{\cong \mathrm{Rel}(B,\mathrm{pt})}$$

defined by

$$R_*(U) \overset{\text{def}}{=} \mathrm{Ran}_R(U),$$

$$\begin{tikzcd} & B \arrow[d, dashed, "\mathrm{Ran}_R(U)"] \arrow[dl, phantom, "\Downarrow"] \\ A \arrow[ur, "R"] \arrow[r, mapsto, "U"'] & \mathrm{pt}, \end{tikzcd}$$

being explicitly computed by

$$R^*(U) \overset{\text{def}}{=} \mathrm{Ran}_R(U)$$

$$\cong \int_{a \in A} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, U_a^{-1}\right),$$

where we have used **??**.

Proof 8.5.4.1.3 ▶ Proof of Remark 8.5.4.1.2

We have

$$\begin{aligned}
\mathrm{Ran}_R(V) &\cong \int_{a \in A} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, U_a^{-1}\right) \\
&= \left\{ b \in B \,\middle|\, \int_{a \in A} \mathrm{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, U_a^\star\right) = \mathsf{true} \right\} \\
&= \left\{ b \in B \,\middle|\, \begin{array}{l} \text{for each } a \in A, \text{ at least one of the} \\ \text{following conditions hold:} \\ \\ \quad 1.\ \text{We have } R_a^b = \mathsf{false} \\ \quad 2.\ \text{The following conditions hold:} \\ \\ \qquad (a)\ \text{We have } R_a^b = \mathsf{true} \\ \qquad (b)\ \text{We have } U_a^\star = \mathsf{true} \end{array} \right\} \\
&= \left\{ b \in B \,\middle|\, \begin{array}{l} \text{for each } a \in A, \text{ at least one of the} \\ \text{following conditions hold:} \\ \\ \quad 1.\ \text{We have } b \notin R(A) \\ \quad 2.\ \text{The following conditions hold:} \\ \\ \qquad (a)\ \text{We have } b \in R(a) \\ \qquad (b)\ \text{We have } a \in U \end{array} \right\} \\
&= \left\{ b \in B \,\middle|\, \begin{array}{l} \text{for each } a \in A, \text{ if we have} \\ b \in R(a), \text{ then } a \in U \end{array} \right\} \\
&= \left\{ b \in B \,\middle|\, R^{-1}(b) \subset U \right\} \\
&\overset{\text{def}}{=} R^{-1}(U).
\end{aligned}$$

This finishes the proof. ∎

Proposition 8.5.4.1.4 ▶ Properties of Codirect Images

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functoriality.* The assignment $U \mapsto R_*(U)$ defines a functor

$$R_*\colon (\mathcal{P}(A), \subset) \to (\mathcal{P}(B), \subset)$$

where

- *Action on Objects.* For each $U \in \mathcal{P}(A)$, we have

$$[R_*](U) \overset{\text{def}}{=} R_*(U).$$

- *Action on Morphisms.* For each $U, V \in \mathcal{P}(A)$:
  - If $U \subset V$, then $R_*(U) \subset R_*(V)$.

2. *Adjointness.* We have an adjunction

$$\left(R^{-1} \dashv R_*\right)\colon \quad \begin{tikzcd} \mathcal{P}(B) \arrow[r, bend left, "R^{-1}"] \arrow[r, phantom, "\perp"] & \mathcal{P}(A), \arrow[l, bend left, "R_*"] \end{tikzcd}$$

witnessed by a bijections of sets

$$\mathrm{Hom}_{\mathcal{P}(A)}\left(R^{-1}(U), V\right) \cong \mathrm{Hom}_{\mathcal{P}(A)}(U, R_*(V)),$$

natural in $U \in \mathcal{P}(A)$ and $V \in \mathcal{P}(B)$, i.e. such that:

($\star$) The following conditions are equivalent:
  - We have $R^{-1}(U) \subset V$.
  - We have $U \subset R_*(V)$.

3. *Lax Preservation of Colimits.* We have an inclusion of sets

$$\bigcup_{i \in I} R_*(U_i) \subset R_*\left(\bigcup_{i \in I} U_i\right),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(A)^{\times I}$. In particular, we have inclusions

$$R_*(U) \cup R_*(V) \subset R_*(U \cup V),$$
$$\emptyset \subset R_*(\emptyset),$$

natural in $U, V \in \mathcal{P}(A)$.

4. *Preservation of Limits.* We have an equality of sets

$$R_*\left(\bigcap_{i\in I} U_i\right) = \bigcap_{i\in I} R_*(U_i),$$

natural in $\{U_i\}_{i\in I} \in \mathcal{P}(A)^{\times I}$. In particular, we have equalities

$$R_*(U \cap V) = R_*(U) \cap R_*(V),$$
$$R_*(A) = B,$$

natural in $U, V \in \mathcal{P}(A)$.

5. *Symmetric Lax Monoidality With Respect to Unions.* The codirect image function of Item 1 has a symmetric lax monoidal structure

$$\left(R_*, R_*^{\otimes}, R_{!|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(A), \cup, \emptyset) \to (\mathcal{P}(B), \cup, \emptyset),$$

being equipped with inclusions

$$R_{!|U,V}^{\otimes} \colon R_*(U) \cup R_*(V) \subset R_*(U \cup V),$$
$$R_{!|\mathbb{1}}^{\otimes} \colon \emptyset \subset R_*(\emptyset),$$

natural in $U, V \in \mathcal{P}(A)$.

6. *Symmetric Strict Monoidality With Respect to Intersections.* The direct image function of Item 1 has a symmetric strict monoidal structure

$$\left(R_*, R_*^{\otimes}, R_{!|\mathbb{1}}^{\otimes}\right) \colon (\mathcal{P}(A), \cap, A) \to (\mathcal{P}(B), \cap, B),$$

being equipped with equalities

$$R_{!|U,V}^{\otimes} \colon R_*(U \cap V) \xrightarrow{=} R_*(U) \cap R_*(V),$$
$$R_{!|\mathbb{1}}^{\otimes} \colon R_*(A) \xrightarrow{=} B,$$

natural in $U, V \in \mathcal{P}(A)$.

7. *Relation to Direct Images.* We have

$$R_*(U) = B \setminus R_!(A \setminus U)$$

for each $U \in \mathcal{P}(A)$.

Proof 8.5.4.1.5 ▶ Proof of Proposition 8.5.4.1.4

Item 1: Functoriality

Clear.

Item 2: Adjointness

This follows from Kan Extensions, **??** of **??**.

Item 3: Lax Preservation of Colimits

Omitted.

Item 4: Preservation of Limits

This follows from Item 2 and **??**, **??** of **??**.

Item 5: Symmetric Lax Monoidality With Respect to Unions

This follows from Item 3.

Item 6: Symmetric Strict Monoidality With Respect to Intersections

This follows from Item 4.

Item 7: Relation to Direct Images

This follows from Item 7 of Proposition 8.5.1.1.3. Alternatively, we may prove it directly as follows, with the proof proceeding in the same way as in the case of functions (Constructions With Sets, Item 16 of Proposition 4.6.3.1.7).

We claim that $R_*(U) = B \setminus R_!(A \setminus U)$:

- *The First Implication.* We claim that

$$R_*(U) \subset B \setminus R_!(A \setminus U).$$

Let $b \in R_*(U)$. We need to show that $b \notin R_!(A \setminus U)$, i.e. that there is no $a \in A \setminus U$ such that $b \in R(a)$.

This is indeed the case, as otherwise we would have $a \in R^{-1}(b)$ and $a \notin U$, contradicting $R^{-1}(b) \subset U$ (which holds since $b \in R_*(U)$).

Thus $b \in B \setminus R_!(A \setminus U)$.

- *The Second Implication.* We claim that

$$B \setminus R_!(A \setminus U) \subset R_*(U).$$

Let $b \in B \setminus R_!(A \setminus U)$. We need to show that $b \in R_*(U)$, i.e. that $R^{-1}(b) \subset U$.

Since $b \notin R_!(A \setminus U)$, there exists no $a \in A \setminus U$ such that $b \in R(a)$, and hence $R^{-1}(b) \subset U$.

Thus $b \in R_*(U)$.

This finishes the proof.

Proposition 8.5.4.1.6 ► Properties of the Codirect Image Function Operation

Let $R\colon A \nrightarrow B$ be a relation.

1. *Functionality I.* The assignment $R \mapsto R_*$ defines a function

$$(-)_*\colon \mathsf{Sets}(A, B) \longrightarrow \mathsf{Sets}(\mathcal{P}(A), \mathcal{P}(B)).$$

2. *Functionality II.* The assignment $R \mapsto R_*$ defines a function

$$(-)_*\colon \mathsf{Sets}(A, B) \longrightarrow \mathsf{Hom}_{\mathsf{Pos}}((\mathcal{P}(A), \subset), (\mathcal{P}(B), \subset)).$$

3. *Interaction With Identities.* For each $A \in \mathsf{Obj}(\mathsf{Sets})$, we have

$$(\mathrm{id}_A)_* = \mathrm{id}_{\mathcal{P}(A)}.$$

4. *Interaction With Composition.* For each pair of composable relations

$R\colon A \nrightarrow B$ and $S\colon B \nrightarrow C$, we have

$$(S \diamond R)_* = S_* \circ R_*, \qquad \begin{tikzcd} \mathcal{P}(A) \arrow[r, "R_*"] \arrow[dr, "(S\diamond R)_*"'] & \mathcal{P}(B) \arrow[d, "S_*"] \\ & \mathcal{P}(C). \end{tikzcd}$$

Proof 8.5.4.1.7 ▶ Proof of Proposition 8.5.4.1.6

Item 1: Functionality I

Clear.

Item 2: Functionality II

Clear.

Item 3: Interaction With Identities

Indeed, we have

$$\begin{aligned} (\chi_A)_*(U) &\overset{\text{def}}{=} \left\{a \in A \,\middle|\, \chi_A^{-1}(a) \subset U\right\} \\ &\overset{\text{def}}{=} \{a \in A \mid \{a\} \subset U\} \\ &= U \end{aligned}$$

for each $U \in \mathcal{P}(A)$. Thus $(\chi_A)_* = \mathrm{id}_{\mathcal{P}(A)}$.

Item 4: Interaction With Composition

Indeed, we have

$$\begin{aligned} (S \diamond R)_*(U) &\overset{\text{def}}{=} \left\{c \in C \,\middle|\, [S \diamond R]^{-1}(c) \subset U\right\} \\ &\overset{\text{def}}{=} \left\{c \in C \,\middle|\, S^{-1}\left(R^{-1}(c)\right) \subset U\right\} \\ &= \left\{c \in C \,\middle|\, R^{-1}(c) \subset S_*(U)\right\} \\ &\overset{\text{def}}{=} R_*(S_*(U)) \\ &\overset{\text{def}}{=} [R_* \circ S_*](U) \end{aligned}$$

for each $U \in \mathcal{P}(C)$, where we used Item 2 of Proposition 8.5.4.1.4, which implies that the conditions

- We have $S^{-1}\left(R^{-1}(c)\right) \subset U$.
- We have $R^{-1}(c) \subset S_*(U)$.

are equivalent. Thus $(S \diamond R)_* = S_* \circ R_*$.

8.5.5 Functoriality of Powersets

Proposition 8.5.5.1.1 ▶ Functoriality of Powersets I

The assignment $X \mapsto \mathcal{P}(X)$ defines functors[1]

$$\begin{aligned}
\mathcal{P}_! &\colon \mathrm{Rel} \longrightarrow \mathrm{Sets}, \\
\mathcal{P}_{-1} &\colon \mathrm{Rel}^{\mathrm{op}} \longrightarrow \mathrm{Sets}, \\
\mathcal{P}^{-1} &\colon \mathrm{Rel}^{\mathrm{op}} \longrightarrow \mathrm{Sets}, \\
\mathcal{P}_* &\colon \mathrm{Rel} \longrightarrow \mathrm{Sets}
\end{aligned}$$

where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathrm{Rel})$, we have

$$\begin{aligned}
\mathcal{P}_!(A) &\stackrel{\text{def}}{=} \mathcal{P}(A), \\
\mathcal{P}_{-1}(A) &\stackrel{\text{def}}{=} \mathcal{P}(A), \\
\mathcal{P}^{-1}(A) &\stackrel{\text{def}}{=} \mathcal{P}(A), \\
\mathcal{P}_*(A) &\stackrel{\text{def}}{=} \mathcal{P}(A).
\end{aligned}$$

- *Action on Morphisms.* For each morphism $R\colon A \nrightarrow B$ of Rel, the images

$$\begin{aligned}
\mathcal{P}_!(R) &\colon \mathcal{P}(A) \longrightarrow \mathcal{P}(B), \\
\mathcal{P}_{-1}(R) &\colon \mathcal{P}(B) \longrightarrow \mathcal{P}(A), \\
\mathcal{P}^{-1}(R) &\colon \mathcal{P}(B) \longrightarrow \mathcal{P}(A), \\
\mathcal{P}_*(R) &\colon \mathcal{P}(A) \longrightarrow \mathcal{P}(B)
\end{aligned}$$

  of R by $\mathcal{P}_!, \mathcal{P}_{-1}, \mathcal{P}^{-1}$, and $\mathcal{P}_*$ are defined by

$$\mathcal{P}_!(R) \stackrel{\text{def}}{=} R_!,$$

$$\mathcal{P}_{-1}(R) \stackrel{\text{def}}{=} R_{-1},$$
$$\mathcal{P}^{-1}(R) \stackrel{\text{def}}{=} R^{-1},$$
$$\mathcal{P}_{*}(R) \stackrel{\text{def}}{=} R_{*},$$

as in Definitions 8.5.1.1.1, 8.5.2.1.1, 8.5.3.1.1 and 8.5.4.1.1.

[1]The functor $\mathcal{P}_! \colon \mathsf{Rel} \to \mathsf{Sets}$ admits a left adjoint; see Item 3 of Proposition 8.2.2.1.2.

Proof 8.5.5.1.2 ▶ Proof of Proposition 8.5.5.1.1

This follows from Items 3 and 4 of Proposition 8.5.1.1.5, Items 3 and 4 of Proposition 8.5.2.1.6, Items 3 and 4 of Proposition 8.5.3.1.6, and Items 3 and 4 of Proposition 8.5.4.1.6. ∎

8.5.6 Functoriality of Powersets: Relations on Powersets

Let A and B be sets and let $R \colon A \nrightarrow B$ be a relation.

Definition 8.5.6.1.1 ▶ The Relation on Powersets Associated to a Relation

The **relation on powersets associated to** R is the relation

$$\mathcal{P}(R) \colon \mathcal{P}(A) \nrightarrow \mathcal{P}(B)$$

defined by[1]

$$\mathcal{P}(R)^V_U \stackrel{\text{def}}{=} \mathbf{Rel}\big(\chi_{\mathrm{pt}}, V \diamond R \diamond U\big)$$

for each $U \in \mathcal{P}(A)$ and each $V \in \mathcal{P}(B)$.

[1]Illustration:

$$\begin{tikzcd} \mathrm{pt} \arrow[r, "U"'] \arrow[rrr, bend left, "\chi_{\mathrm{pt}}"] & A \arrow[r, "R"'] & B \arrow[r, "V"'] & \mathrm{pt}. \end{tikzcd}$$

Remark 8.5.6.1.2 ▶ Unwinding Definition 8.5.6.1.1

In detail, we have $U \sim_{\mathcal{P}(R)} V$ iff the following equivalent conditions hold:

- We have $\chi_{\mathrm{pt}} \subset V \diamond R \diamond U$.

- We have $(V \diamond R \diamond U)^{\star}_{\star} = \mathsf{true}$, i.e. we have

$$\int^{a \in A} \int^{b \in B} V^{\star}_{b} \times R^{b}_{a} \times U^{a}_{\star} = \mathsf{true}.$$

- There exists some $a \in A$ and some $b \in B$ such that:
  - We have $U^{a}_{\star} = \mathsf{true}$.
  - We have $R^{b}_{a} = \mathsf{true}$.
  - We have $V^{\star}_{b} = \mathsf{true}$.
- There exists some $a \in A$ and some $b \in B$ such that:
  - We have $a \in U$.
  - We have $a \sim_R b$.
  - We have $b \in V$.

Proposition 8.5.6.1.3 ▶ Functoriality of Powersets II

The assignment $R \mapsto \mathcal{P}(R)$ defines a functor

$$\mathcal{P} \colon \mathrm{Rel} \longrightarrow \mathrm{Rel}.$$

Proof 8.5.6.1.4 ▶ Proof of Proposition 8.5.6.1.3

Omitted. ∎

8.6 The Left Skew Monoidal Structure on $\mathrm{Rel}(A, B)$

8.6.1 The Left Skew Monoidal Product

Let A and B be sets and let $J \colon A \nrightarrow B$ be a relation.

DEFINITION 8.6.1.1.1 ► THE LEFT J-SKEW MONOIDAL PRODUCT OF $\mathbf{Rel}(A, B)$

The **left J-skew monoidal product of $\mathbf{Rel}(A, B)$** is the functor

$$\triangleleft_J \colon \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \to \mathbf{Rel}(A, B)$$

where

- *Action on Objects.* For each $R, S \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, we have

$$S \triangleleft_J R \overset{\text{def}}{=} S \diamond \mathrm{Rift}_J(R),$$

$$\begin{tikzcd}
 & A \arrow[r, "S", "|" marking] \arrow[d, "J", "|" marking] \arrow[dl, dashed, "\mathrm{Rift}_J(R)"'] \arrow[dl, phantom, shift right=3ex, "\Downarrow"] & B. \\
A \arrow[r, "R", "|" marking] & B &
\end{tikzcd}$$

- *Action on Morphisms.* For each $R, S, R', S' \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, the action on Hom-sets

$$(\triangleleft_J)_{(G,F),(G',F')} \colon \mathrm{Hom}_{\mathbf{Rel}(A,B)}(S, S') \times \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, R') \to \mathrm{Hom}_{\mathbf{Rel}(A,B)}(S \triangleleft_J R, S' \triangleleft_J R')$$

of $\triangleleft_J$ at $((R, S), (R', S'))$ is defined by[1]

$$\beta \triangleleft_J \alpha \overset{\text{def}}{=} \beta \diamond \mathrm{Rift}_J(\alpha),$$

$$\begin{tikzcd}
 & & A \arrow[r, bend left, "S", "|" marking] \arrow[r, bend right, "S'"', "|" marking] \arrow[r, phantom, "\beta\Downarrow"] \arrow[dd, "J", "|" marking] \arrow[ddl, dashed, bend right, "\mathrm{Rift}_J(R)"', "|" marking] \arrow[ddl, dashed, "\mathrm{Rift}_J(R')", "|" marking] \arrow[ddl, phantom, shift right=3ex, "\Downarrow"] & B \\
 & & & \\
A \arrow[r, "R", "|" marking] \arrow[rr, bend right, "R'"', "|" marking] \arrow[rr, phantom, shift right=3ex, "\alpha\Downarrow"] & R \arrow[rr] & & B
\end{tikzcd}$$

for each $\beta \in \mathrm{Hom}_{\mathbf{Rel}(A,B)}(S, S')$ and each $\alpha \in \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, R')$.

[1] Since $\mathbf{Rel}(A, B)$ is posetal, this is to say that if $S \subset S'$ and $R \subset R'$, then $S \triangleleft_J R \subset S' \triangleleft_J R'$.

8.6.2 The Left Skew Monoidal Unit

Let A and B be sets and let $J\colon A \nrightarrow B$ be a relation.

Definition 8.6.2.1.1 ▶ The Left J-Skew Monoidal Unit of $\mathbf{Rel}(A, B)$

The **left J-skew monoidal unit of $\mathbf{Rel}(A, B)$** is the functor

$$\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleleft_J}\colon \mathsf{pt} \to \mathbf{Rel}(A, B)$$

picking the object

$$\mathbb{1}^{\triangleleft_J}_{\mathbf{Rel}(A,B)} \overset{\text{def}}{=} J$$

of $\mathbf{Rel}(A, B)$.

8.6.3 The Left Skew Associators

Let A and B be sets and let $J\colon A \nrightarrow B$ be a relation.

Definition 8.6.3.1.1 ▶ The Left J-Skew Associator of $\mathbf{Rel}(A, B)$

The **left J-skew associator of $\mathbf{Rel}(A, B)$** is the natural transformation

$$\alpha^{\mathbf{Rel}(A,B),\triangleleft_J}\colon \triangleleft_J \circ (\triangleleft_J \times \mathsf{id}) \Longrightarrow \triangleleft_J \circ (\mathsf{id} \times \triangleleft_J) \circ \boldsymbol{\alpha}^{\mathsf{Cats}}_{\mathbf{Rel}(A,B),\mathbf{Rel}(A,B),\mathbf{Rel}(A,B)},$$

as in the diagram

$$\begin{tikzcd}
 & \mathbf{Rel}(A, B) \times (\mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B)) \arrow[dr, "\mathsf{id}\times\triangleleft_J"] \arrow[dl, dashed, "\boldsymbol{\alpha}^{\mathsf{Cats}}_{\mathbf{Rel}(A,B),\mathbf{Rel}(A,B),\mathbf{Rel}(A,B)}"'] & \\
(\mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B)) \times \mathbf{Rel}(A, B) \arrow[dr, "\triangleleft_J\times\mathsf{id}"'] \arrow[rr, Rightarrow, "\alpha^{\mathbf{Rel}(A,B),\triangleleft_J}"] & & \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \arrow[dl, "\triangleleft_J"] \\
 & \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \underset{\triangleleft_J}{\times} \mathbf{Rel}(A, B), &
\end{tikzcd}$$

whose component

$$\alpha^{\mathbf{Rel}(A,B),\triangleleft_J}_{T,S,R}\colon \underbrace{(T \triangleleft_J S) \triangleleft_J R}_{\overset{\text{def}}{=} T \diamond \mathrm{Rift}_J(S) \diamond \mathrm{Rift}_J(R)} \hookrightarrow \underbrace{T \triangleleft_J (S \triangleleft_J R)}_{\overset{\text{def}}{=} T \diamond \mathrm{Rift}_J(S \diamond \mathrm{Rift}_J(R))}$$

at (T, S, R) is given by

$$\alpha^{\mathbf{Rel}(A,B),\triangleleft_J}_{T,S,R} \overset{\text{def}}{=} \mathrm{id}_T \diamond \gamma,$$

where

$$\gamma\colon \mathrm{Rift}_J(S) \diamond \mathrm{Rift}_J(R) \hookrightarrow \mathrm{Rift}_J\big(S \diamond \mathrm{Rift}_J(R)\big)$$

is the inclusion adjunct to the inclusion

$$\epsilon_S \star \mathrm{id}_{\mathrm{Rift}_J(R)}\colon \underbrace{J \diamond \mathrm{Rift}_J(S) \diamond \mathrm{Rift}_J(R)}_{\overset{\text{def}}{=} J_!\left(\mathrm{Rift}_J(S) \diamond \mathrm{Rift}_J(R)\right)} \hookrightarrow S \diamond \mathrm{Rift}_J(R)$$

under the adjunction $J_! \dashv \mathrm{Rift}_J$, where $\epsilon\colon J \diamond \mathrm{Rift}_J \Longrightarrow \mathrm{id}_{\mathbf{Rel}(A,B)}$ is the counit of the adjunction $J_! \dashv \mathrm{Rift}_J$.

8.6.4 The Left Skew Left Unitors

Let A and B be sets and let $J\colon A \nrightarrow B$ be a relation.

Definition 8.6.4.1.1 ▶ The Left J-Skew Left Unitor of $\mathbf{Rel}(A, B)$

The **left J-skew left unitor of $\mathbf{Rel}(A, B)$** is the natural transformation

$$\lambda^{\mathbf{Rel}(A,B),\triangleleft_J}\colon \triangleleft_J \circ \left(\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleleft_J} \times \mathsf{id}\right) \Longrightarrow \boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)}$$

as in the diagram

$$\begin{tikzcd}
\mathsf{pt} \times \mathbf{Rel}(A,B) \arrow[r, "\mathbb{1}_{\triangleleft_J}^{\mathbf{Rel}(A,B)} \times \mathsf{id}"] \arrow[dr, dashed, bend right, "\lambda_{\mathbf{Rel}(A,B)}^{\mathsf{Cats}_2}"'] & \mathbf{Rel}(A,B) \times \mathbf{Rel}(A,B) \arrow[d, "\triangleleft_J"] \arrow[dl, Rightarrow, "\lambda^{\mathbf{Rel}(A,B),\triangleleft_J}"'] \\
& \mathbf{Rel}(A,B),
\end{tikzcd}$$

whose component

$$\lambda_R^{\mathbf{Rel}(A,B),\triangleleft_J} : \underbrace{J \triangleleft_J R}_{\overset{\text{def}}{=} J \diamond \mathrm{Rift}_J(R)} \hookrightarrow R$$

at R is given by

$$\lambda_R^{\mathbf{Rel}(A,B),\triangleleft_J} \overset{\text{def}}{=} \epsilon_R,$$

where $\epsilon : J_! \diamond \mathrm{Rift}_J \Longrightarrow \mathrm{id}_{\mathbf{Rel}(A,B)}$ is the counit of the adjunction $J_! \dashv \mathrm{Rift}_J$.

8.6.5 The Left Skew Right Unitors

Let A and B be sets and let $J : A \nrightarrow B$ be a relation.

Definition 8.6.5.1.1 ► The Left J-Skew Right Unitor of $\mathbf{Rel}(A,B)$

The **left J-skew right unitor of $\mathbf{Rel}(A,B)$** is the natural transformation

$$\rho^{\mathbf{Rel}(A,B),\triangleleft_J} : \boldsymbol{\rho}_{\mathbf{Rel}(A,B)}^{\mathsf{Cats}_2} \Longrightarrow \triangleleft_J \circ \left(\mathsf{id} \times \mathbb{1}_{\triangleleft_J}^{\mathbf{Rel}(A,B)}\right)$$

as in the diagram

$$\begin{tikzcd}
\mathbf{Rel}(A,B) \times \mathsf{pt} \arrow[r, "\mathsf{id} \times \mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleleft_J}"] \arrow[dr, dashed, bend right, "\rho^{\mathrm{Cats}_2}_{\mathbf{Rel}(A,B)}"'] & \mathbf{Rel}(A,B) \times \mathbf{Rel}(A,B), \arrow[d, "\triangleleft_J"] \arrow[dl, phantom, "\Nearrow \rho^{\mathbf{Rel}(A,B),\triangleleft_J}"] \\
& \mathbf{Rel}(A,B)
\end{tikzcd}$$

whose component

$$\rho^{\mathbf{Rel}(A,B),\triangleleft_J}_R : R \hookrightarrow \underbrace{R \triangleleft_J J}_{\overset{\text{def}}{=} R \diamond \mathrm{Rift}_J(J)}$$

at R is given by the composition

$$\begin{aligned}
R &\overset{\sim}{\Longrightarrow} R \diamond \chi_A \\
&\overset{\mathrm{id}_R \diamond \eta_{\chi_A}}{\Longrightarrow} R \diamond \mathrm{Rift}_J(J_!(\chi_A)) \\
&\overset{\text{def}}{=} R \diamond \mathrm{Rift}_J(J \diamond \chi_A) \\
&\overset{\sim}{\Longrightarrow} R \diamond \mathrm{Rift}_J(J) \\
&\overset{\text{def}}{=} R \triangleleft_J J,
\end{aligned}$$

where $\eta : \mathrm{id}_{\mathbf{Rel}(A,A)} \Longrightarrow \mathrm{Rift}_J \circ J_!$ is the unit of the adjunction $J_! \dashv \mathrm{Rift}_J$.

8.6.6 The Left Skew Monoidal Structure on $\mathbf{Rel}(A, B)$

Proposition 8.6.6.1.1 ▶ The Left J-Skew Monoidal Structure on $\mathbf{Rel}(A, B)$

The category $\mathbf{Rel}(A, B)$ admits a left skew monoidal category structure consisting of

- *The Underlying Category.* The posetal category associated to the

poset $\mathbf{Rel}(A, B)$ of relations from A to B of **??** of **??**.

- *The Left Skew Monoidal Product.* The left J-skew monoidal product

$$\triangleleft_J \colon \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \to \mathbf{Rel}(A, B)$$

of Definition 8.6.1.1.1.

- *The Left Skew Monoidal Unit.* The functor

$$\mathbb{1}^{\mathbf{Rel}(A,B),\triangleleft_J} \colon \mathsf{pt} \to \mathbf{Rel}(A, B)$$

of Definition 8.6.2.1.1.

- *The Left Skew Associators.* The natural transformation

$$\alpha^{\mathbf{Rel}(A,B),\triangleleft_J} \colon \triangleleft_J \circ (\triangleleft_J \times \mathsf{id}) \Longrightarrow \triangleleft_J \circ (\mathsf{id} \times \triangleleft_J) \circ \boldsymbol{\alpha}^{\mathsf{Cats}}_{\mathbf{Rel}(A,B),\mathbf{Rel}(A,B),\mathbf{Rel}(A,B)}$$

of Definition 8.6.3.1.1.

- *The Left Skew Left Unitors.* The natural transformation

$$\lambda^{\mathbf{Rel}(A,B),\triangleleft_J} \colon \triangleleft_J \circ \left(\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleleft_J} \times \mathsf{id}\right) \Longrightarrow \boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)}$$

of Definition 8.6.4.1.1.

- *The Left Skew Right Unitors.* The natural transformation

$$\rho^{\mathbf{Rel}(A,B),\triangleleft_J} \colon \boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)} \Longrightarrow \triangleleft_J \circ \left(\mathsf{id} \times \mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleleft_J}\right)$$

of Definition 8.6.5.1.1.

Proof 8.6.6.1.2 ▶ Proof of Proposition 8.6.6.1.1

Since $\mathbf{Rel}(A, B)$ is posetal, the commutativity of the pentagon identity, the left skew left triangle identity, the left skew right triangle identity, the left skew middle triangle identity, and the zigzag identity is automatic

(Categories, Item 4 of Proposition 11.2.7.1.2), and thus $\mathbf{Rel}(A, B)$ together with the data in the statement forms a left skew monoidal category. ∎

8.7 The Right Skew Monoidal Structure on $\mathbf{Rel}(A, B)$

Let A and B be sets and let $J\colon A \nrightarrow B$ be a relation.

8.7.1 The Right Skew Monoidal Product

DEFINITION 8.7.1.1.1 ▶ THE RIGHT J-SKEW MONOIDAL PRODUCT OF $\mathbf{Rel}(A, B)$

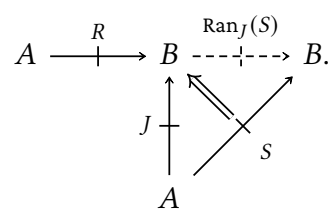
The **right J-skew monoidal product of $\mathbf{Rel}(A, B)$** is the functor

$$\triangleright_J\colon \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \longrightarrow \mathbf{Rel}(A, B)$$

where

- *Action on Objects.* For each $R, S \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, we have

$$S \triangleright_J R \overset{\text{def}}{=} \mathrm{Ran}_J(S) \diamond R,$$

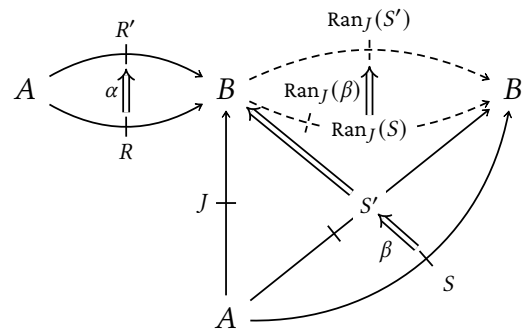


- *Action on Morphisms.* For each $R, S, R', S' \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, the action on Hom-sets

$$(\triangleright_J)_{(S,R),(S',R')}\colon \mathrm{Hom}_{\mathbf{Rel}(A,B)}(S, S') \times \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, R') \longrightarrow \mathrm{Hom}_{\mathbf{Rel}(A,B)}(S \triangleright_J R, S' \triangleright_J R')$$

of $\triangleright_J$ at $((S,R),(S',R'))$ is defined by[1]

$$\beta \triangleright_J \alpha \overset{\text{def}}{=} \mathrm{Ran}_J(\beta) \diamond \alpha,$$



for each $\beta \in \mathrm{Hom}_{\mathbf{Rel}(A,B)}(S,S')$ and each $\alpha \in \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R,R')$.

[1]Since $\mathbf{Rel}(A,B)$ is posetal, this is to say that if $S \subset S'$ and $R \subset R'$, then $S \triangleright_J R \subset S' \triangleright_J R'$.

8.7.2 The Right Skew Monoidal Unit

Definition 8.7.2.1.1 ▶ The Right J-Skew Monoidal Unit of $\mathbf{Rel}(A,B)$

The **right J-skew monoidal unit of $\mathbf{Rel}(A,B)$** is the functor

$$\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleright_J} : \mathsf{pt} \longrightarrow \mathbf{Rel}(A,B)$$

picking the object

$$\mathbb{1}^{\triangleright_J}_{\mathbf{Rel}(A,B)} \overset{\text{def}}{=} J$$

of $\mathbf{Rel}(A,B)$.

8.7.3 The Right Skew Associators

Definition 8.7.3.1.1 ▶ The Right J-Skew Associator of $\mathbf{Rel}(A,B)$

The **right J-skew associator of $\mathbf{Rel}(A,B)$** is the natural transformation

$$\alpha^{\mathbf{Rel}(A,B),\triangleright_J} : \triangleright_J \circ (\mathsf{id} \times \triangleright_J) \Longrightarrow \triangleright_J \circ (\triangleright_J \times \mathsf{id}) \circ \boldsymbol{\alpha}^{\mathsf{Cats},-1}_{\mathbf{Rel}(A,B),\mathbf{Rel}(A,B),\mathbf{Rel}(A,B)},$$

as in the diagram

$$\begin{tikzcd}
 & (\mathbf{Rel}(A,B)\times\mathbf{Rel}(A,B))\times\mathbf{Rel}(A,B) \arrow[dr, "\rhd_J\times\mathrm{id}"] \arrow[dl, dashed, "\boldsymbol{\alpha}^{\mathrm{Cats},-1}_{\mathbf{Rel}(A,B),\mathbf{Rel}(A,B),\mathbf{Rel}(A,B)}"'] \arrow[d, phantom, "\Uparrow\alpha^{\mathbf{Rel}(A,B),\rhd_J}"] \\
\mathbf{Rel}(A,B)\times(\mathbf{Rel}(A,B)\times\mathbf{Rel}(A,B)) \arrow[dr, "\mathrm{id}\times\rhd_J"'] \arrow[rr, phantom, "\Downarrow"] & & \mathbf{Rel}(A,B)\times\mathbf{Rel}(A,B) \arrow[dl, "\rhd_J"] \\
 & \mathbf{Rel}(A,B)\times\mathbf{Rel}(A,B) \underset{\rhd_J}{\times} \mathbf{Rel}(A,B), &
\end{tikzcd}$$

whose component

$$\alpha^{\mathbf{Rel}(A,B),\rhd_J}_{T,S,R}\colon \underbrace{T\rhd_J\big(S\rhd_J R\big)}_{\overset{\mathrm{def}}{=}\mathrm{Ran}_J(T)\diamond\mathrm{Ran}_J(S)\diamond R} \hookrightarrow \underbrace{\big(T\rhd_J S\big)\rhd_J R}_{\overset{\mathrm{def}}{=}\mathrm{Ran}_J\big(\mathrm{Ran}_J(T)\diamond S\big)\diamond R}$$

at (T,S,R) is given by

$$\alpha^{\mathbf{Rel}(A,B),\rhd}_{T,S,R} \overset{\mathrm{def}}{=} \gamma\diamond\mathrm{id}_R,$$

where

$$\gamma\colon\ \mathrm{Ran}_J(T)\diamond\mathrm{Ran}_J(S)\hookrightarrow\mathrm{Ran}_J\big(\mathrm{Ran}_J(T)\diamond S\big)$$

is the inclusion adjunct to the inclusion

$$\mathrm{id}_{\mathrm{Ran}_J(T)}\diamond\epsilon_S\colon\ \underbrace{\mathrm{Ran}_J(T)\diamond\mathrm{Ran}_J(S)\diamond J}_{\overset{\mathrm{def}}{=}J^*\left(\mathrm{Ran}_J(T)\diamond\mathrm{Ran}_J(S)\right)}\hookrightarrow\mathrm{Ran}_J(T)\diamond S$$

under the adjunction $J^*\dashv\mathrm{Ran}_J$, where $\epsilon\colon\ \mathrm{Ran}_J\diamond J\Longrightarrow\mathrm{id}_{\mathbf{Rel}(A,B)}$ is the counit of the adjunction $J^*\dashv\mathrm{Ran}_J$.

8.7.4 The Right Skew Left Unitors

Definition 8.7.4.1.1 ▶ The Right J-Skew Left Unitor of $\mathbf{Rel}(A, B)$

The **right J-skew left unitor of $\mathbf{Rel}(A, B)$** is the natural transformation

$$\lambda^{\mathbf{Rel}(A,B),\triangleright_J} \colon \boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)} \Longrightarrow \triangleright_J \circ \left(\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleright} \times \mathsf{id}\right),$$

as in the diagram

$$\begin{tikzcd}
\mathsf{pt} \times \mathbf{Rel}(A, B) \arrow[r, "\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleright_J} \times \mathsf{id}"] \arrow[dr, dashed, bend right, "\boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)}"'] & \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \arrow[d, "\triangleright_J"] \arrow[dl, Rightarrow, "\lambda^{\mathbf{Rel}(A,B),\triangleright_J}"] \\
& \mathbf{Rel}(A, B),
\end{tikzcd}$$

whose component

$$\lambda^{\mathbf{Rel}(A,B),\triangleright_J}_R \colon R \hookrightarrow \underbrace{J \triangleright_J R}_{\overset{\text{def}}{=} \mathrm{Ran}_J(J) \diamond R}$$

at R is given by the composition

$$\begin{aligned}
R &\overset{\sim}{\Longrightarrow} \chi_B \diamond R \\
&\overset{\eta_{\chi_B}}{\Longrightarrow} \diamond \mathrm{id}_R \mathrm{Ran}_J(J^*(\chi_A)) \diamond R \\
&\overset{\text{def}}{=} \mathrm{Ran}_J(J^* \diamond \chi_A) \diamond R \\
&\overset{\sim}{\Longrightarrow} \mathrm{Ran}_J(J) \diamond R \\
&\overset{\text{def}}{=} R \triangleright_J J,
\end{aligned}$$

where $\eta \colon \mathrm{id}_{\mathbf{Rel}(B,B)} \Longrightarrow \mathrm{Ran}_J \circ J^*$ is the unit of the adjunction $J^* \dashv \mathrm{Ran}_J$.

8.7.5 The Right Skew Right Unitors

Definition 8.7.5.1.1 ▶ The Right J-Skew Right Unitor of $\mathbf{Rel}(A, B)$

The **right J-skew right unitor of $\mathbf{Rel}(A, B)$** is the natural transformation

$$\rho^{\mathbf{Rel}(A,B),\triangleright_J} \colon \triangleright_J \circ \left(\mathsf{id} \times \mathbb{1}_{\triangleright}^{\mathbf{Rel}(A,B)}\right) \Longrightarrow \boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)},$$

as in the diagram

$$\begin{tikzcd}
\mathbf{Rel}(A,B) \times \mathsf{pt} \arrow[r, "\mathsf{id}\times\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleright_J}"] \arrow[dr, dashed, bend right, "\boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)}"'] & \mathbf{Rel}(A,B) \times \mathbf{Rel}(A,B), \arrow[d, "\triangleright_J"] \arrow[dl, Rightarrow, "\rho^{\mathbf{Rel}(A,B),\triangleright_J}"] \\
& \mathbf{Rel}(A,B)
\end{tikzcd}$$

whose component

$$\rho_S^{\mathbf{Rel}(A,B),\triangleright_J} \colon \underbrace{S \triangleright_J J}_{\overset{\text{def}}{=}\mathrm{Ran}_J(S)\diamond J} \hookrightarrow S$$

at S is given by

$$\rho_S^{\mathbf{Rel}(A,B),\triangleright_J} \overset{\text{def}}{=} \epsilon_R,$$

where $\epsilon \colon J^* \circ \mathrm{Ran}_J \Longrightarrow \mathrm{id}_{\mathbf{Rel}(A,B)}$ is the counit of the adjunction $J^* \dashv \mathrm{Ran}_J$.

8.7.6 The Right Skew Monoidal Structure on $\mathbf{Rel}(A, B)$

Proposition 8.7.6.1.1 ▶ The Right J-Skew Monoidal Structure on $\mathbf{Rel}(A, B)$

The category $\mathbf{Rel}(A, B)$ admits a right skew monoidal category structure consisting of

- *The Underlying Category.* The posetal category associated to the poset $\mathbf{Rel}(A, B)$ of relations from A to B of **??** of **??**.
- *The Right Skew Monoidal Product.* The right J-skew monoidal product
$$\triangleleft_J \colon \mathbf{Rel}(A, B) \times \mathbf{Rel}(A, B) \to \mathbf{Rel}(A, B)$$
of Definition 8.7.1.1.1.
- *The Right Skew Monoidal Unit.* The functor
$$\mathbb{1}^{\mathbf{Rel}(A,B),\triangleleft_J} \colon \mathsf{pt} \to \mathbf{Rel}(A, B)$$
of Definition 8.7.2.1.1.
- *The Right Skew Associators.* The natural transformation
$$\alpha^{\mathbf{Rel}(A,B),\triangleright_J} \colon \triangleright_J \circ (\mathsf{id} \times \triangleright_J) \Longrightarrow \triangleright_J \circ (\triangleright_J \times \mathsf{id}) \circ \boldsymbol{\alpha}^{\mathsf{Cats},-1}_{\mathbf{Rel}(A,B),\mathbf{Rel}(A,B),\mathbf{Rel}(A,B)}$$
of Definition 8.7.3.1.1.
- *The Right Skew Left Unitors.* The natural transformation
$$\lambda^{\mathbf{Rel}(A,B),\triangleright_J} \colon \boldsymbol{\lambda}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)} \Longrightarrow \triangleright_J \circ \left(\mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleright} \times \mathsf{id}\right)$$
of Definition 8.7.4.1.1.
- *The Right Skew Right Unitors.* The natural transformation
$$\rho^{\mathbf{Rel}(A,B),\triangleright_J} \colon \triangleright_J \circ \left(\mathsf{id} \times \mathbb{1}^{\mathbf{Rel}(A,B)}_{\triangleright}\right) \Longrightarrow \boldsymbol{\rho}^{\mathsf{Cats}_2}_{\mathbf{Rel}(A,B)}$$
of Definition 8.7.5.1.1.

Proof 8.7.6.1.2 ▶ Proof of Proposition 8.7.6.1.1

Since $\mathbf{Rel}(A, B)$ is posetal, the commutativity of the pentagon identity, the right skew left triangle identity, the right skew right triangle identity, the right skew middle triangle identity, and the zigzag identity is automatic (Categories, Item 4 of Proposition 11.2.7.1.2), and thus $\mathbf{Rel}(A, B)$ together with the data in the statement forms a right skew monoidal category.

Appendices

A Other Chapters

Preliminaries

1. Introduction
2. A Guide to the Literature

Sets

3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
7. Tensor Products of Pointed Sets

Relations

8. Relations
9. Constructions With Relations
10. Conditions on Relations

Categories

11. Categories

Monoidal Categories

12. Constructions With Monoidal Categories

Bicategories

13. Types of Morphisms in Bicategories

Extra Part

14. Notes

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